

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J17 MLB

LAST_MODIFIED=Fri May 31 13:43:58 2013

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
A	0002056619	PRODUCTION RELEASED	2013-05-30

Page	Contents	Sync	Date
1	Table of Contents	MASTER	N/A
2	BOM Configuration	317_DINI	02/06/2013
3	DEBUG LEDS	317_ULTIMATE	05/16/2013
4	Holes/PD parts	317_ULTIMATE	05/16/2013
5	CPU DMI/PEG/FDI/RSVD	317_ULTIMATE	05/16/2013
6	CPU Clock/Misc/JTAG/CFG	317_ULTIMATE	05/16/2013
7	CPU DDR3 Interfaces	317_ULTIMATE	05/16/2013
8	CPU Power	317_ULTIMATE	05/16/2013
9	CPU Ground	317_ULTIMATE	05/16/2013
10	CPU DECOUPLING	317_ULTIMATE	05/16/2013
11	PCH RTC/HDA/JTAG/SATA/CLK	317_ULTIMATE	05/16/2013
12	PCH DMI/FDI/PM/GFX/PCI	317_ULTIMATE	05/16/2013
13	PCH PCI-E/USB	317_ULTIMATE	05/16/2013
14	PCH GPIO/MISC/NCTF	317_ULTIMATE	05/16/2013
15	PCH Power	317_ULTIMATE	05/16/2013
16	PCH Grounds	317_ULTIMATE	05/16/2013
17	PCH DECOUPLING	317_ULTIMATE	05/16/2013
18	CPU & PCH XDP	317_ULTIMATE	05/16/2013
19	Chipset Support	317_ULTIMATE	05/16/2013
20	Project Chipset Support	317_ULTIMATE	05/16/2013
21	CPU Memory S3 Support	317_ULTIMATE	05/16/2013
22	DDR3 VREF MARGINING	317_ULTIMATE	05/16/2013
23	DDR3 SO-DIMM Connector A Slot0	317_ULTIMATE	05/16/2013
24	DDR3 SO-DIMM Connector A Slot1	317_ULTIMATE	05/16/2013
25	DDR3 SO-DIMM CONNECTOR B SLOT0	317_ULTIMATE	05/16/2013
26	DDR3 SO-DIMM CONNECTOR B SLOT1	317_ULTIMATE	05/16/2013
27	DDR3 ALIASES AND BITSWAPS	317_ULTIMATE	05/16/2013
28	Thunderbolt Host (1 of 2)	317_ULTIMATE	05/16/2013
29	Thunderbolt Host (2 of 2)	317_ULTIMATE	05/16/2013
30	Thunderbolt Power Support	317_ULTIMATE	05/16/2013
31	Thunderbolt Connector A	317_ULTIMATE	05/16/2013
32	Thunderbolt Connector B	317_ULTIMATE	05/16/2013
33	TBT DDC Crossbar	317_ULTIMATE	05/16/2013
34	AIRPORT/BT	317_ULTIMATE	05/16/2013
35	SATA/SSD Connectors	317_ULTIMATE	05/16/2013
36	HDD/SSD Temp Sense	317_ULTIMATE	05/16/2013
37	ETHERNET PHY (CAESAR IV)	317_ULTIMATE	05/16/2013
38	Ethernet Support & Connector	317_ULTIMATE	05/16/2013
39	SD READER CONNECTOR	317_ULTIMATE	05/16/2013
40	Camera Controller	317_ULTIMATE	05/16/2013
41	Camera Controller Support	317_ULTIMATE	05/16/2013
42	Internal DP Support	317_ULTIMATE	05/16/2013
43	Internal DP MUXing	317_ULTIMATE	05/16/2013
44	EXTERNAL USB PORTS A & B	317_ULTIMATE	05/16/2013
45	EXTERNAL USB PORTS C & D	317_ULTIMATE	05/16/2013
46	SMC	317_ULTIMATE	05/16/2013
47	SMC Support	317_ULTIMATE	05/16/2013
48	SPI and Debug Connector	317_ULTIMATE	05/16/2013
49	SMBus Connections	317_ULTIMATE	05/16/2013
50	I and V Sense	317_ULTIMATE	05/16/2013
51	I and V Sense(Continued)	317_ULTIMATE	05/16/2013
52	Temperature Sensors	317_ULTIMATE	05/16/2013
53	System Fan	317_ULTIMATE	05/16/2013
54	AUDIO: CODEC/REGULATORS	317_ULTIMATE	05/16/2013
55	AUDIO: HEADPHONE AMP	317_ULTIMATE	05/16/2013
56	AUDIO: LEFT SPKR AMP	317_ULTIMATE	05/16/2013
57	AUDIO: RIGHT SPKR AMP	317_ULTIMATE	05/16/2013
58	AUDIO: Jack, Mikey, CHS Switch	317_ULTIMATE	05/16/2013
59	Audio: Spkr/Mic Conn.	317_ULTIMATE	05/16/2013

Page	Contents	Sync	Date
60	AUDIO: Detects/Grounding	J17_ULTIMATE	05/16/2013
61	AUDIO: Speaker ID	J17_ULTIMATE	05/16/2013
62	Power Connectors / VReg G3Hot	J17_ULTIMATE	05/16/2013
63	VReg CPU VCC Cntl	J17_ULTIMATE	05/16/2013
64	VReg CPU VCC Phases	J17_ULTIMATE	05/16/2013
65	VReg CPU VCC Phase 4	J17_ULTIMATE	05/16/2013
66	VReg VDDQ S3	J17_ULTIMATE	05/16/2013
67	VReg 3.3V S5/5V S4	J17_ULTIMATE	05/16/2013
68	GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS	J17_ULTIMATE	05/16/2013
69	VReg GPU Core Phases	J17_ULTIMATE	05/16/2013
70	VReg GPU Core Phases	J17_ULTIMATE	05/16/2013
71	VREG GPU CORE PHASE 4	J17_ULTIMATE	05/16/2013
72	BACKLIGHT LED DRIVER	J17_ULTIMATE	05/16/2013
73	Backlight LED Driver	J17_ULTIMATE	05/16/2013
74	Backlight Controller	J17_ULTIMATE	05/16/2013
75	FET-Controlled S0 and S4	J17_ULTIMATE	05/16/2013
76	PM Regulator Enables	J17_ULTIMATE	05/16/2013
77	PM Power Good	J17_ULTIMATE	05/16/2013
78	KEPLER PCI-E	J17_PETE	01/21/2013
79	KEPLER CORE/FB POWER	J17_PETE	01/21/2013
80	KEPLER FRAME BUFFER I/F	J17_PETE	11/02/2012
81	GDDR5 Frame Buffer A	J17_PETE	11/02/2012
82	GDDR5 Frame Buffer B	J17_PETE	11/02/2012
83	KEPLER EDP/DP/GPIO	J17_PETE	11/02/2012
84	KEPLER GPIOs,CLK & STRAPS	J17_PETE	11/02/2012
85	GPU SIGNAL & POWER ALIASES	J17_PETE	01/21/2013
86	KEPLER PEX PWR/GNDS	J17_PETE	01/21/2013
87	Power Connectors/Aliases	J17_ULTIMATE	05/16/2013
88	Signal Aliases	J17_ULTIMATE	05/16/2013
89	Unused Signal Aliases	J17_ULTIMATE	05/16/2013
90	Functional / ICT Test	J17_ULTIMATE	05/16/2013
91	J17 RULE DEFINITIONS	J17_ULTIMATE	05/16/2013
92	DDR3 Constraints	J17_ULTIMATE	05/16/2013
93	CPU PCIe Constraints	J17_NICK	12/14/2012
94	PCH PCIe/DMI Constraints	J17_ULTIMATE	05/16/2013
95	SATA/FDI/XDP Constraints	J17_ULTIMATE	05/16/2013
96	PCH and BR Constraints	J17_ULTIMATE	05/16/2013
97	USB/Ethernet/SD Constraints	J17_ULTIMATE	05/16/2013
98	SMBus/Sensor Constraints	J17_ULTIMATE	05/16/2013
99	VReg Constraints	J17_ULTIMATE	05/16/2013
100	CPU VReg Constraints	J17_ULTIMATE	05/16/2013
101	Platform VReg Constraints	J17_ULTIMATE	05/16/2013
102	TBT/DP Constraints	J17_ULTIMATE	05/16/2013
103	GDDR5/GPU Constraints	DB_AARON	03/13/2012
104	BLC Constraints	J17_ULTIMATE	05/16/2013
105	GPU VREG CONSTRAINTS	J17_ULTIMATE	05/16/2013
106	MISC-CONSTRAINTS 1	J17_ULTIMATE	05/16/2013

DRAWING
TITLE=J17
ABBREV=DRAWING
LAST_MODIFIED=Fri May 31 13:43:58 2013

DRAWING TITLE		SCHEM, MLB, J17	
	Apple Inc.		DRAWING NUMBER
			051-9884
		SIZE	D
		REVISION	A.0.0
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		SHEET 1 OF 106	

BOM VARIANTS

BOM NUMBER	BOM NAME	BOM OPTIONS
639-4206	PCBA, MLB, GOOD, HYNIX, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_HYNIX, FBA, FBB, SSD:Y, EEEE:F8YL
639-4418	PCBA, MLB, GOOD, ELPIDA, J17	J17_COMMON, J17, CPU:GOOD, GPU:GK187EGX, FB:2G_ELPIDA, FBA, FBB, SSD:Y, EEEE:FCTC
985-0047	PCBA, MLB, DEV, J17	DEVELOPMENT, J17_DEVEL

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754532	1	H5W, SR14E, PRQ, C0, 3.20, 84W, 4+2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD
33754505	1	H5W, QE70, Q5, C0, 3.26, 84W, 4+2, 1.15, 6M, LGA	CPU	CRITICAL	CPU:GOOD_TDP

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0080	1	SOCKET, MOLEX, LGA1150, CPU- HF	U0500	CRITICAL	

BOM Groups

BOM GROUP	BOM OPTIONS
J17_COMMON	COMMON, ALTERNATE, J17_COMMON1, J17_PROGPARTS
J17_COMMON1	XDP, SPEAKERID, TBTHV:P12V, CPUVCC:4PHASE, EXT_GPU:YES, VDDQ:P1V5
J17_PROGPARTS	SMC:PROG, BOOTROM:PROG, TBTRM:PROG, CIVROM:PROG, CAMROM:PROG, BLCMCU:PROG
J17_DEVEL	XDP_CONN, LPCPLUS, DDRVREF_DAC
DEVEL_SENSORS	TEMPSNSDEV, AP_ISNS:Y
J17_PRODUCTION	AP_ISNS:N

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM_OPTION
33754541	1	1C, SR198, 287, LPT-D, C2, PRQ, 23X22MM, FCBG4708	U1100	CRITICAL	
33754562	1	1C, QEW, LPT-D, C2, SQ5, 23X22MM, FCBG4708	U1100	CRITICAL	PCH_TDP
33851113	1	1C, T87, CR-4C, B1, PRQ, 288 FCBG4, 12X12MM	U2800	CRITICAL	
335S0616	1	1C, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL	

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3880	1	IC, PROGRMD, EFI ROM, V0095, J17	U5210	CRITICAL	BOOTROM:PROG
335S0807	1	IC, 64 MBIT SPI SERIAL FLASH	U5210	CRITICAL	BOOTROM:BLANK
341S3874	1	IC, PROG, SMC-A3, V2.14F9, PVT, J17	U5000	CRITICAL	SMC:PROG
338S1159	1	IC, SMC, LM4FS1BH5BBCIGR, SCPL	U5000	CRITICAL	SMC:BLANK
341S3860	1	IC, EEPROM, CR, V23.10, J17	U2890	CRITICAL	TBTROM:PROG
335S0865	1	IC, EEPROM, SERIAL, 8KB, MLP8	U2890	CRITICAL	TBTROM:BLANK
341S3735	1	IC, ENET 1MBIT, SPI, ROM, V1.13 J17	U3990	CRITICAL	CIVROM:PROG
335S0862	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK
341S3778	1	IC, CAMERA FLASH, V7230, J16/17	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM:BLANK
341S3674	1	IC, BLC, MCU, PROPROGRAMMED, V9204, DB	U8100	CRITICAL	BLCMCU:PROG
337S3978	1	IC, BLC MCU LCU2132FBD64/01, LQFP64	U8100	CRITICAL	BLCMCU:BLANK

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7896	1	LABEL, MLB, 2D	EEEE_F8YL	CRITICAL	EEEE:F8YL
825-7896	1	LABEL, MLB, 2D	EEEE_FCTC	CRITICAL	EEEE:FCTC

J17 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
951-9884	1	SCH, MLB, J17	SCH1	CRITICAL	J17
820-3478	1	PCBF, MLB, J17	PCB1	CRITICAL	J17

J17 ALTERNATES

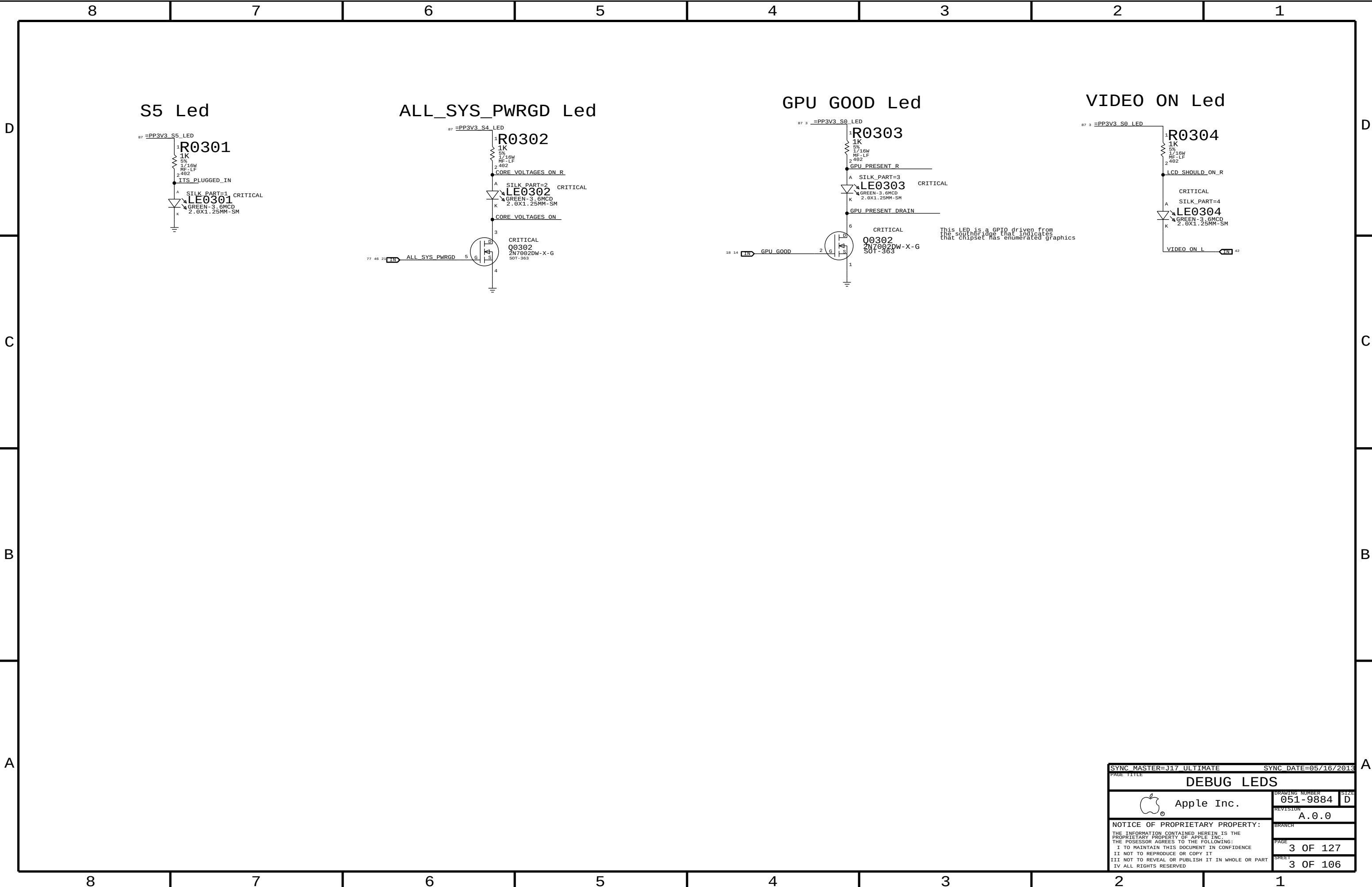
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB Diodes Array
377S0155	377S0104		ALL	USB Diode
157S0084	157S0058		ALL	Enet Magnetics
376S0975	376S1081		ALL	P/NCH DUAL FET
128S0365	128S0368		ALL	150UF CAPS BLK
138S0803	138S0804		ALL	2.2UF CAPS SOFT
126-0161	126-0160		ALL	56UF CAPS 10X12 THD
197S0481	197S0480		Y1950	25MHz PCH Xtal
197S0479	197S0478		ALL	12MHz Cam Xtal
372S0186	372S0185		ALL	Alt temp diode
107S0251	107S0249		R5520	Alt 2mOhm sense
107S0254	107S0241		R5450, R5480	Alt 5mOhm sense
107S0255	107S0240		R5450, R5480	Alt 1mOhm sense
341S3747	341S3735		U3990	ENET ROM,ATMEL,V1.13
138S0681	138S0638		ALL	10uF Caps
376S1106	376S0969		ALL	N-Ch FET
138S0860	138S0775		ALL	Single-source 1uF
138S0859	138S0788		ALL	Single-source 10uF

Y4200 and Y8105

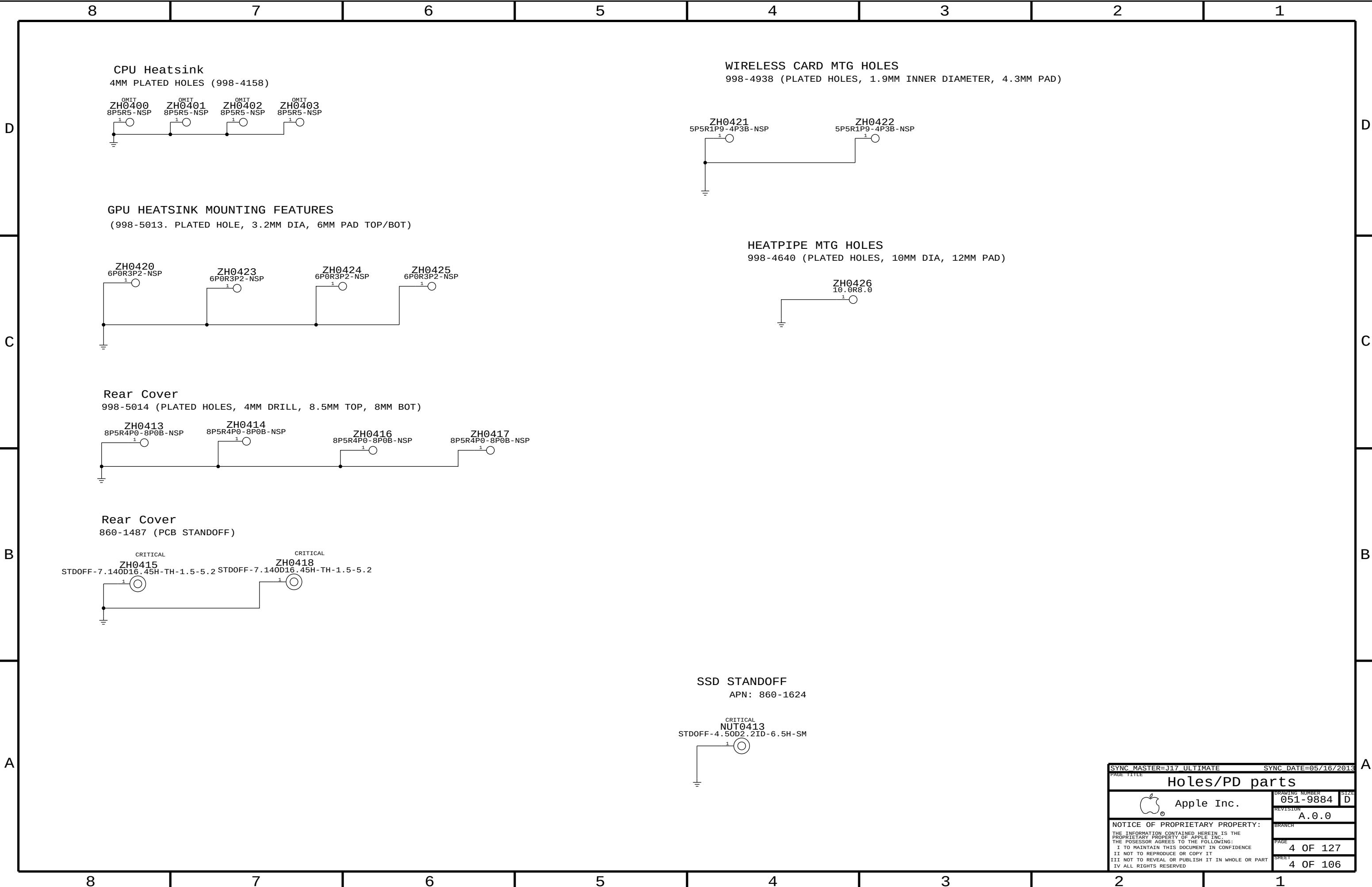
GPU and VRAM

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4438	1	IC, GPU, NV, GK107-EXX, PS-A2	U8700	CRITICAL	GPU: GK107EGX
333S0638	4	IC, GDDR5, 2GBIT1, 64KX32, GEMMA-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB: 2G_HYNIX
333S0695	4	IC, GDDR5, 2GBIT1, 64KX32, B-DIE, 1708 FBGA	U9200, U9250, U9300, U9350	CRITICAL	FB: 2G_ELPIDA

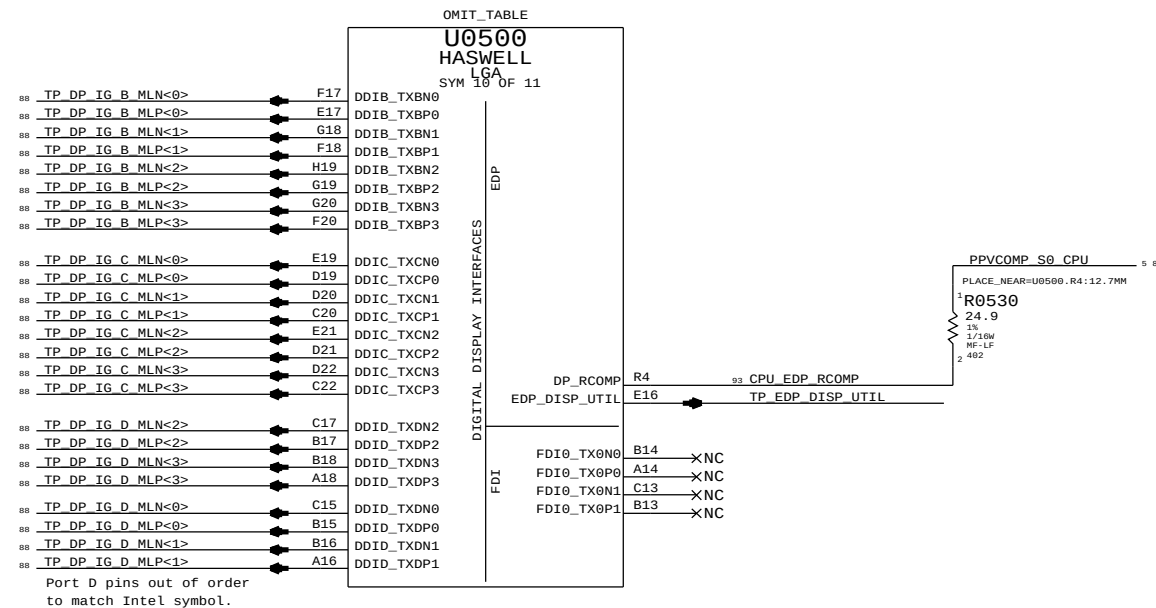
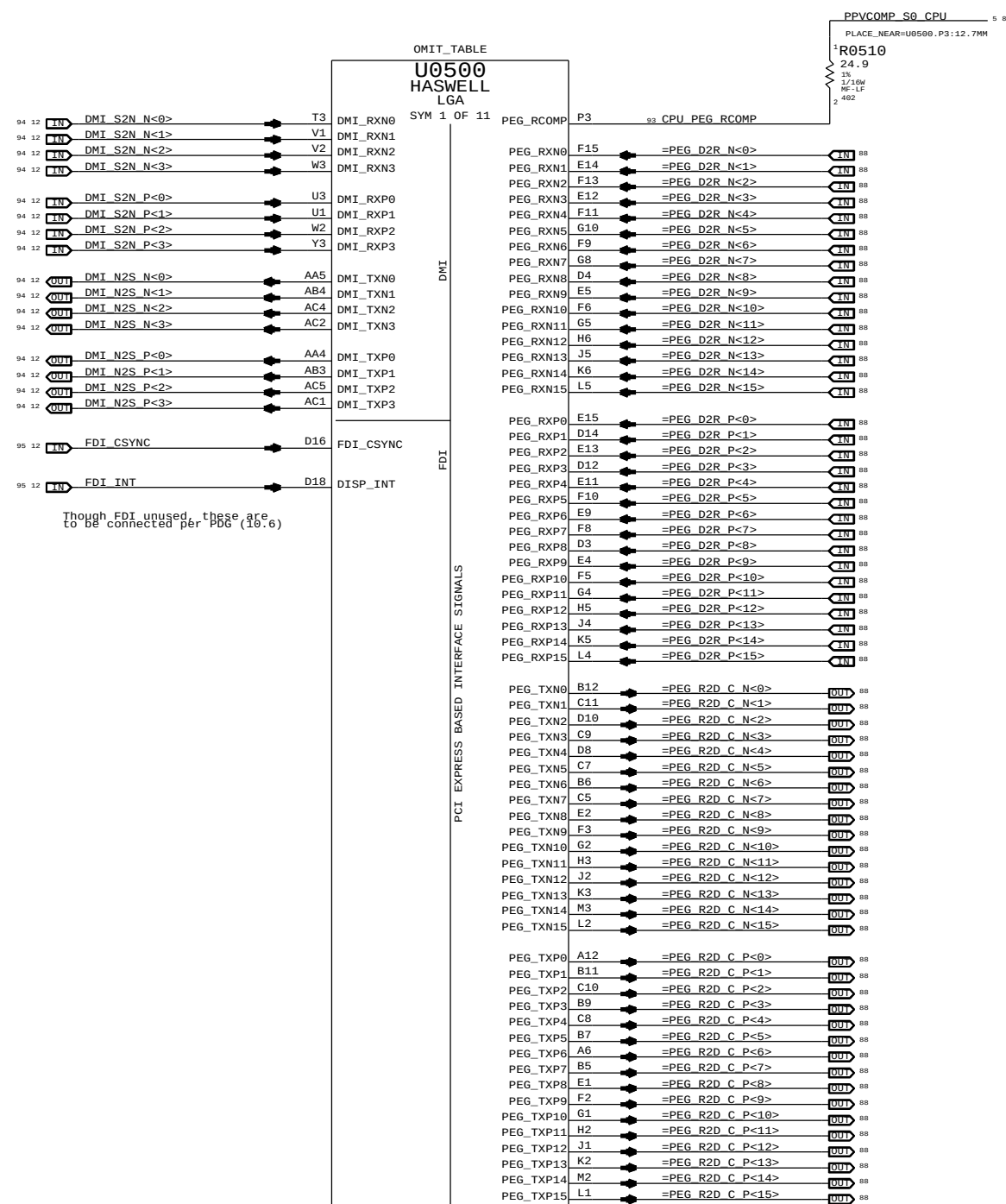
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BRANCH		PAGE	
		2 OF 127	
SHEET		PAGE	
2 OF 106			



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
DEBUG LEDS			
 Apple Inc.		DRAWING NUMBER	051-9884
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		BRANCH	
		PAGE	3 OF 127
		SHEET	3 OF 106



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
Holes/PD parts			
 Apple Inc.		DRAWING NUMBER	051-9884
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		PAGE	4 OF 127
		SHEET	4 OF 106



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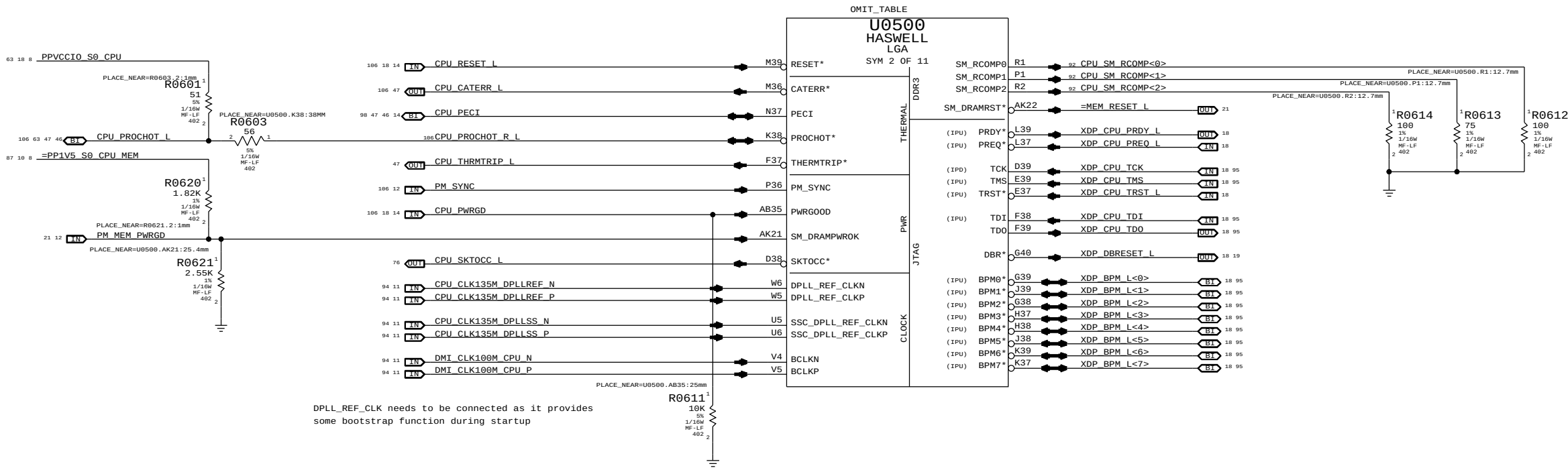
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B

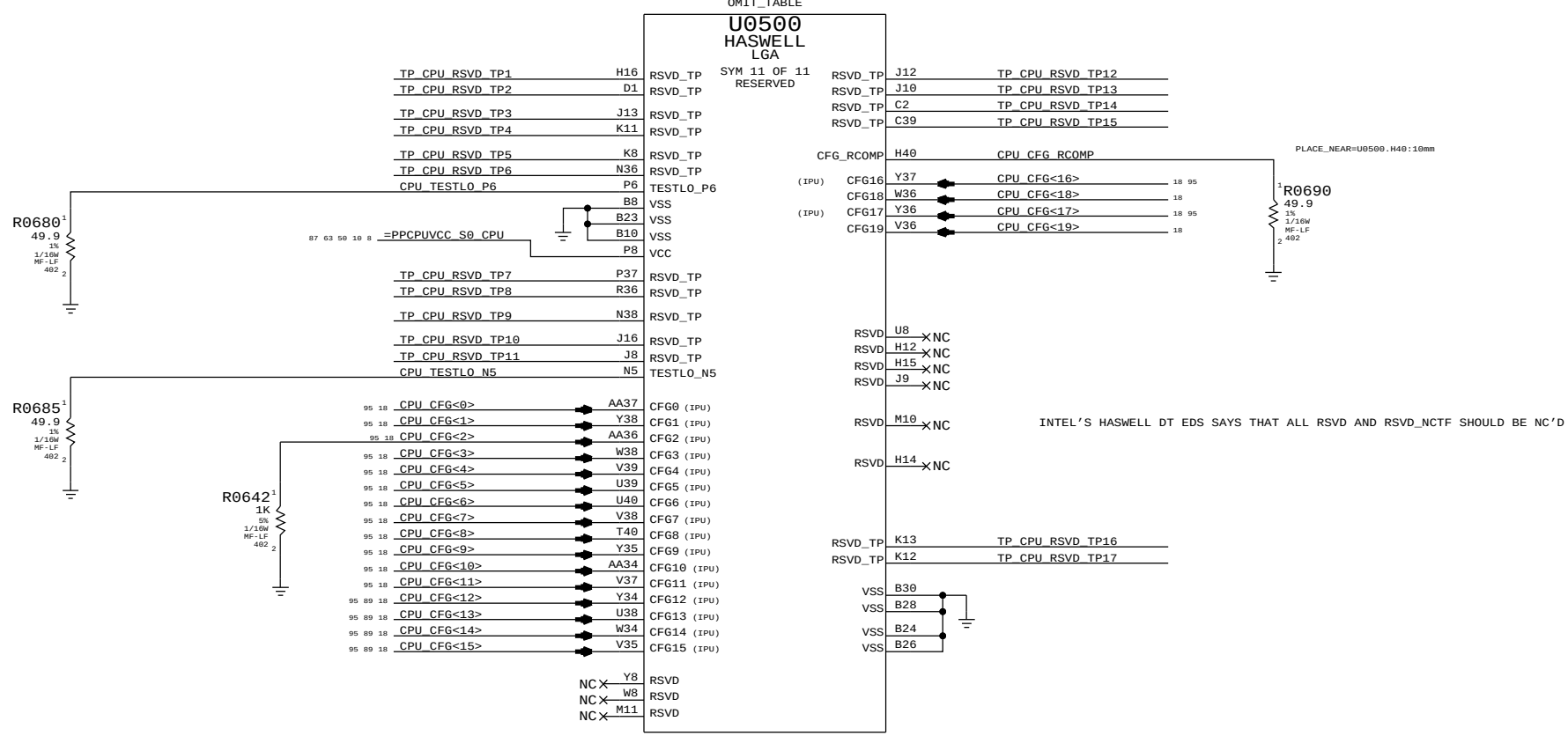
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CFG [7] :PEG DEFER TRAINING	1 = (DEFAULT) IMMEDIATELY AFTER xRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION	11 = 1 X16 (default) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE	1 = DISABLED(default) 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL	1 = NORMAL OPERATION(default) 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL	1 = NORMAL OPERATION(default) 0 = LANES REVERSED



DPLL_REF_CLK needs to be connected as it provides some bootstrap function during startup

TESTLO_X pins to be individually connected to GND via a resistor matching the nominal trace impedance +/-20%



INTEL'S HASWELL DT EDS SAYS THAT ALL RSVD AND RSVD_NCTF SHOULD BE NC'D

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

CPU Clock/Misc/JTAG/CFG

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051-9884

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6 OF 127

6 OF 106



INTEL'S HASWELL DT EDS SAYS THAT ALL RSVD AND RSVD_NCTF SHOULD BE NC'D

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

CPU DDR3 Interfaces



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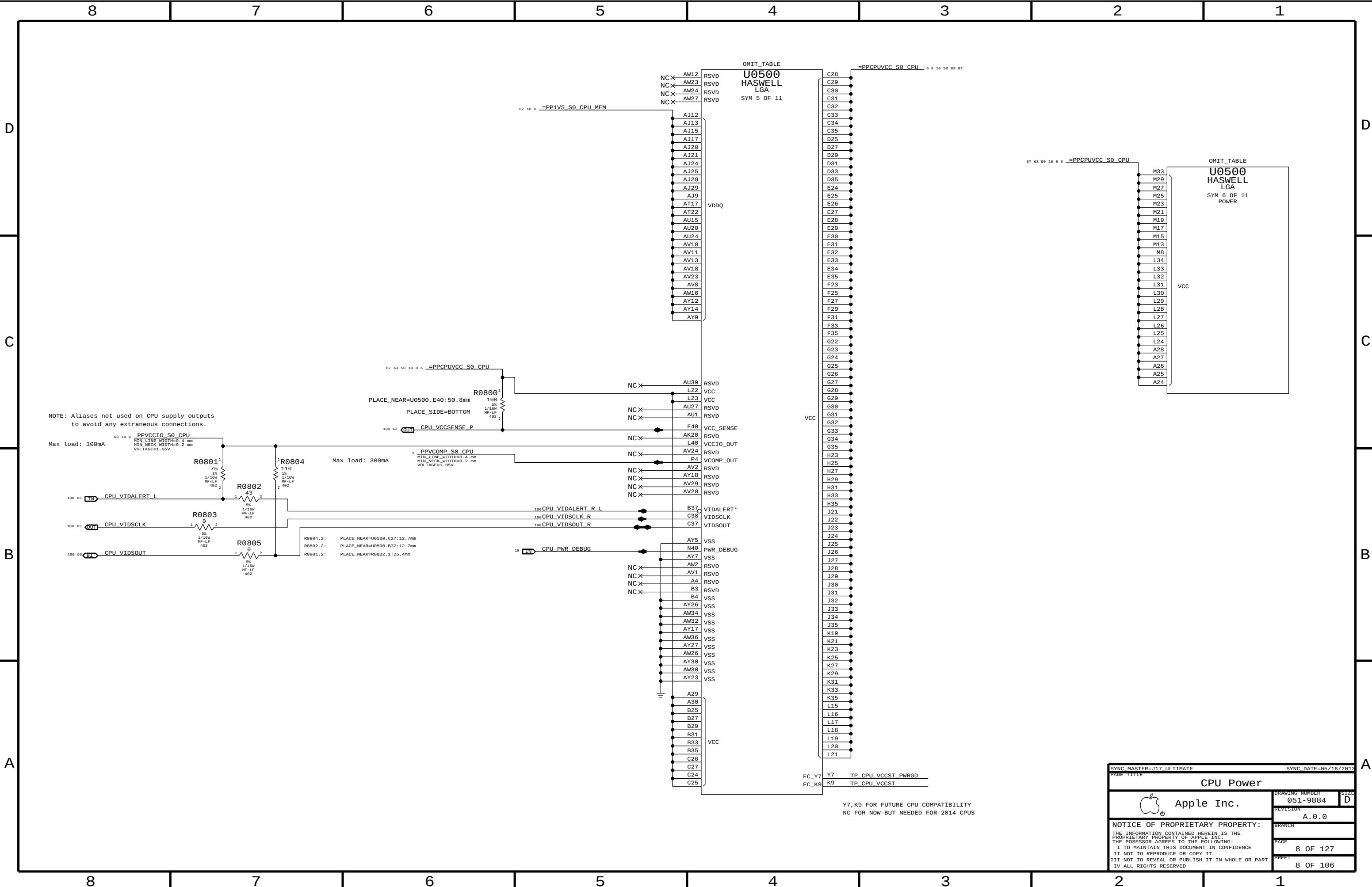
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PAGE

7 OF 127

SHEET

7 OF 106



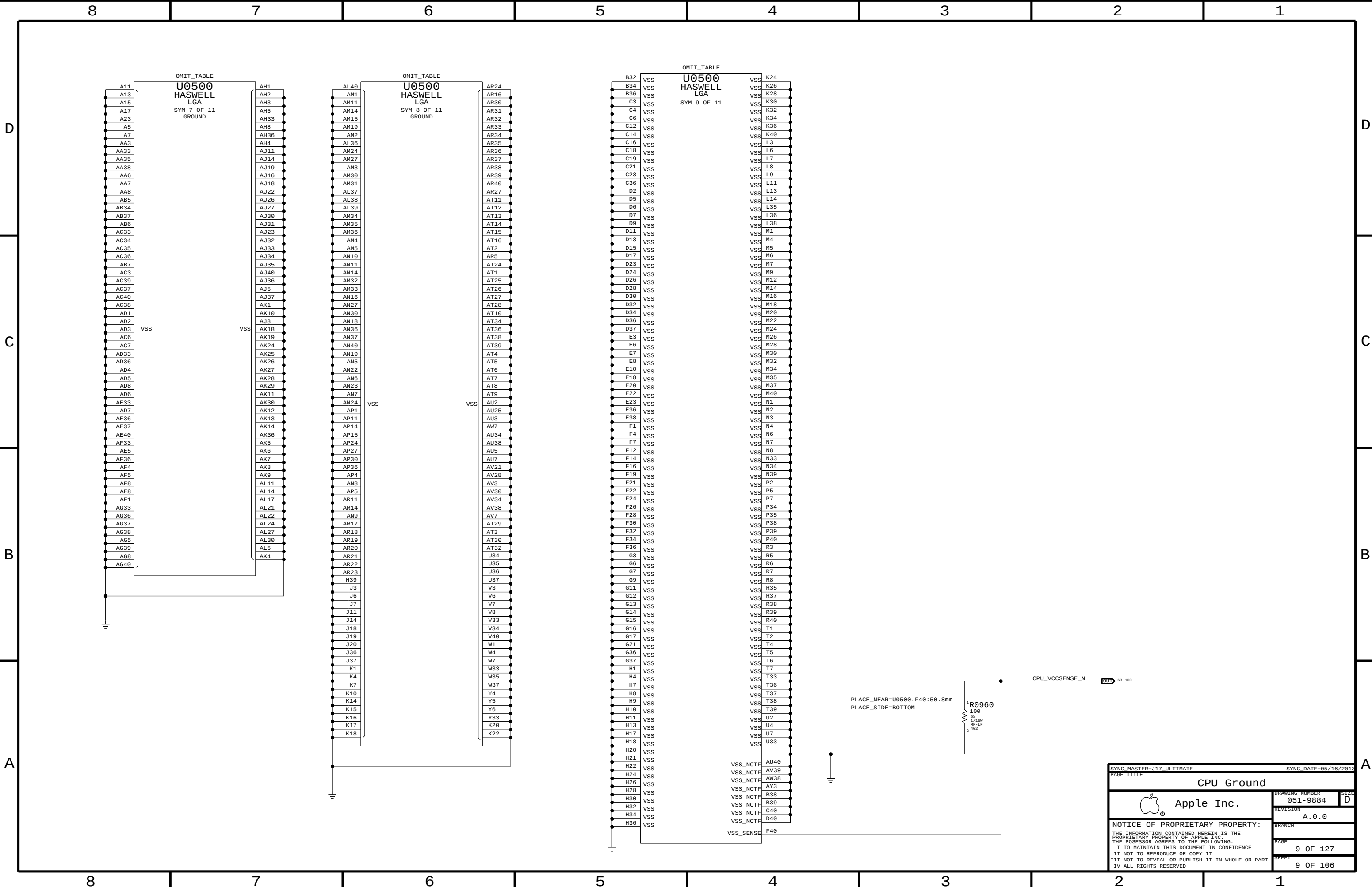
NOTE: Aliases not used on CPU supply outputs to avoid any extraneous connections.

Max load: 300mA

Max load: 300mA

Y7,K9 FOR FUTURE CPU COMPATIBILITY
NC FOR NOW BUT NEEDED FOR 2014 CPUS

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		CPU Power	
		DRAWING NUMBER	051-9884
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		PAGE	8 OF 127
		SHEET	8 OF 106

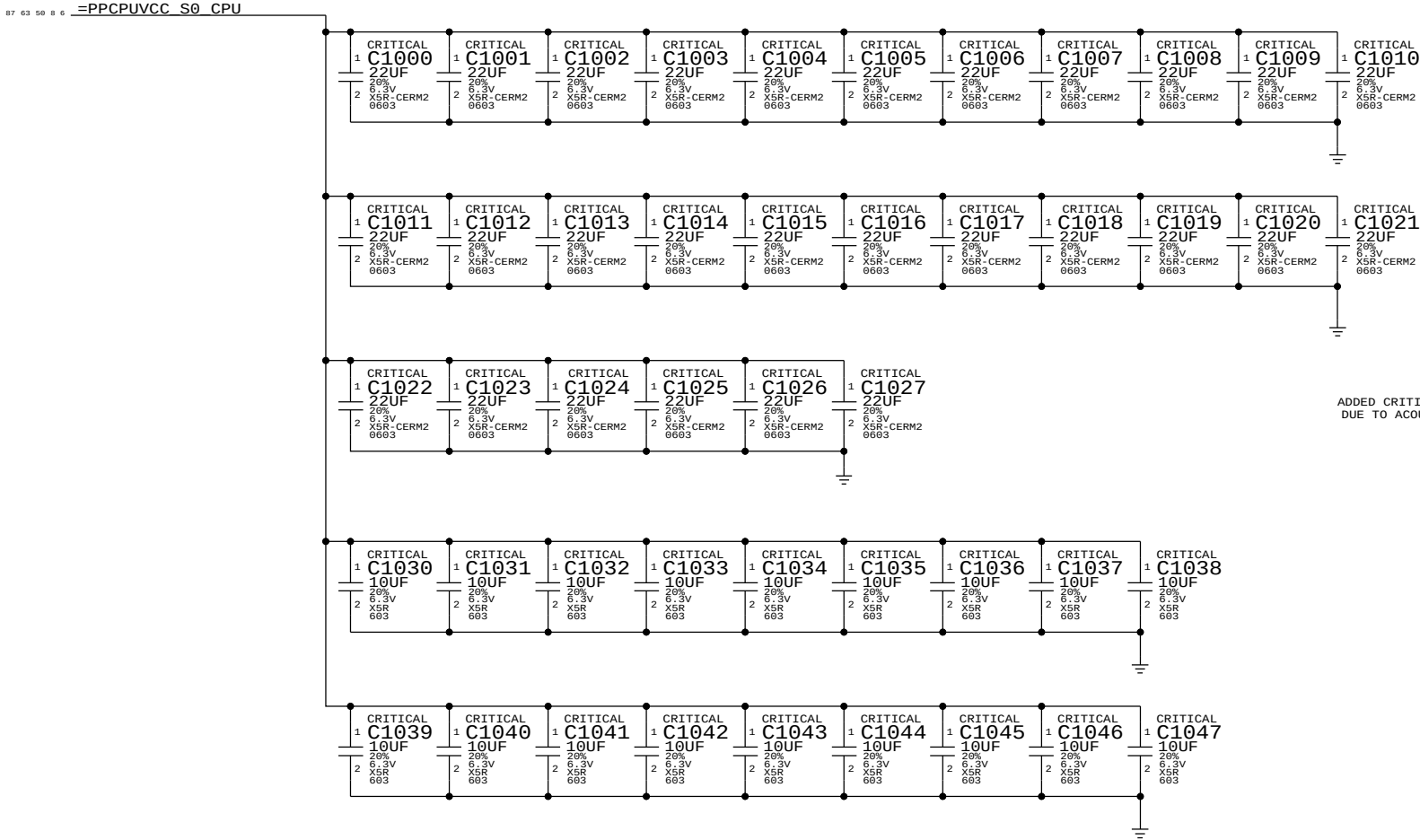


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CPU Ground		DRAWING NUMBER	051-9884
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CPU VCORE DECOUPLING

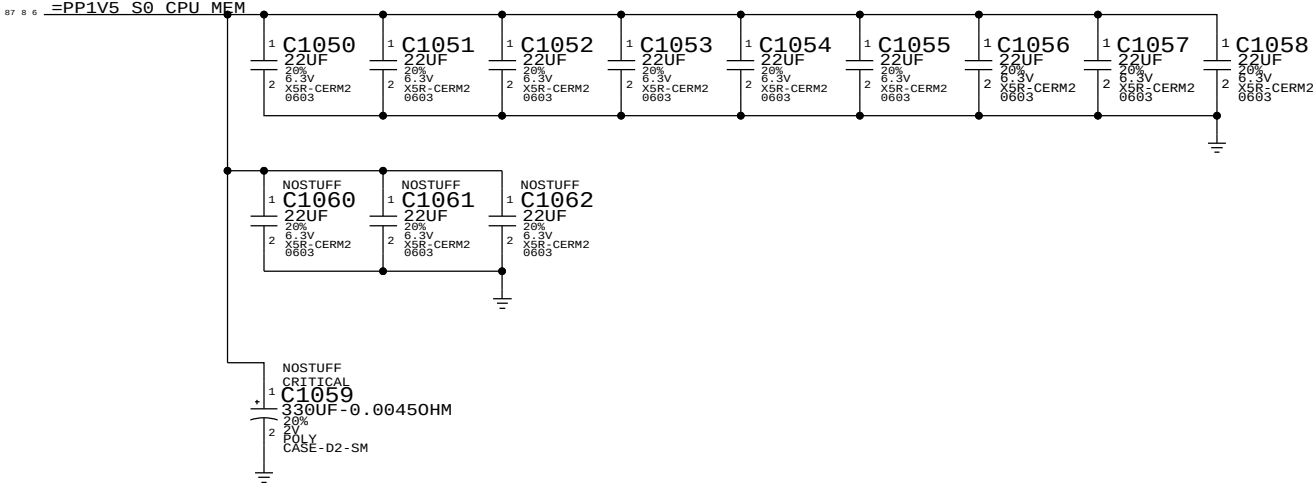
Intel Recommendation:22x 22UF 0805,topside (18 inside cavity, 4 north of processor),8x 470uF bulk caps(5 stuffed,3 no-stuffed)
Apple Implementation:28x 22UF 0603 per Harold
18x 10UF 0603 placed inside socket cavity

Layout Note: These caps should be placed symmetrically on Top and Bottom sides.
BULK CAPS ON CPU VREG PAGE 71



Memory (CPU VCCDDR) DECOUPLING

Intel Recommendation:9x 22UF 0805 near CPU power pins
Apple Implementation:9x 22UF 0603 per Harold
Layout Note: These caps should be placed symmetrically on Top and Bottom sides.



SYNC_MASTER=J17_ULTIMATE		SYNC_DATE=05/16/2013	
PAGE TITLE		CPU DECOUPLING	
		DRAWING NUMBER	051-9884
		SIZE	D
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		BRANCH	
		PAGE	10 OF 127
		SHEET	10 OF 106



NOTE: SSC control is ganged on PCIe 0-3 and 4-7 clocks.
PEG-attached (CPU) PCIe devices must use one set,
while PCH-attached PCIe devices use the other set.
If 2 or less devices are attached to PEG the
CLKOUT_PEG outputs can be used for those devices.

SYNC MASTER=117 ULTIMATE

SYNC DATE=05/16/2013

PAGE 11111

051-9884

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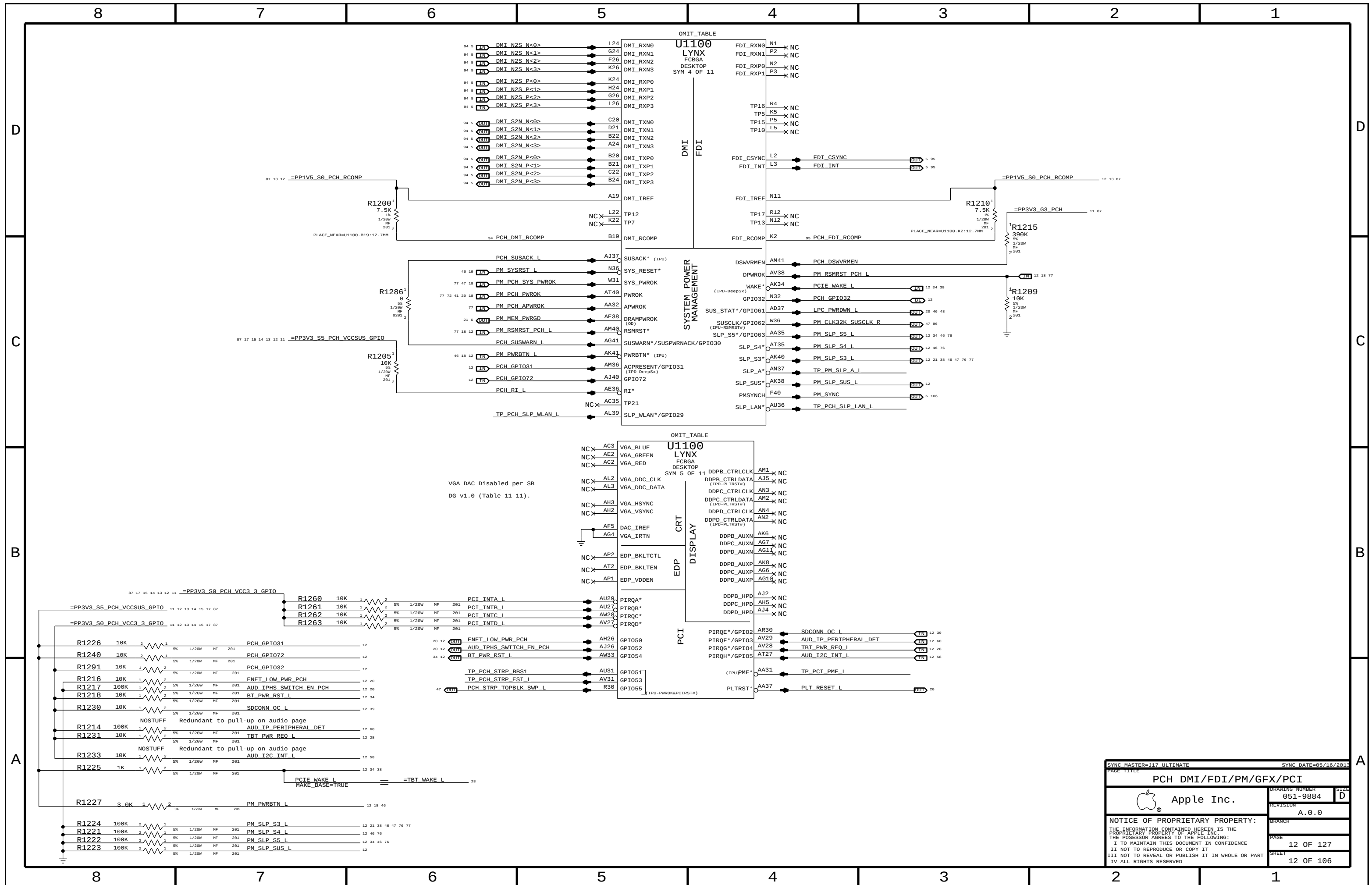
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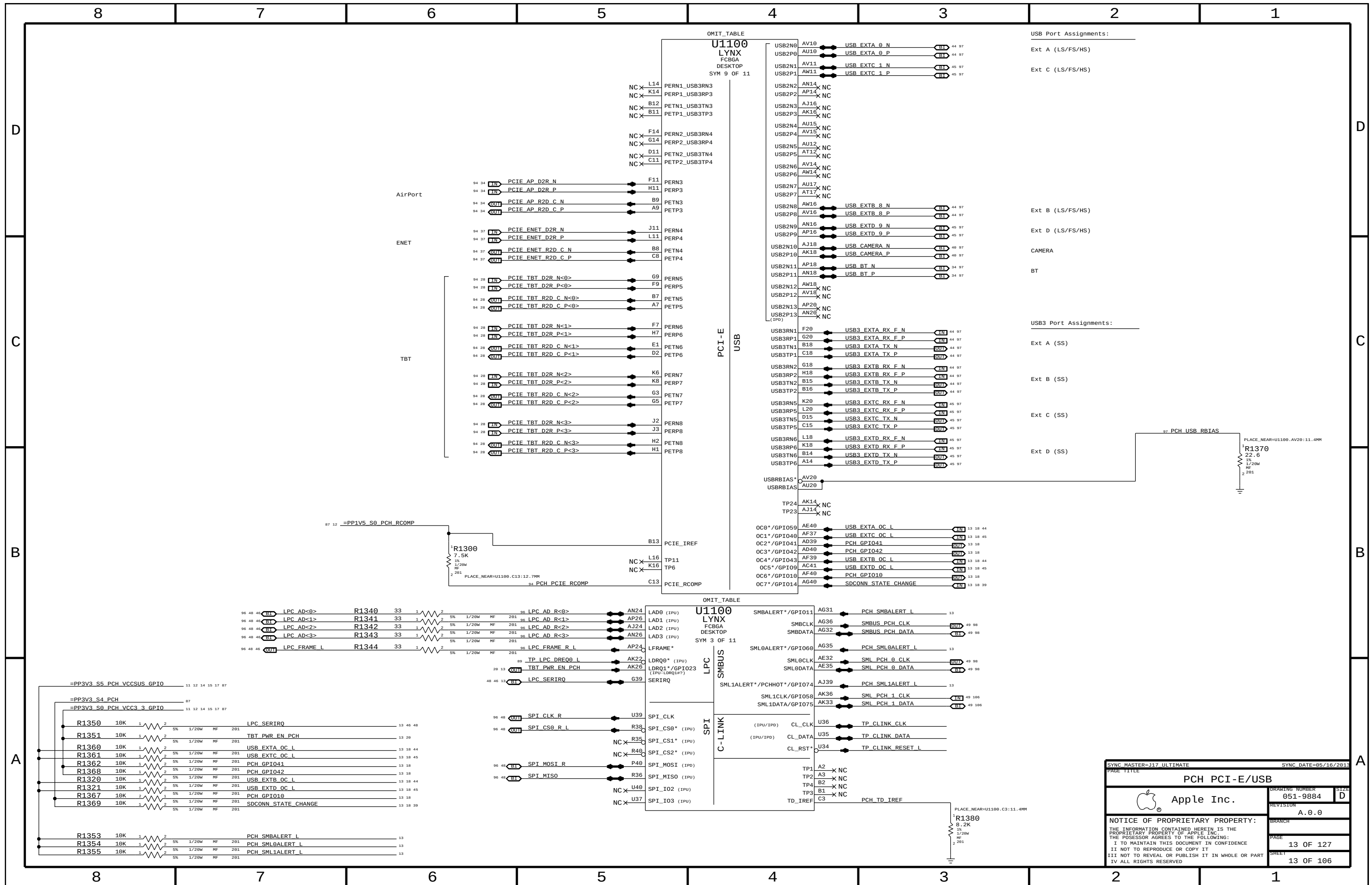
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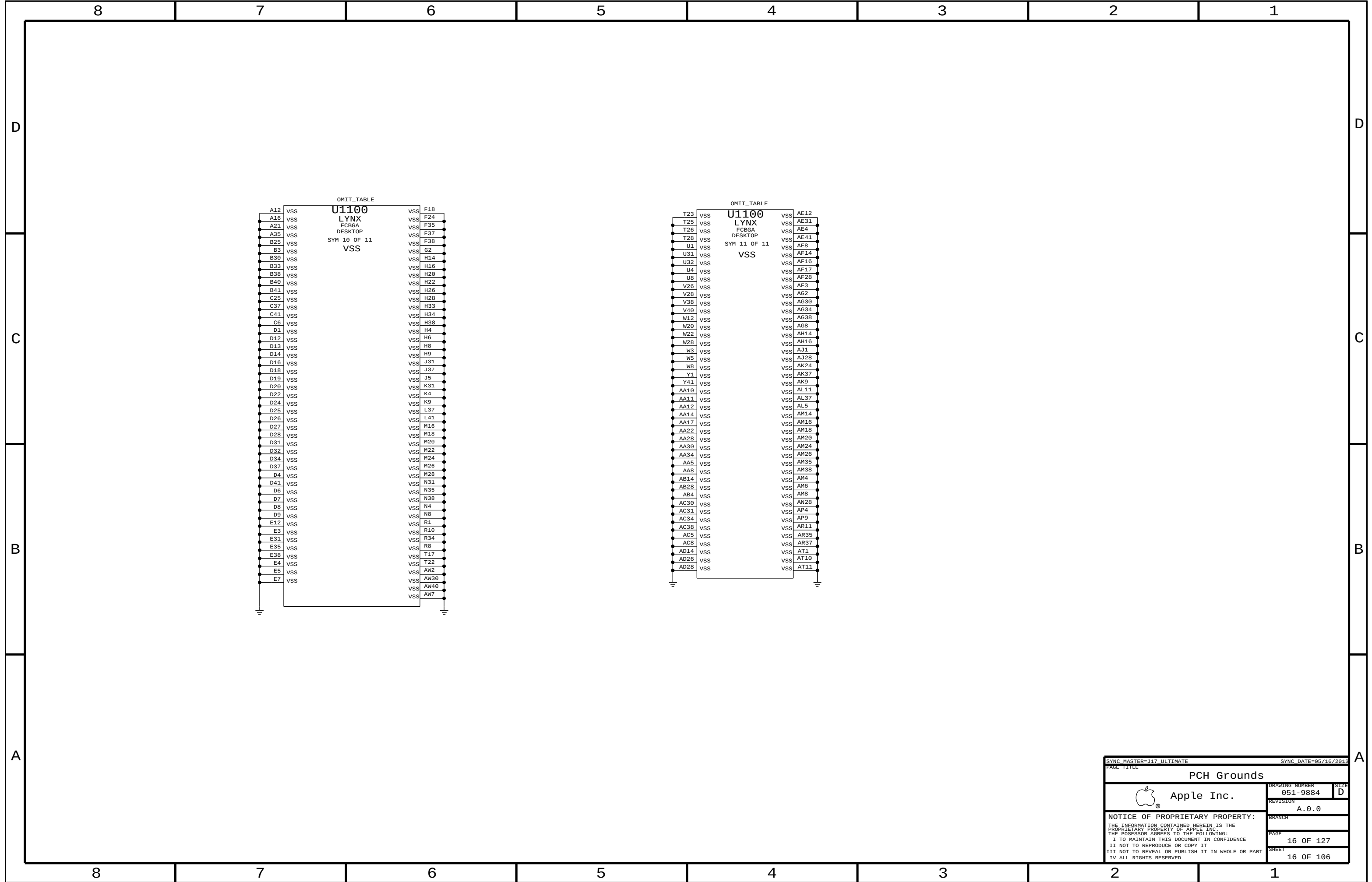
11 OF 127

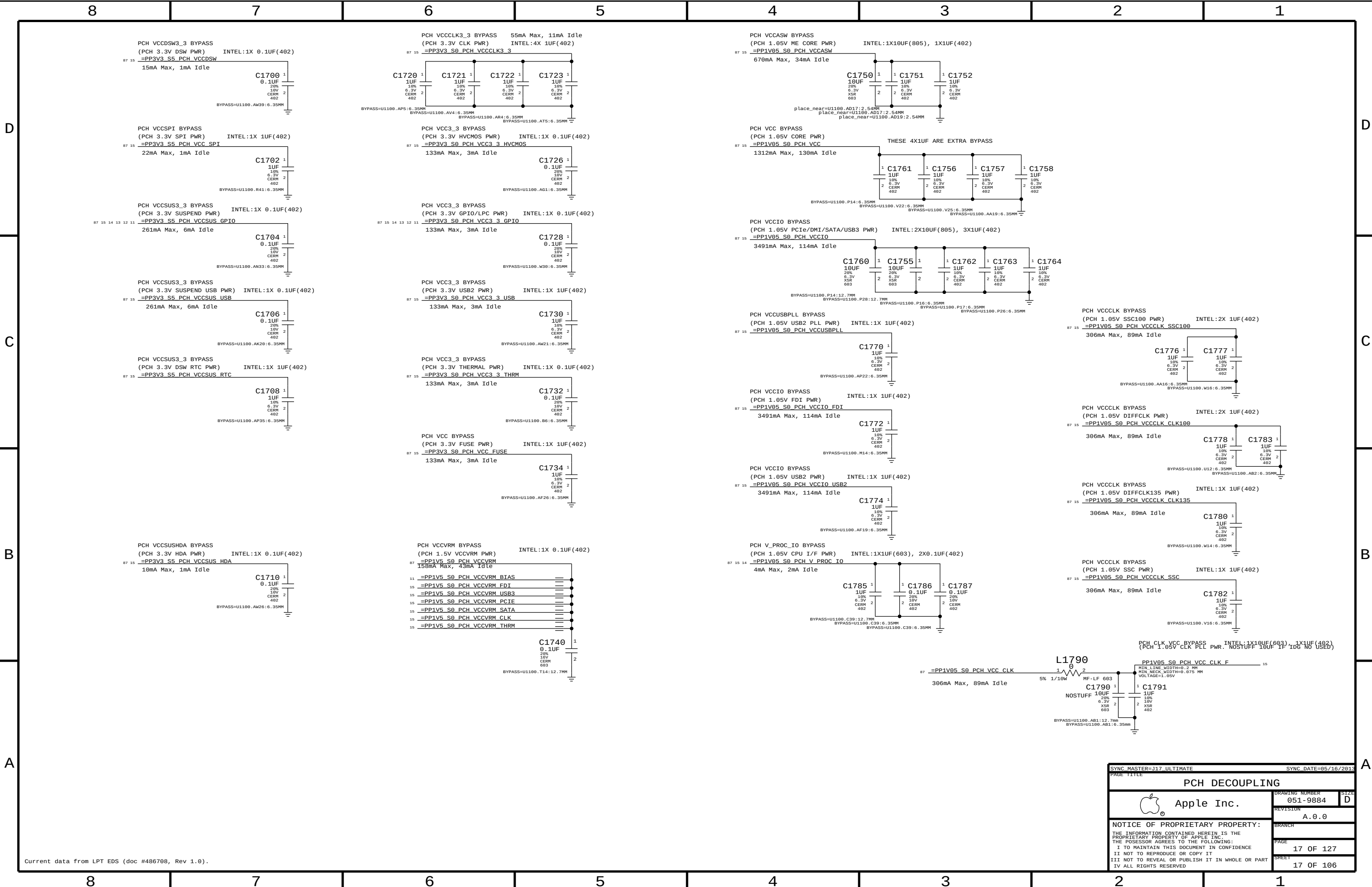
SHEET

11 OF 106





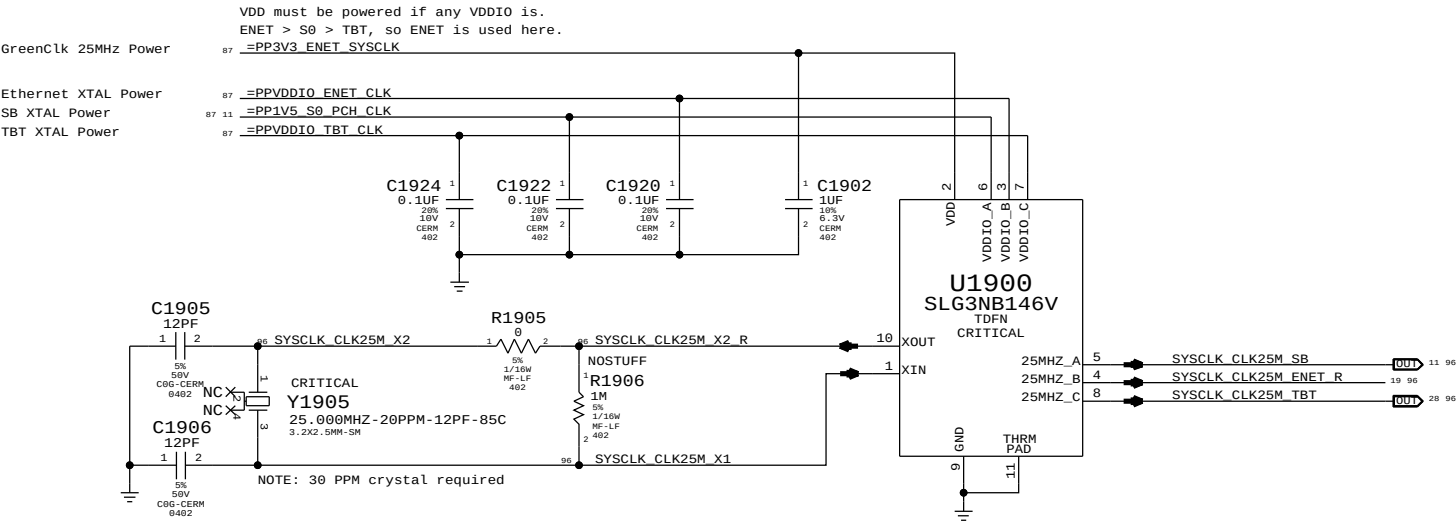




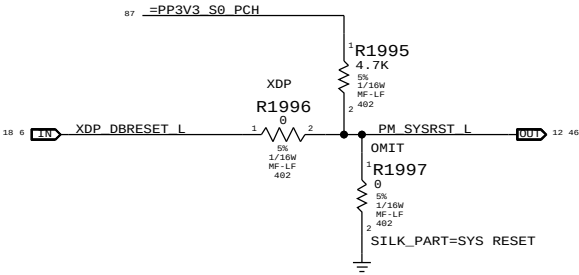
Current data from LPT EDS (doc #486708, Rev 1.0).

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		PCH DECOUPLING	
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		PAGE	17 OF 127
		SHEET	17 OF 106

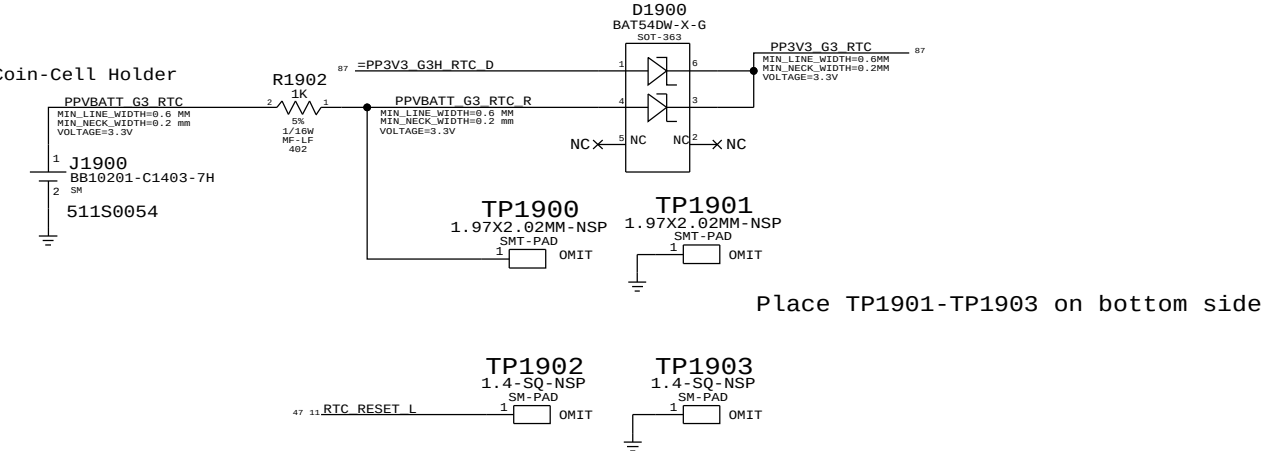
System 25MHz Clock Generator



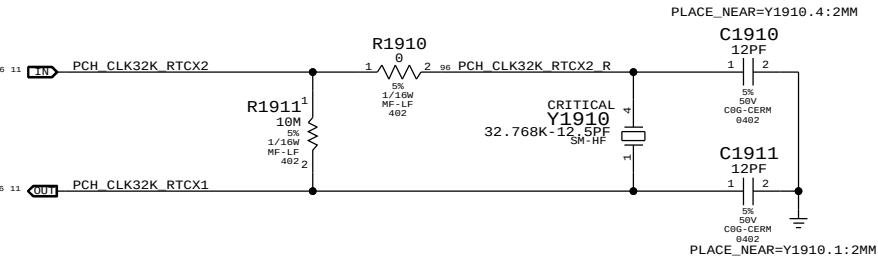
PCH Reset Button



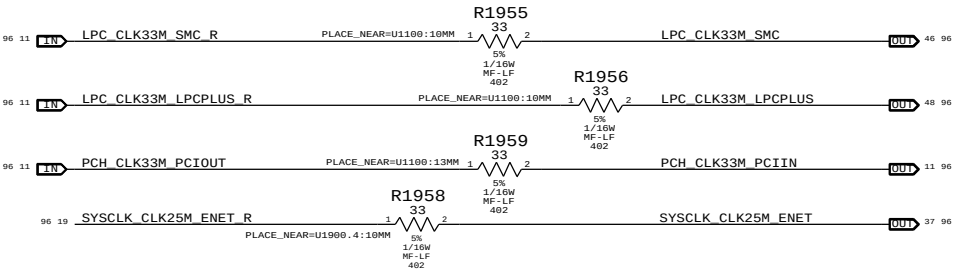
RTC Power Sources



PCH RTC Crystal

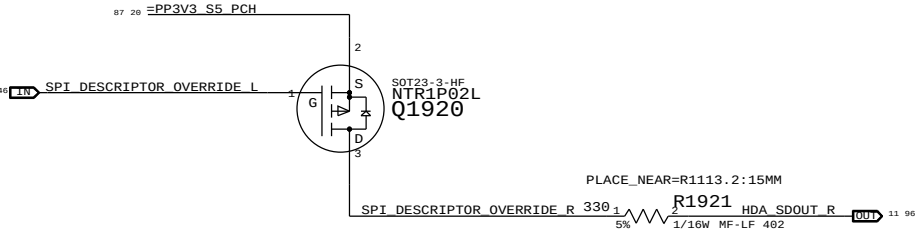


Clock series termination

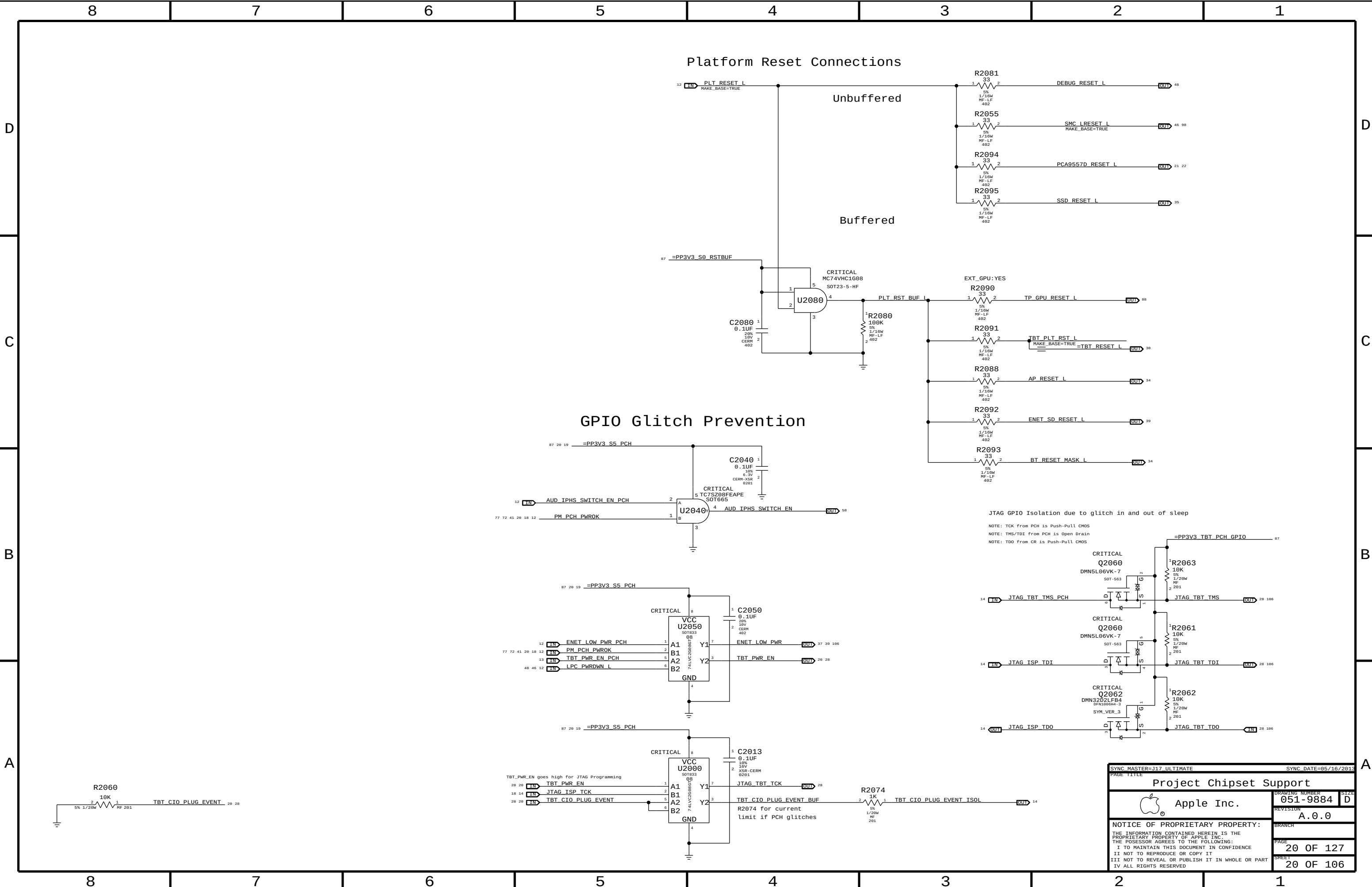


PCH ME Disable Strap

PCH uses HDA_SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting.



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
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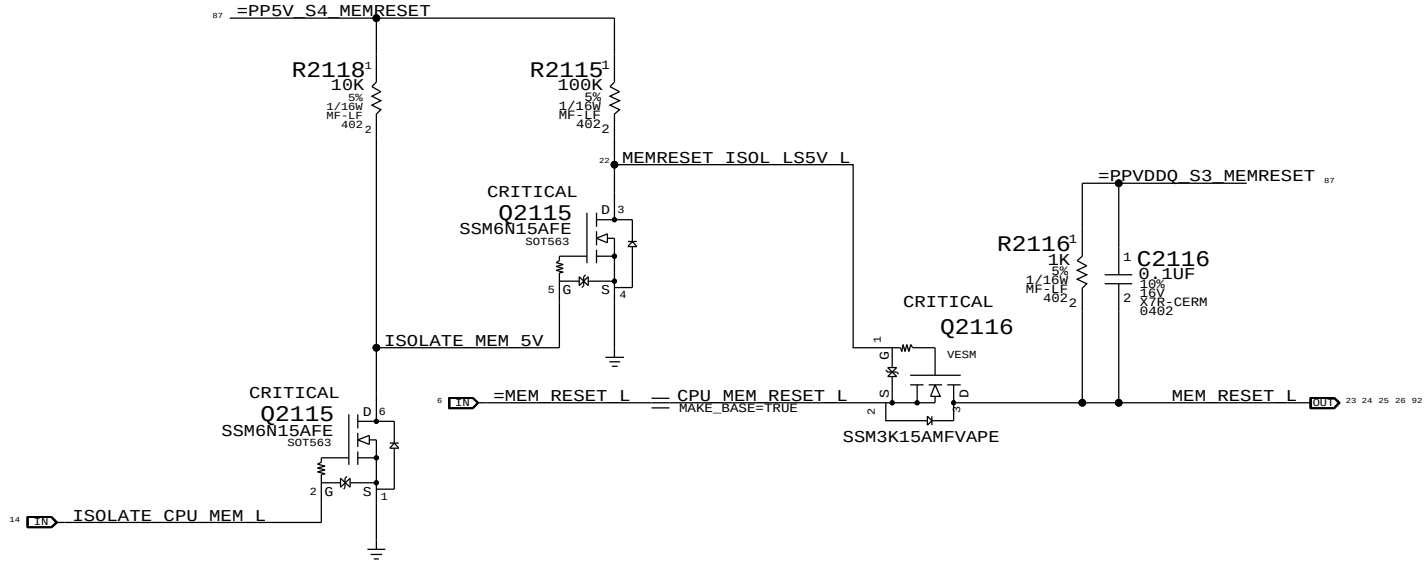
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.

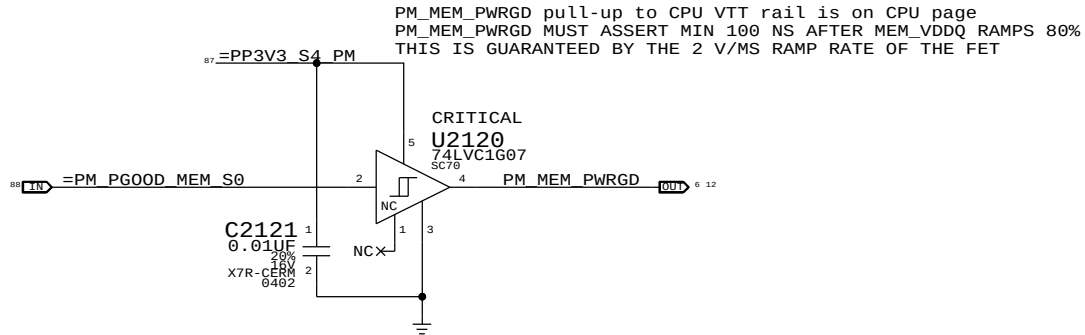
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

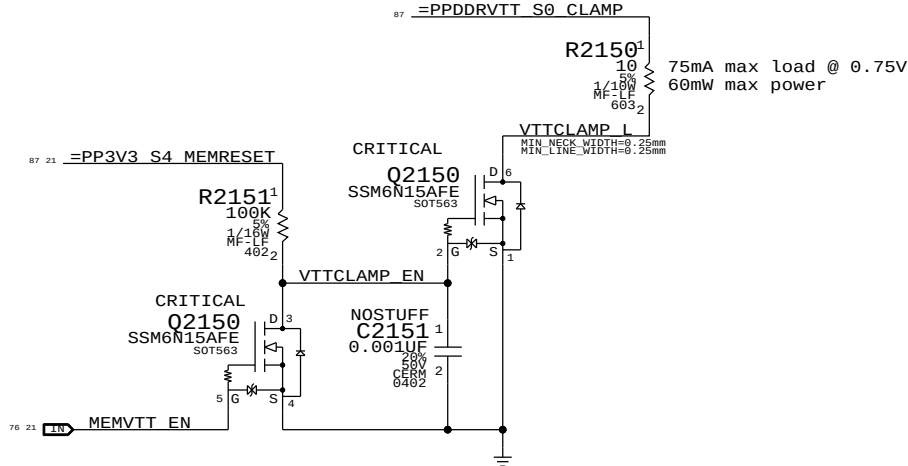
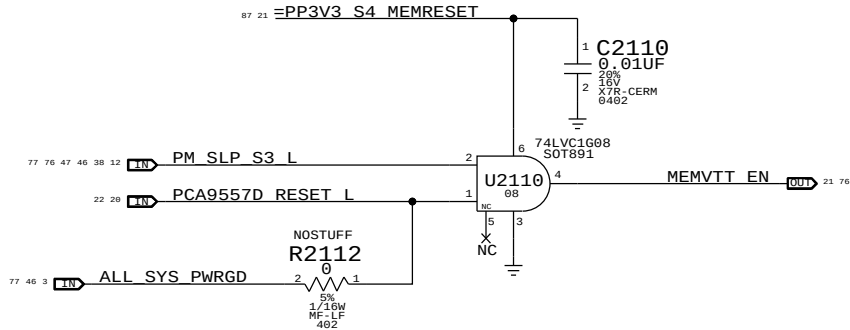
MEMVTT_EN = PLT_RESET_L * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L



MEM S0 "PGOOD" FOR CPU



MEMVTT Clamp
Ensures CKE signals are held low in S3



Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	CPU_MEM_RESET_L	1
1	1	0	1	1	1	1
2	0	0	1	1	1	0
3	0	0	0	X	1	0
4	0	0	1	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

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SYNC DATE=05/16/2013

CPU Memory S3 Support

Apple Inc.

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REVISION

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PAGE

21 OF 127

SHEET

21 OF 106

Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

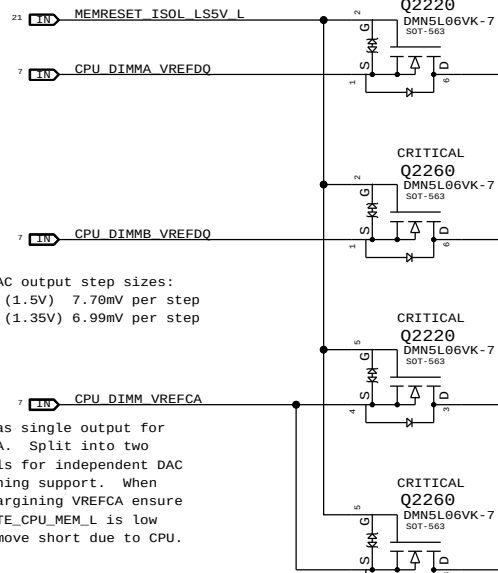
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

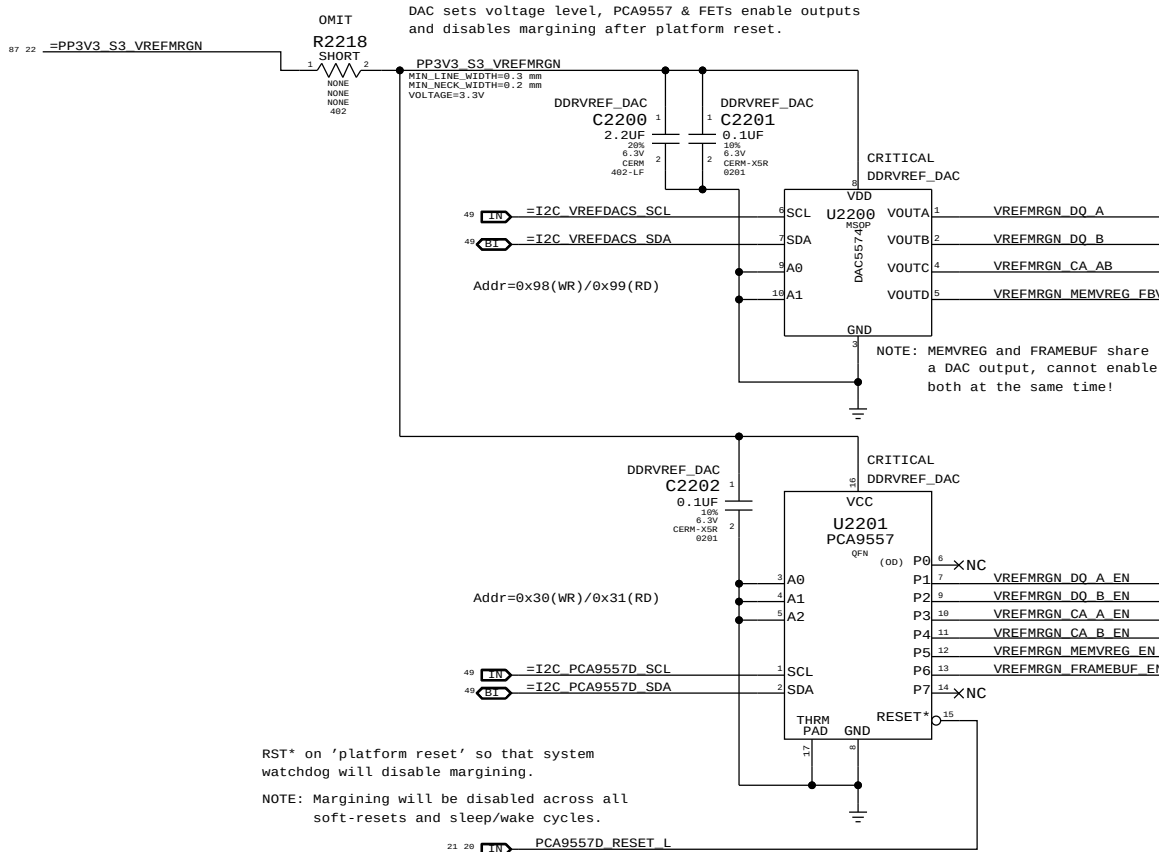
CPU-Based Margining

FETs for CPU isolation during S3



DAC-Based Margining

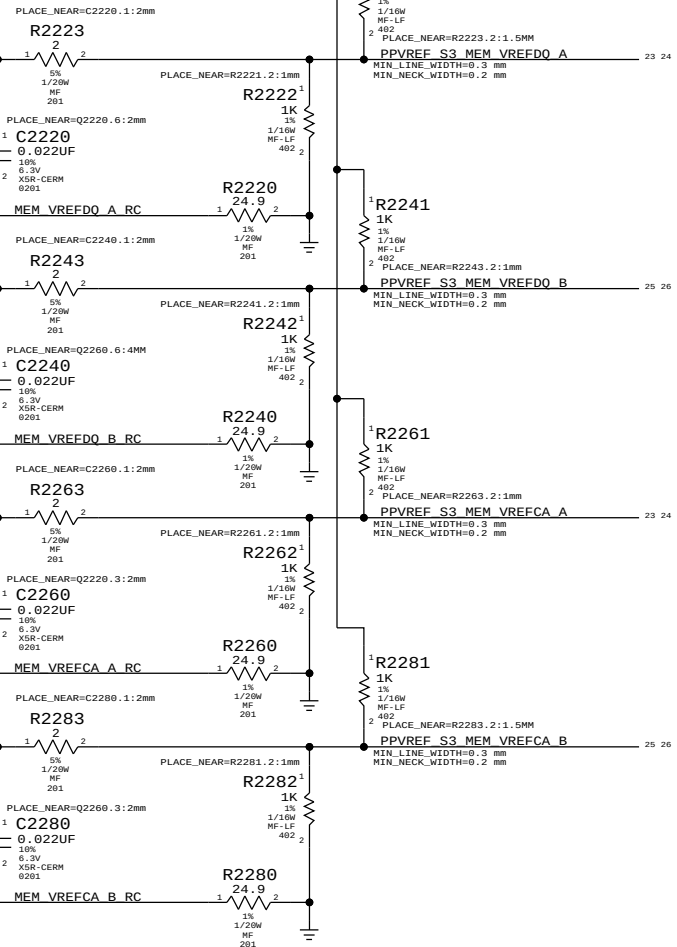
DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.



VRef Dividers

Always used, regardless of margining option.

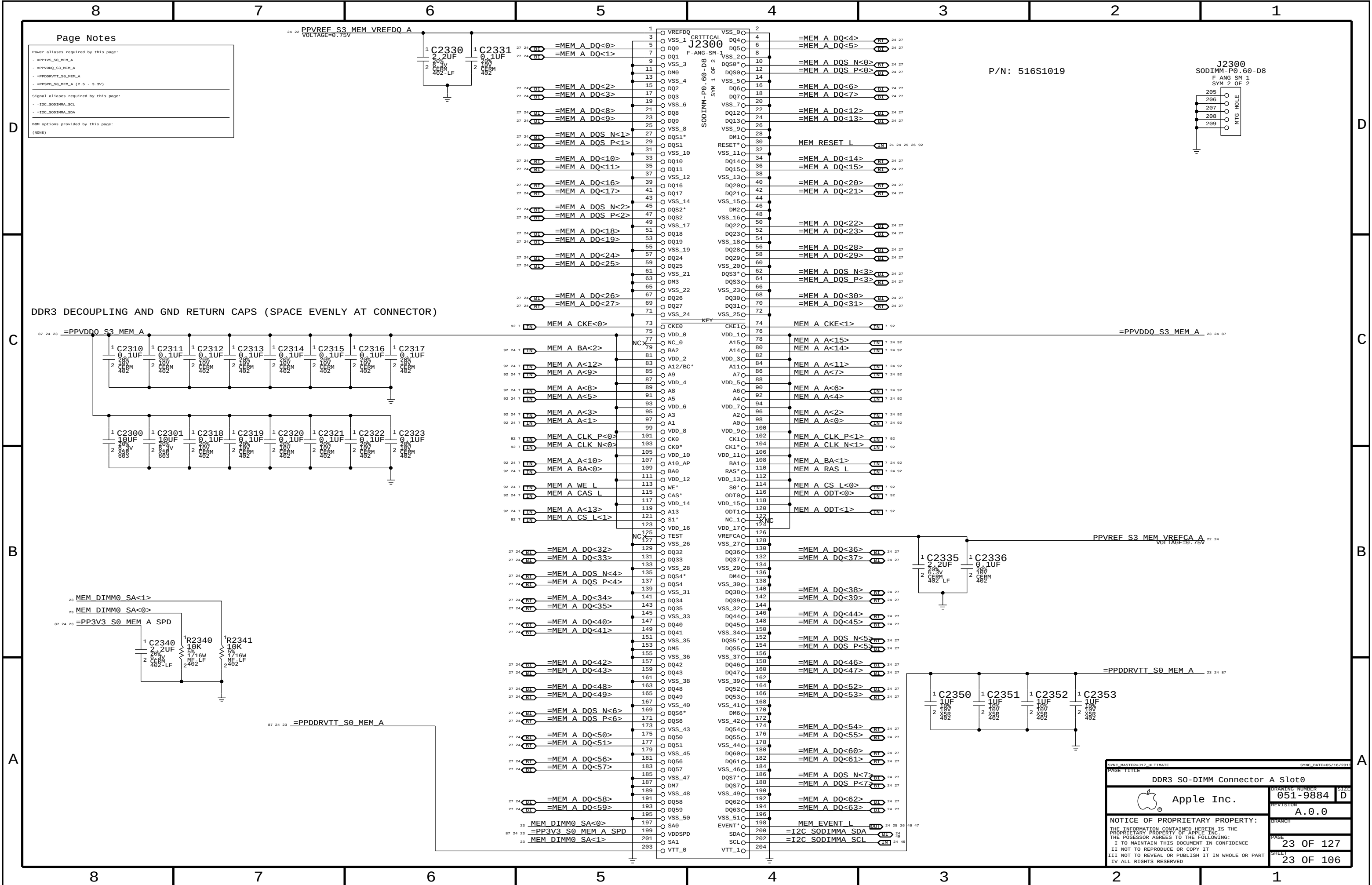
=PPDDR_S3_MEMVREF



	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	DDR3 (1.5V)		DDR3L (1.35V)		
Nominal value	0.750V (DAC: 0x3A = 0.747mV)		0.675V (DAC: 0x34 = 0.670mV)		1.500V (DAC: 0x74 = 1.495V)
Margined target:	0.300V - 1.200V (+/- 450mV)		0.275V - 1.075V (+/- 400mV)		1.200V - 1.800V (+/- 300mV)
DAC range:	0.000V - 1.508V (0x00 - 0x75)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 3.004V (0x00 - 0xE9)
Margined range:	0.299V - 1.206V (+/- 453mV)		0.269V - 1.083V (+/- 406mV)		1.199V - 1.801V (+/- 301mV)
VRef current:	+901uA - 911uA (- = sourced)		+811uA - -36uA (- = sourced)		+36uA - -36uA (- = sourced)
DAC step size:	7.68mV / step @ output		7.67mV / step @ output		2.575mV / step @ output

NOTE: DDR3 assumes TPS51916 supply with 10.0k/49.9k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

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Page Notes

Power aliases required by this page:

- =PP3V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVT_S0_MEM_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

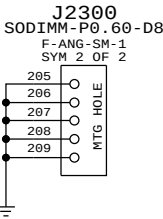
Signal aliases required by this page:

- =I2C_S0DIMMA_SCL
- =I2C_S0DIMMA_SDA

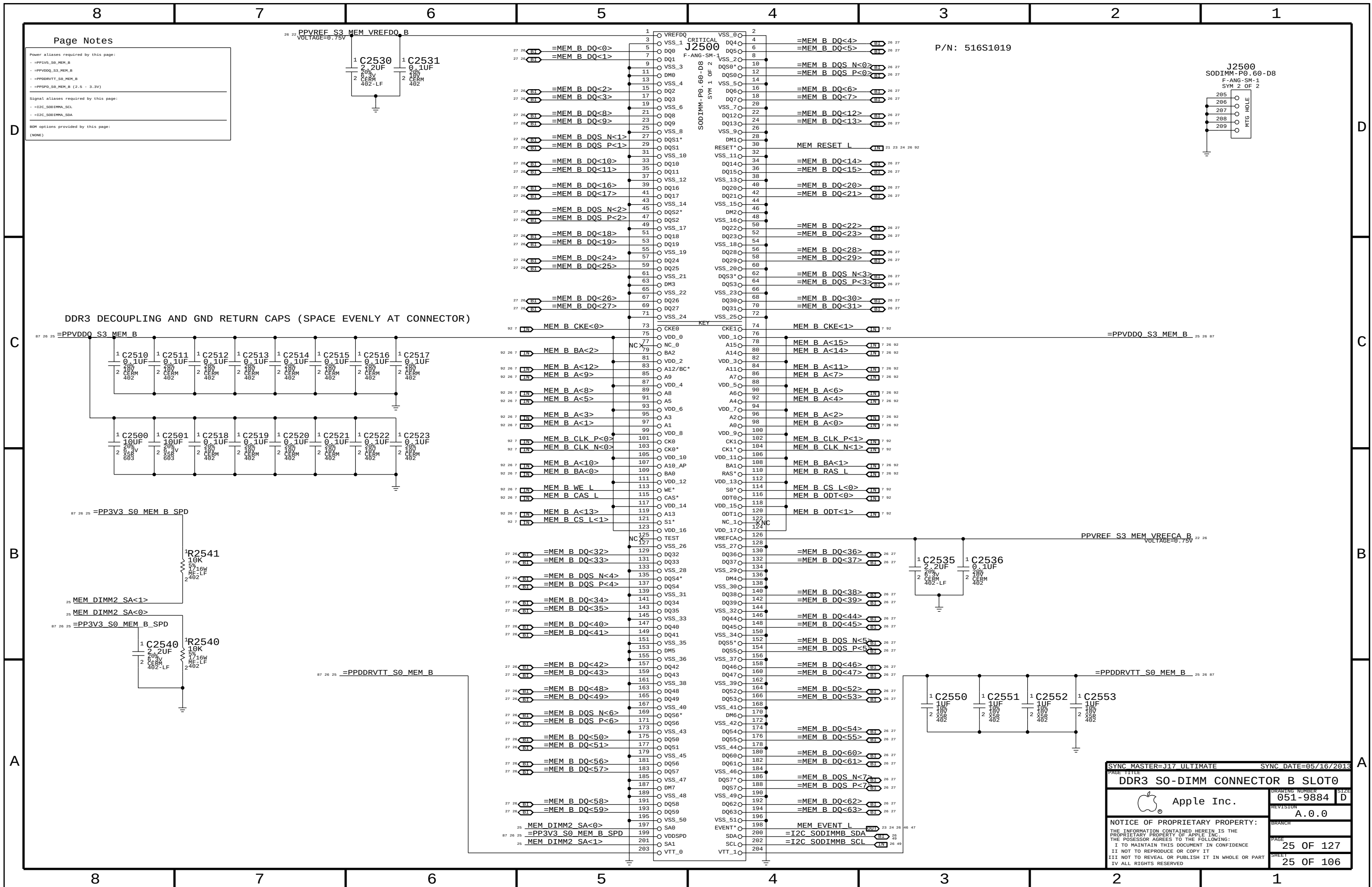
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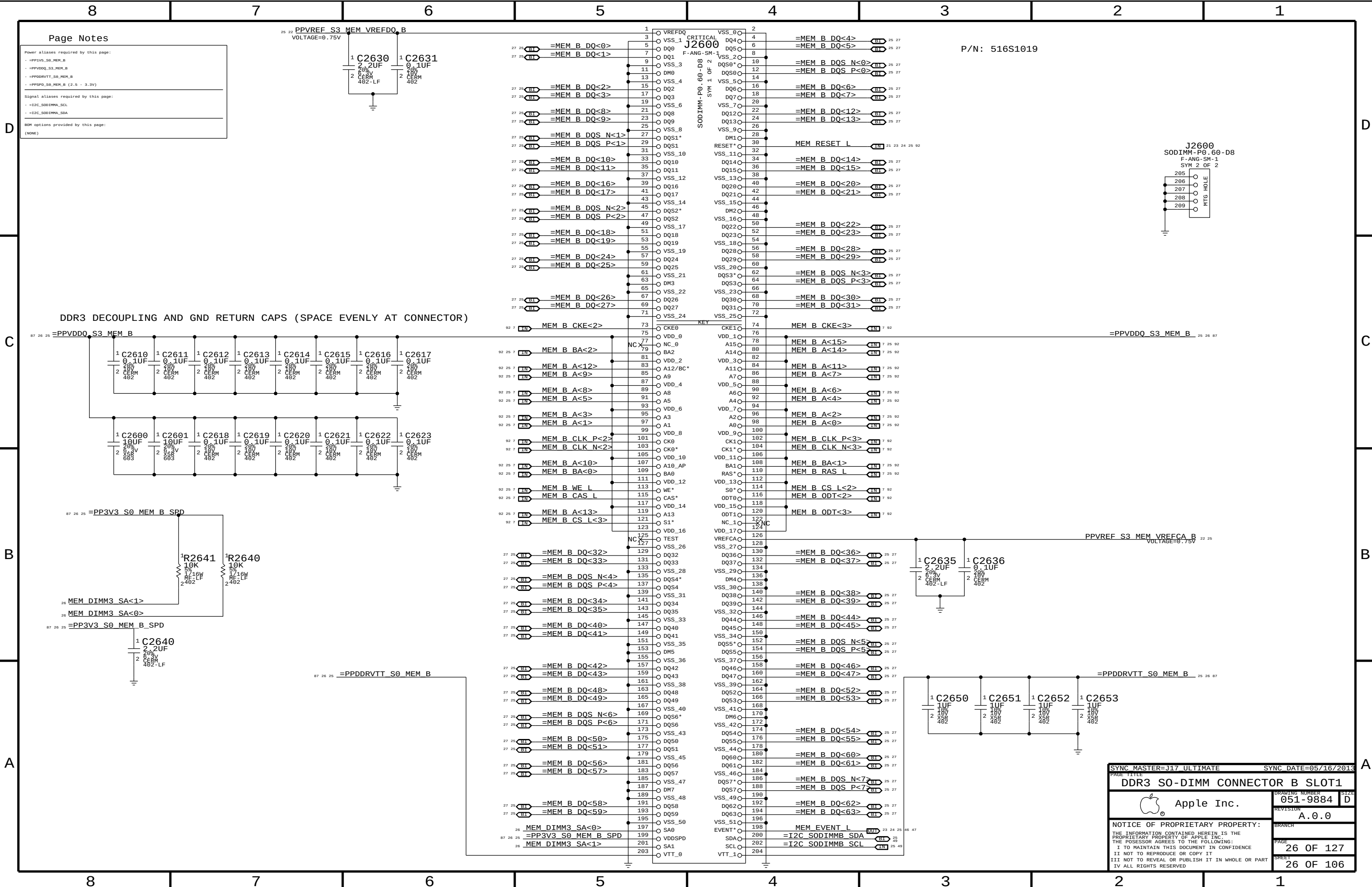
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P/N: 516S1019



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PAGE TITLE			
DDR3 S0-DIMM Connector A Slot0			
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PAGE	23 OF 127		
SHEET	23 OF 106		





Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

P/N: 516S1019

J2600
SODIMM-P0.60-D8
F-ANG-SM-1
SYM 2 OF 2

SYNC MASTER=J17 ULTIMATE SYNC DATE=05/16/2013

PAGE TITLE
DDR3 SO-DIMM CONNECTOR B SLOT1



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PAGE
26 OF 127

SHEET
26 OF 106

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THERE ARE NO PIN SWAPS															
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SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

PAGE TITLE

DDR3 ALIASES AND BITSWAPS

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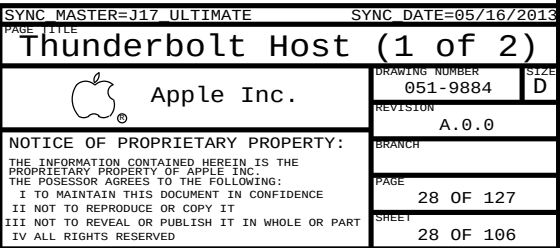
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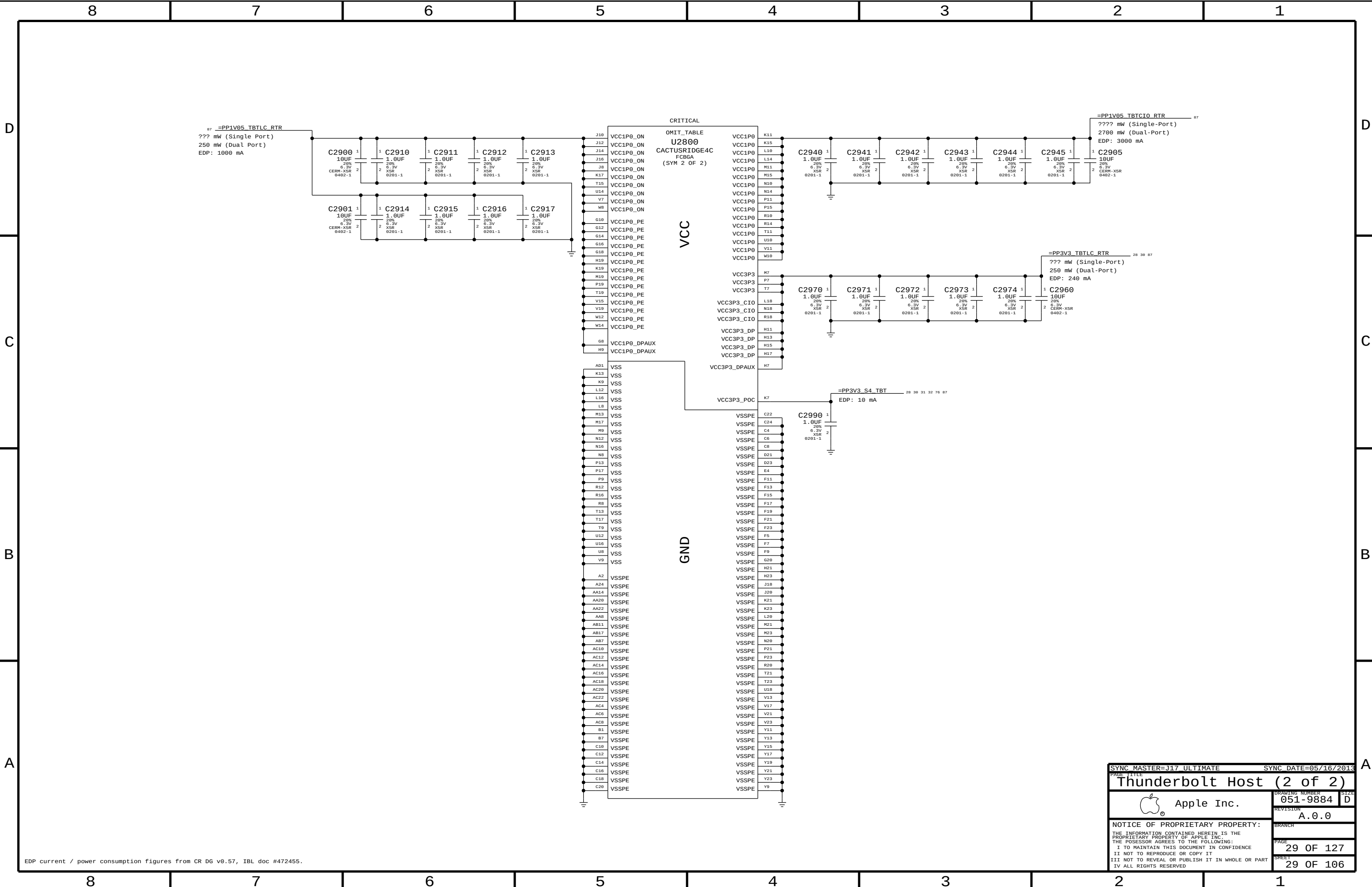
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PAGE
27 OF 127

SHEET
27 OF 106





EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

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Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTTPWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

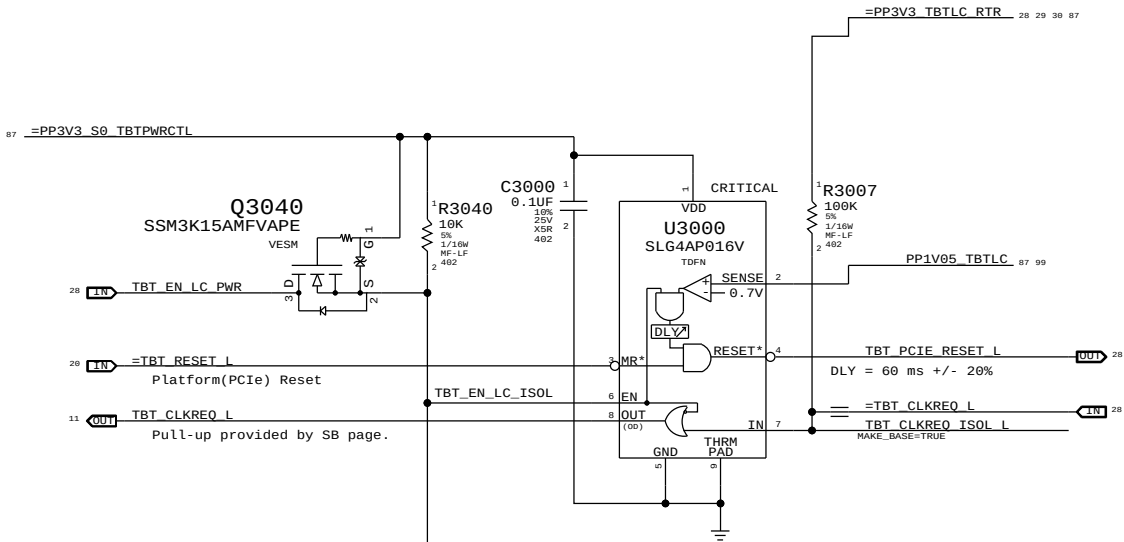
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

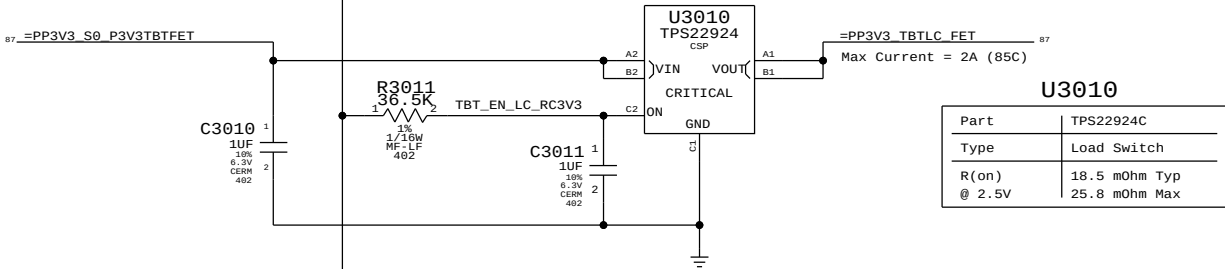
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

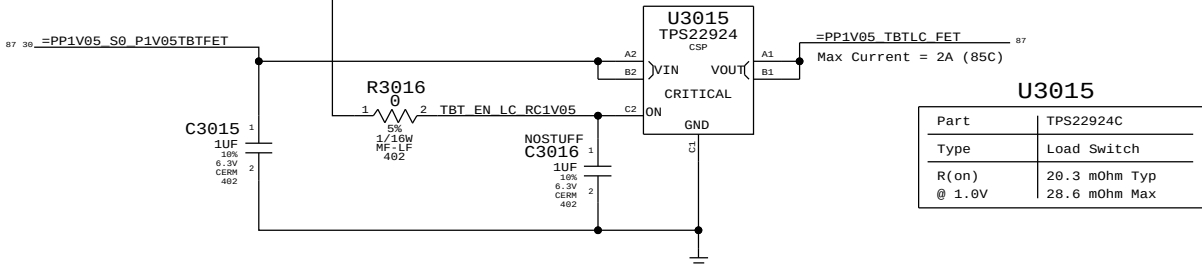
Supervisor & CLKREQ# Isolation



3.3V TBT "LC" Switch

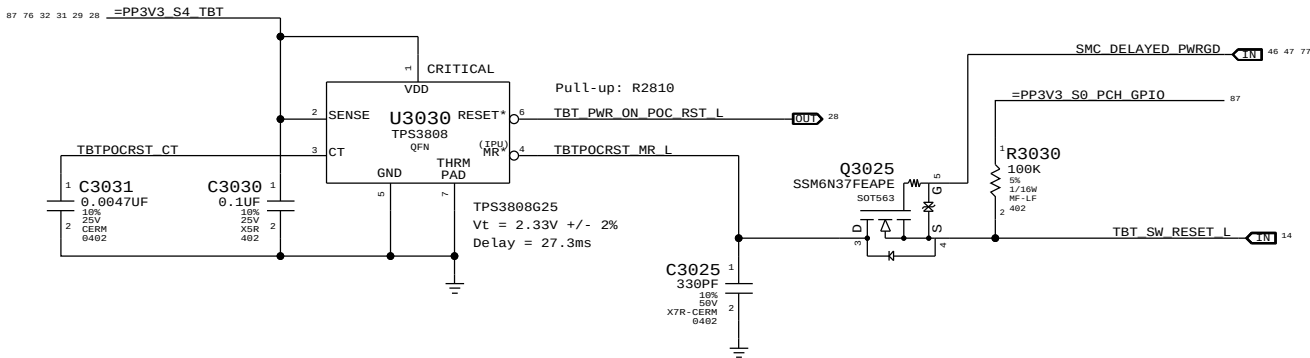


1.05V TBT "LC" Switch

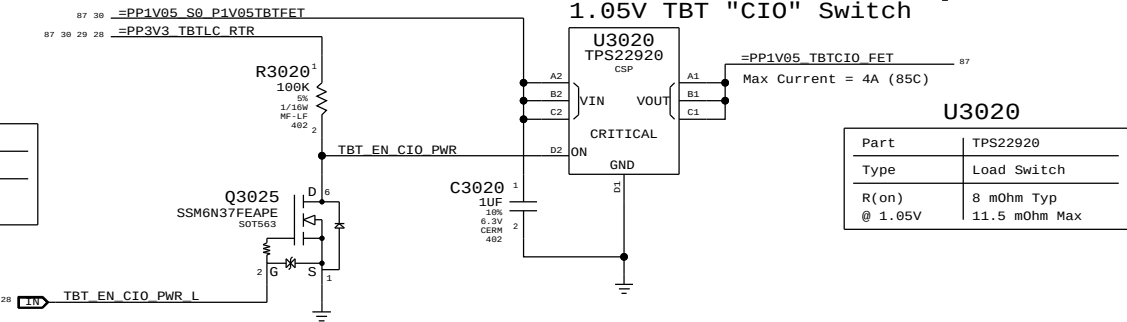


TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



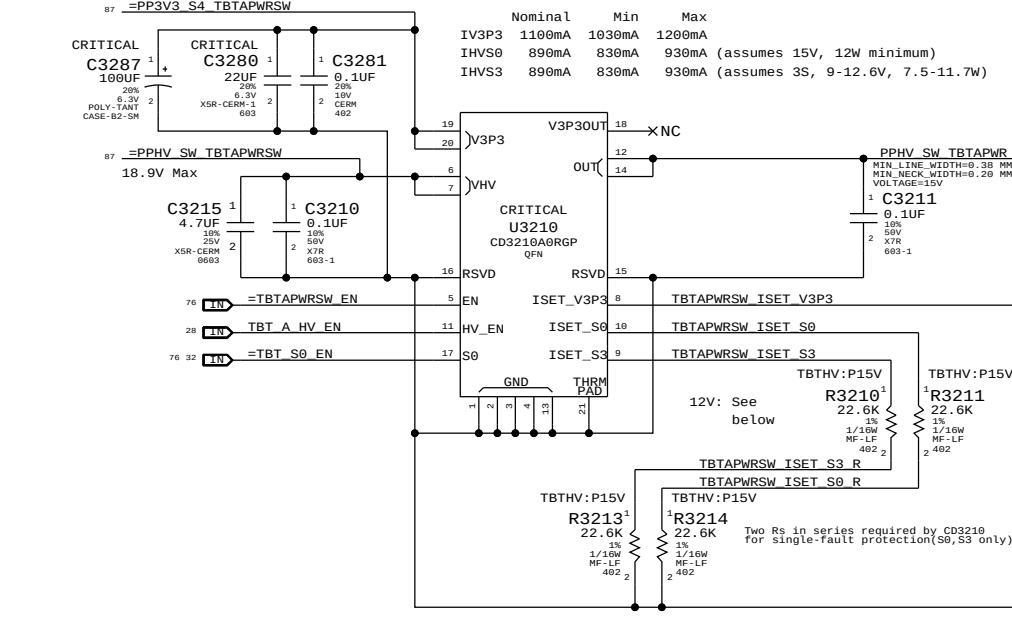
1.05V TBT "CIO" Switch



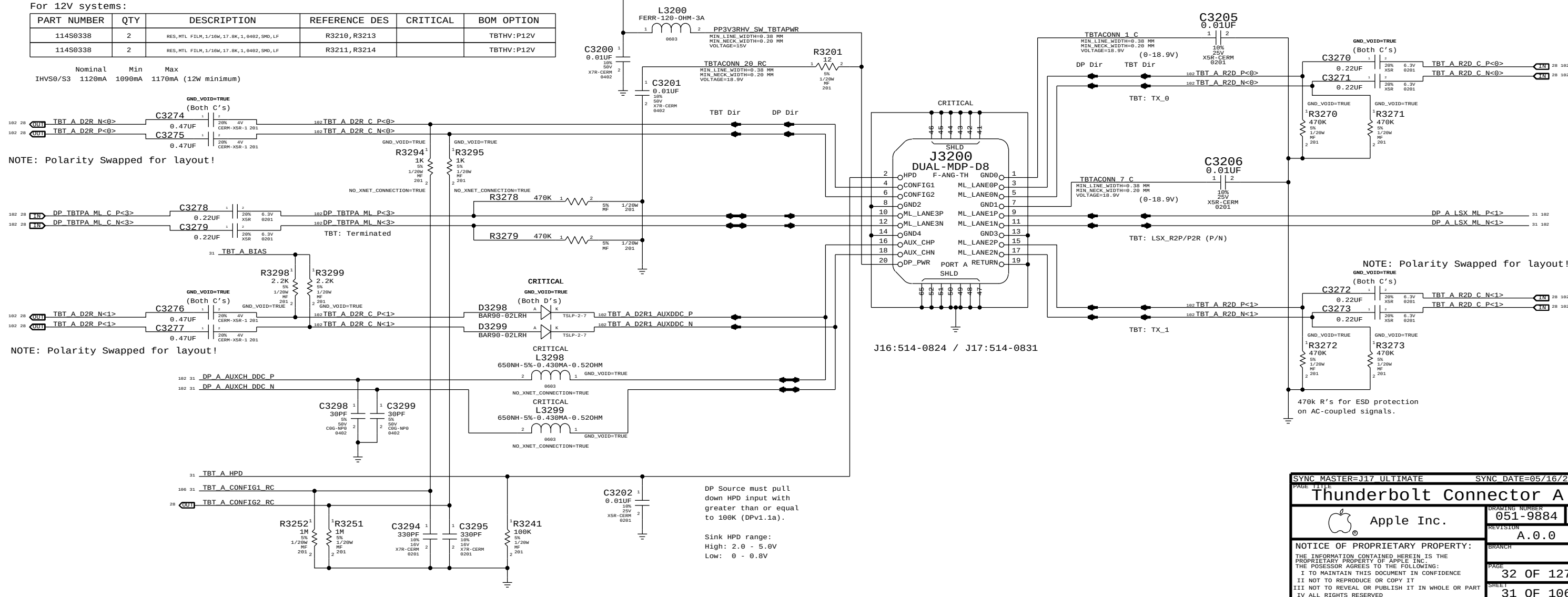
3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
311S0596	311S0593		ALL	T1 Alternate
128S0398	128S0229		ALL	3.3V INPUT CAP



Thunderbolt Connector A

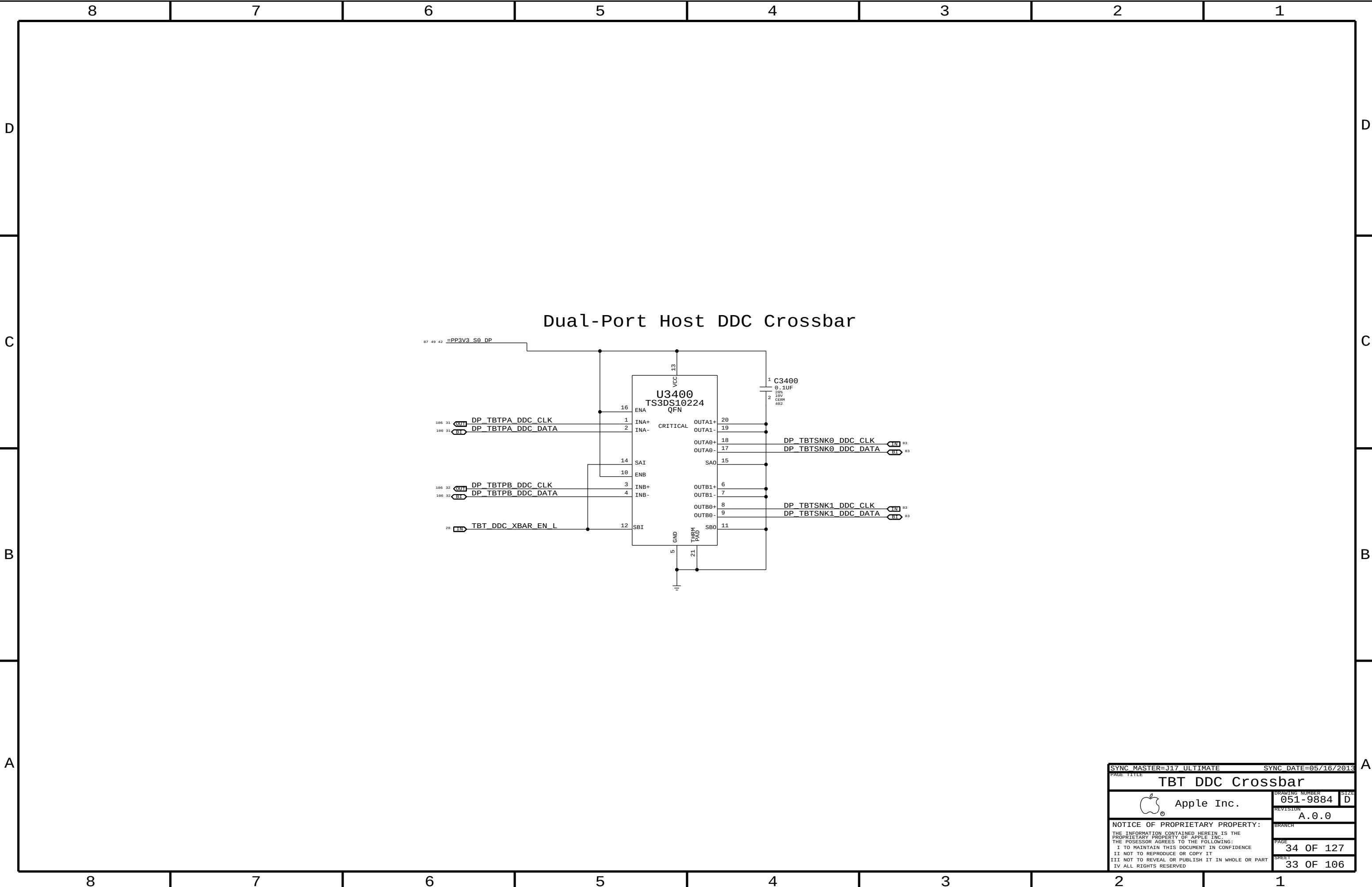


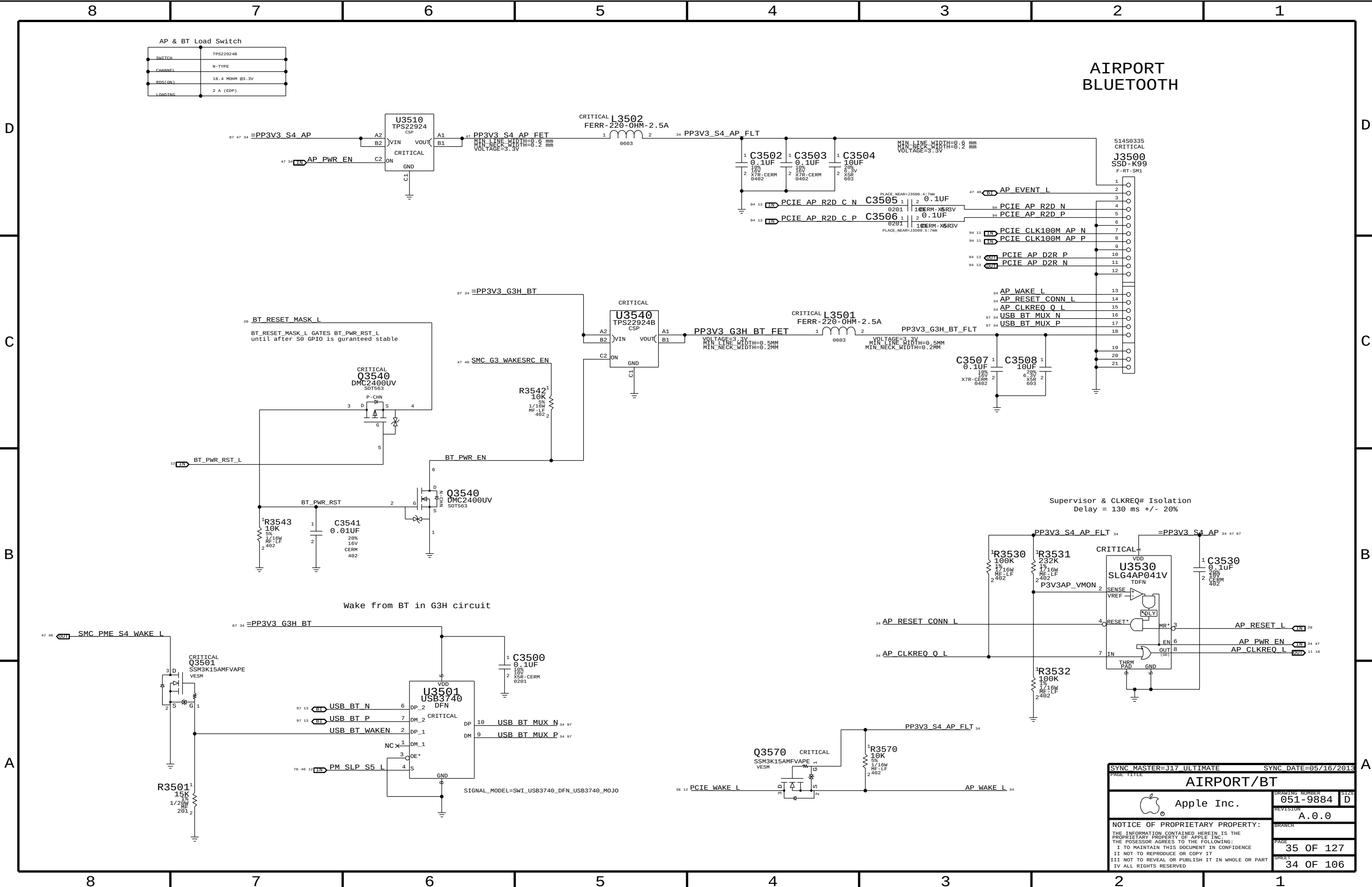
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PAGE TITLE		Thunderbolt Connector A	
Apple Inc.		DRAWING NUMBER	051-9884
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B

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PAGE TITLE			
Thunderbolt Connector B			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9884	D
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		A.0.0	
BRANCH		PAGE	
		33 OF 127	
		SHEET	
		32 OF 106	

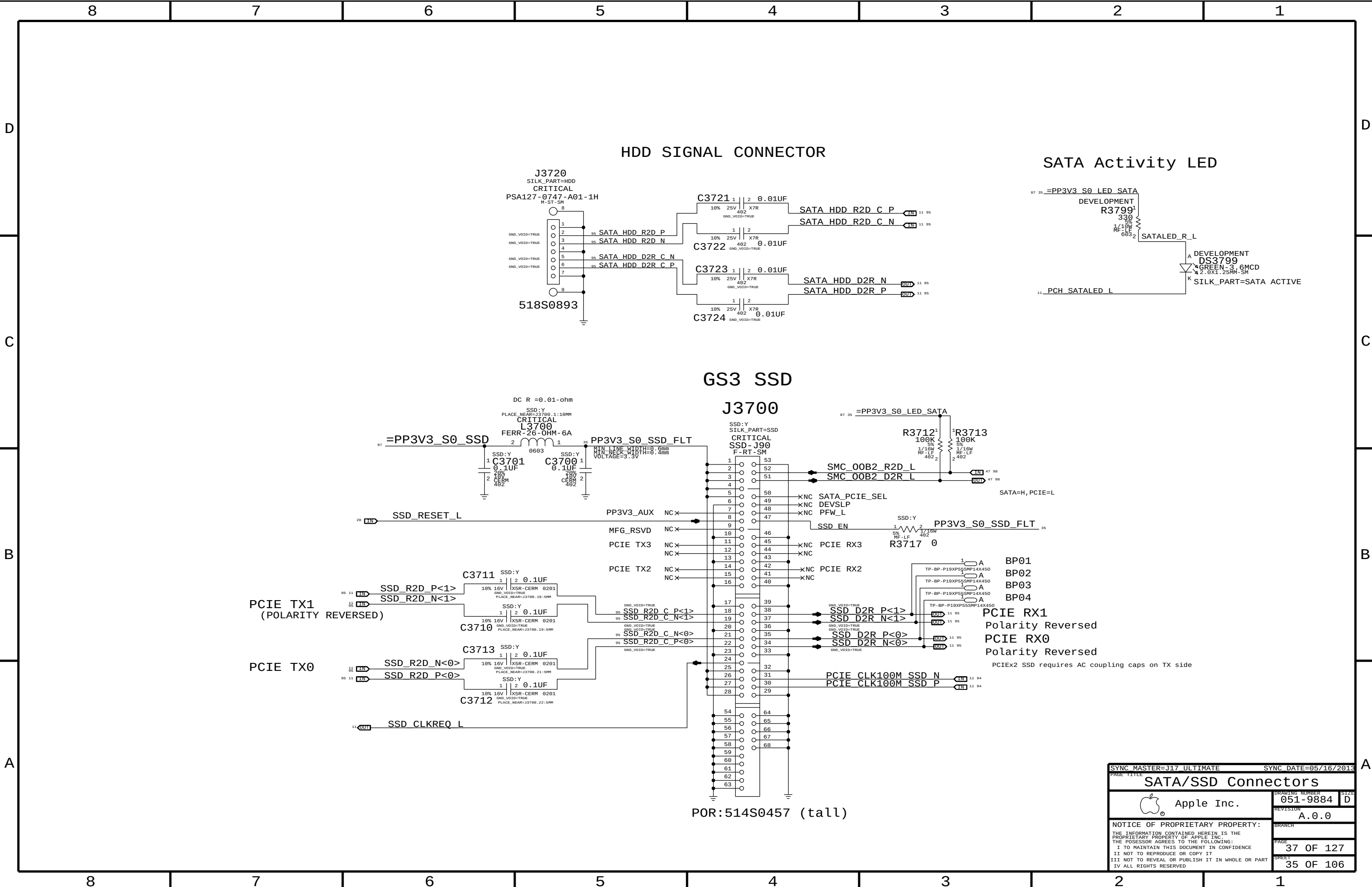




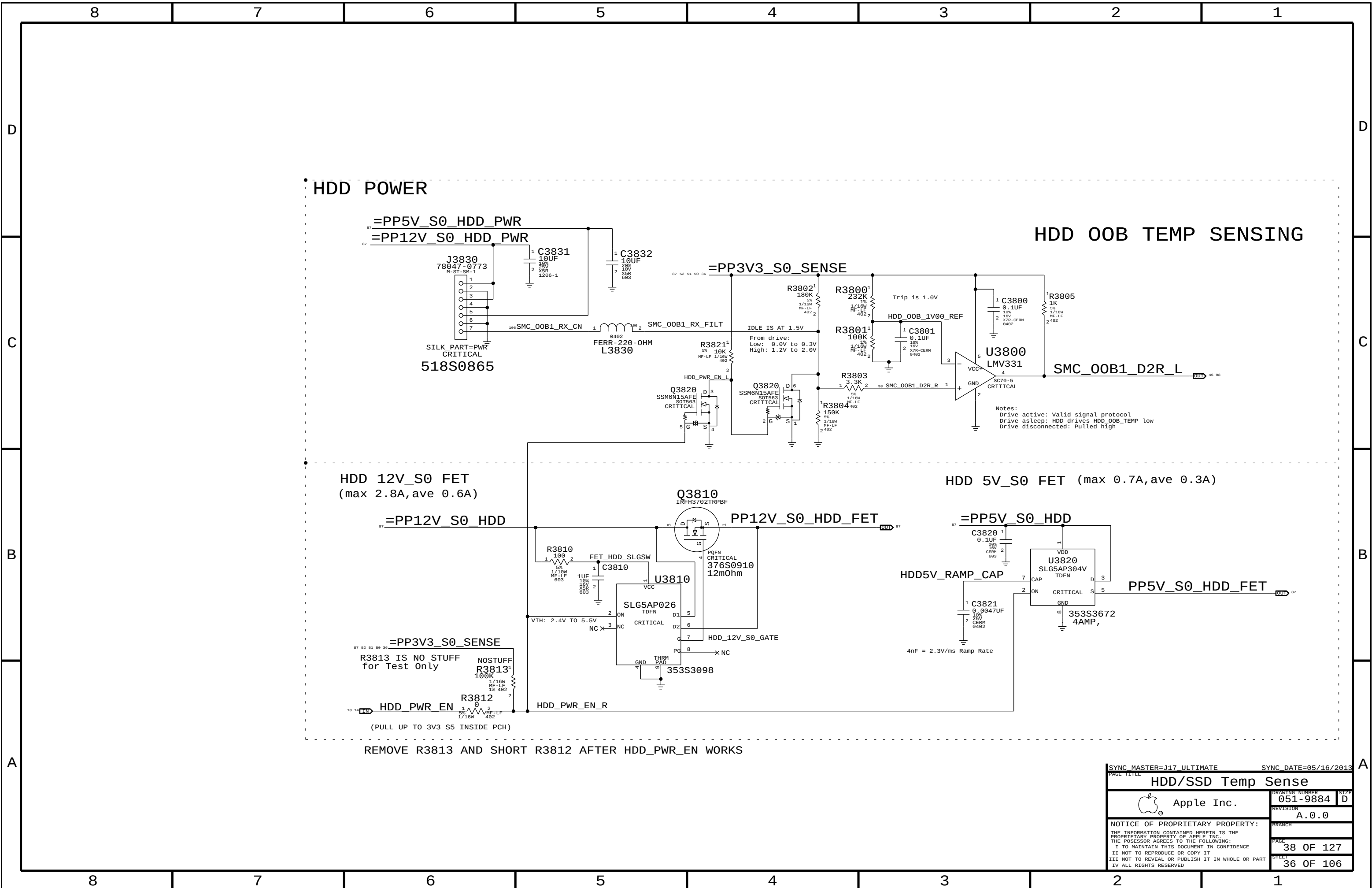
AP & BT Load Switch	
SWITCH	TPS22924B
CHANNEL	N-TYPE
RDS(ON)	18.4 MOHM @3.3V
LOADING	2 A (EDP)

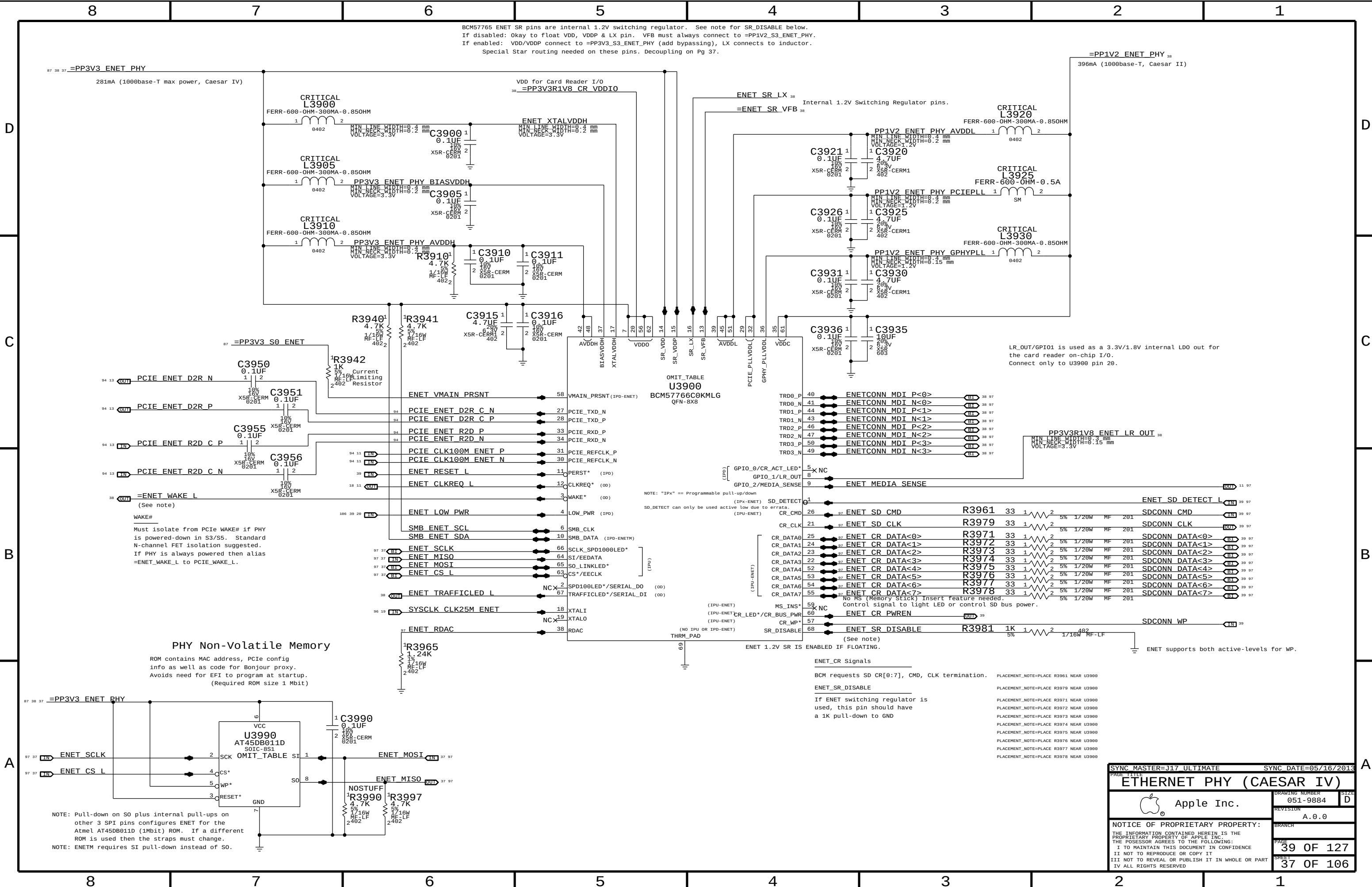
AIRPORT BLUETOOTH

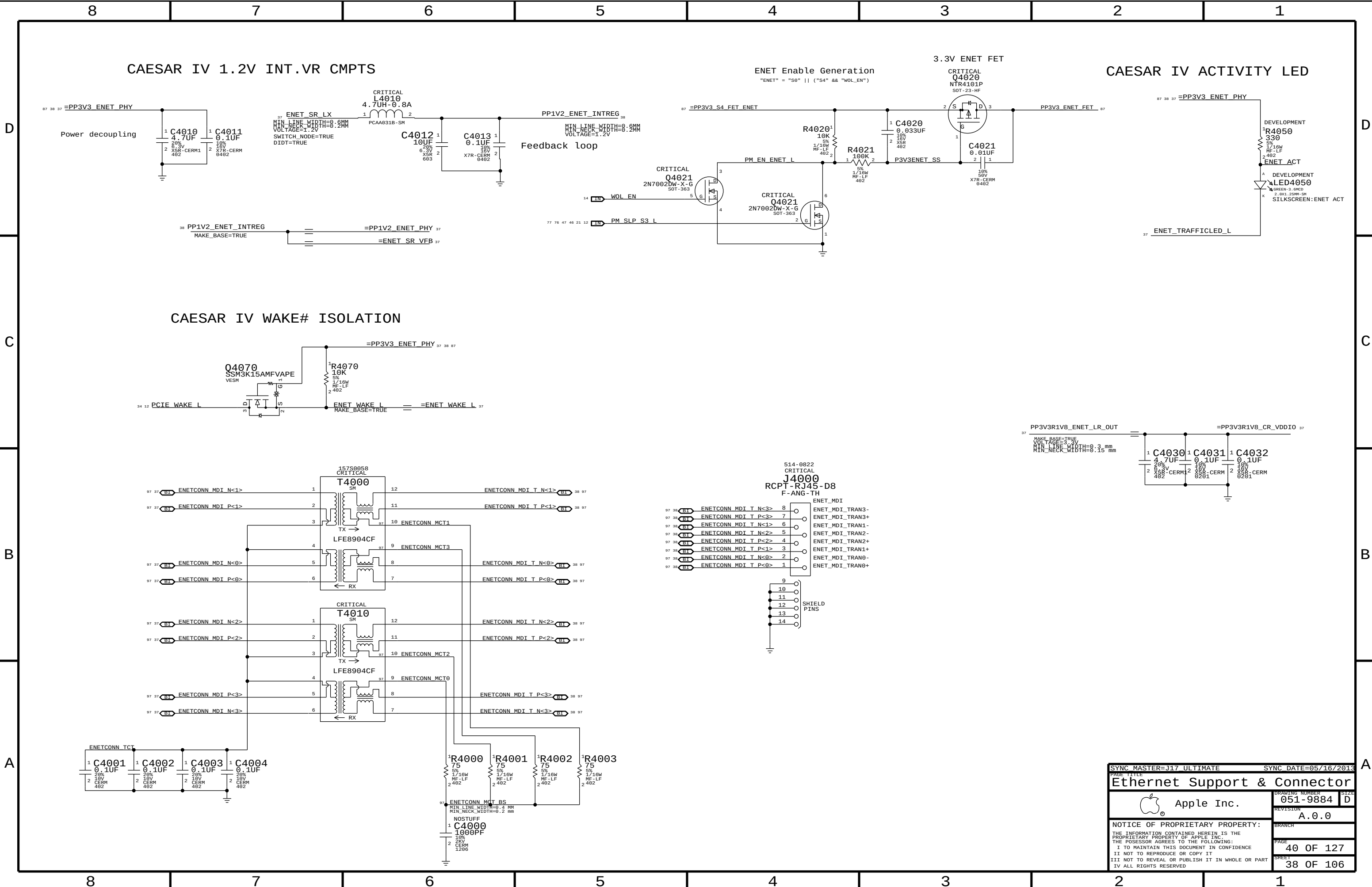
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PAGE TITLE		AIRPORT/BT	
Apple Inc.		DRAWING NUMBER	051-9884
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		BRANCH	
		PAGE	35 OF 127
		SHEET	34 OF 106

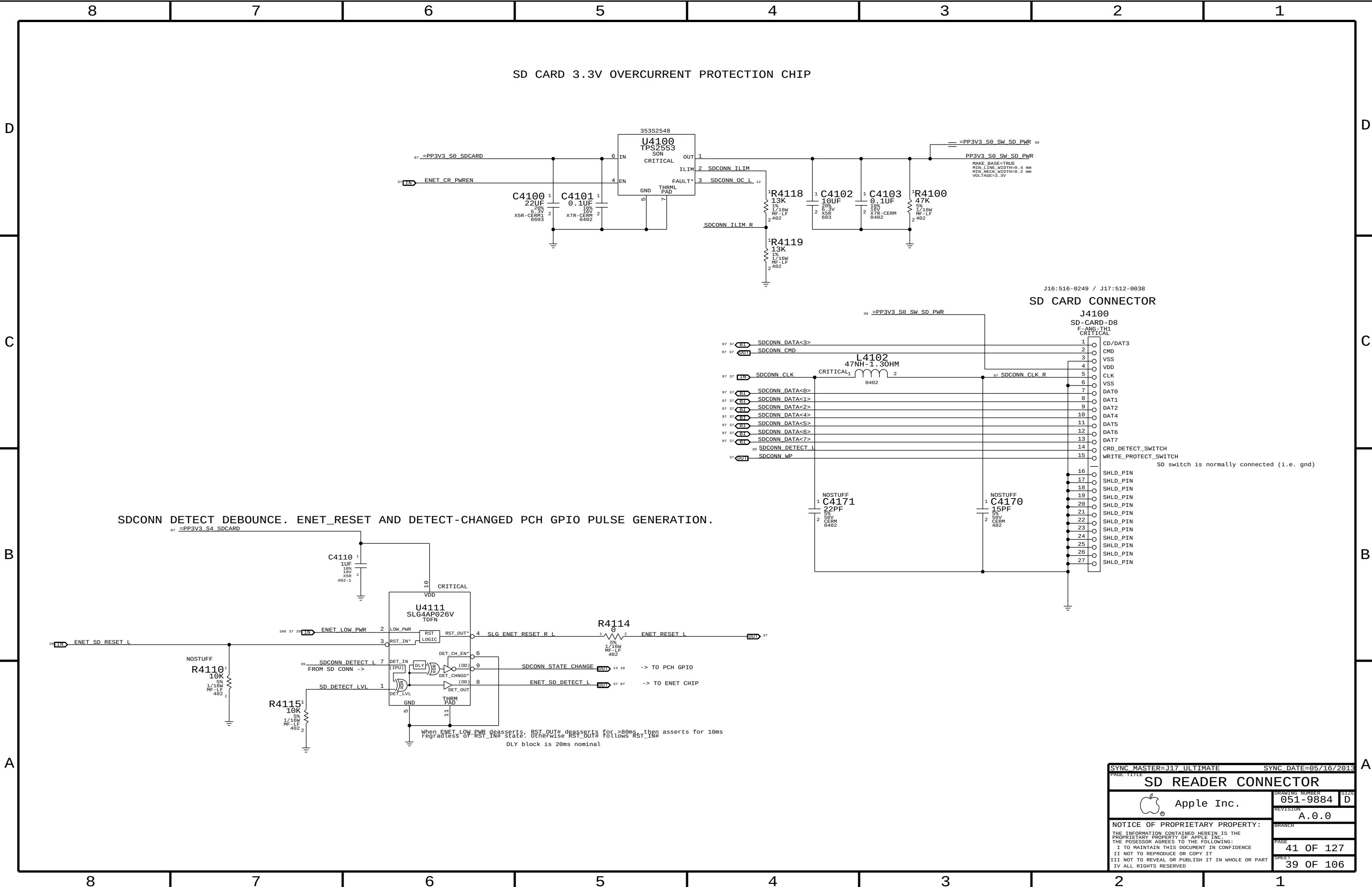


SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
SATA/SSD Connectors			
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		PAGE	37 OF 127
		SHEET	35 OF 106





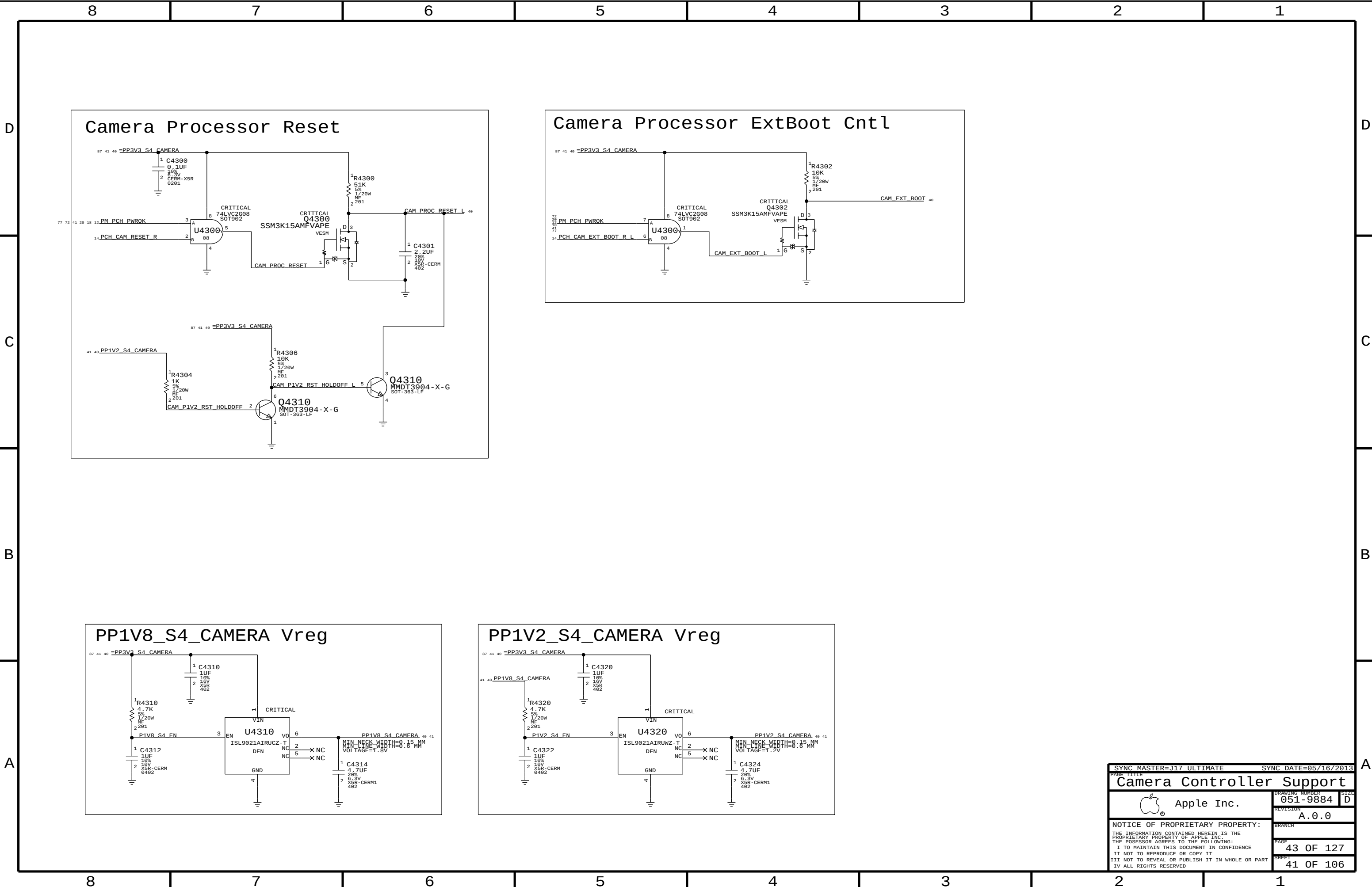




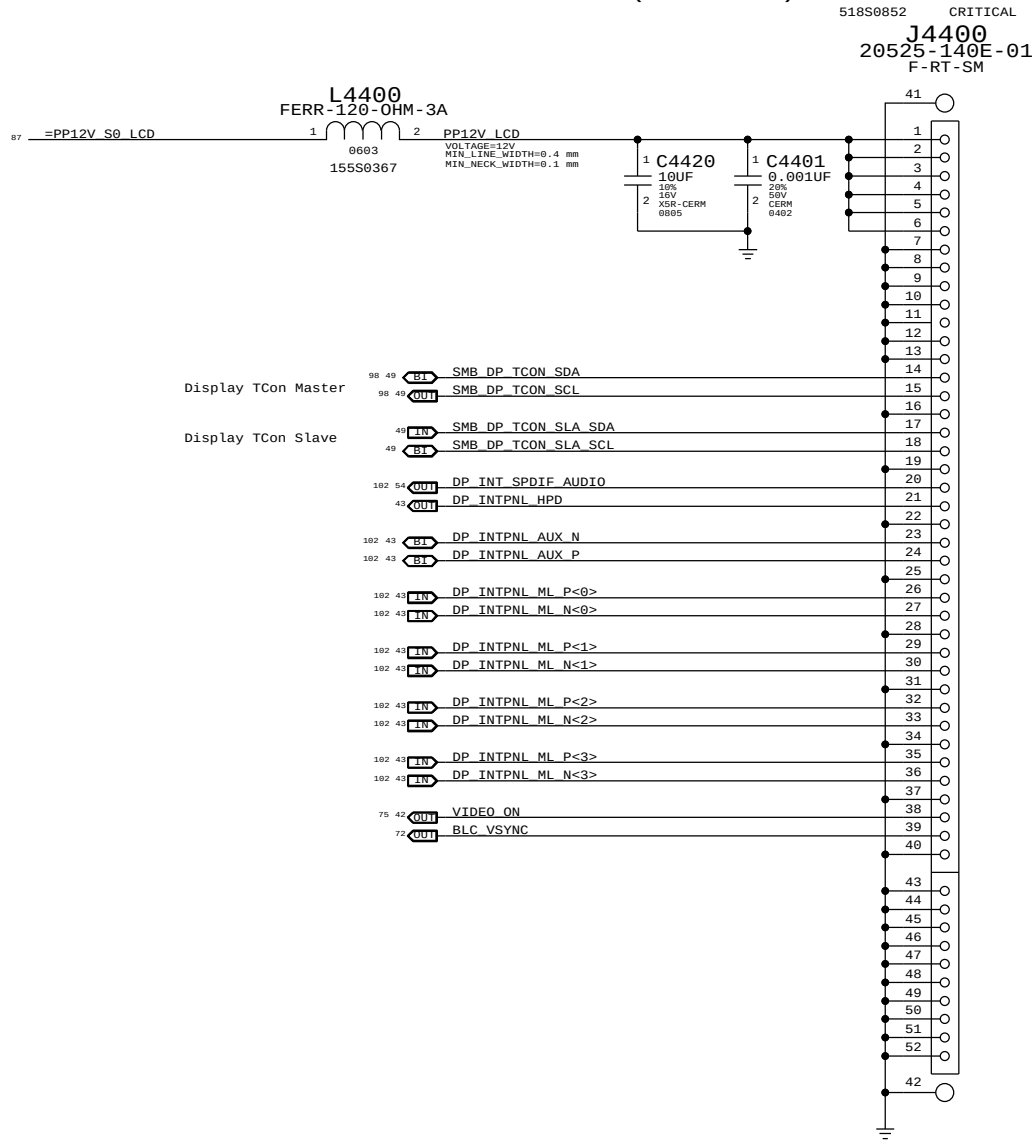
PAGE TITLE		PAGE TITLE	
SD READER CONNECTOR		SD READER CONNECTOR	
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A

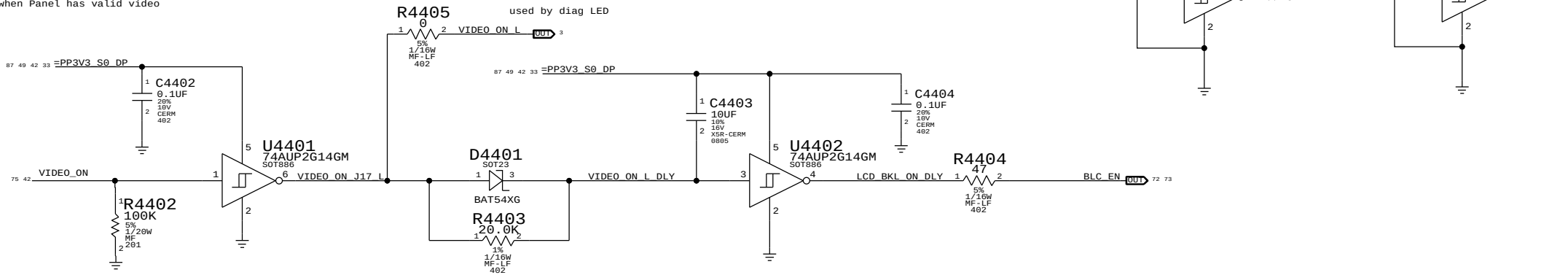


INTERNAL DP (STRAIGHT)

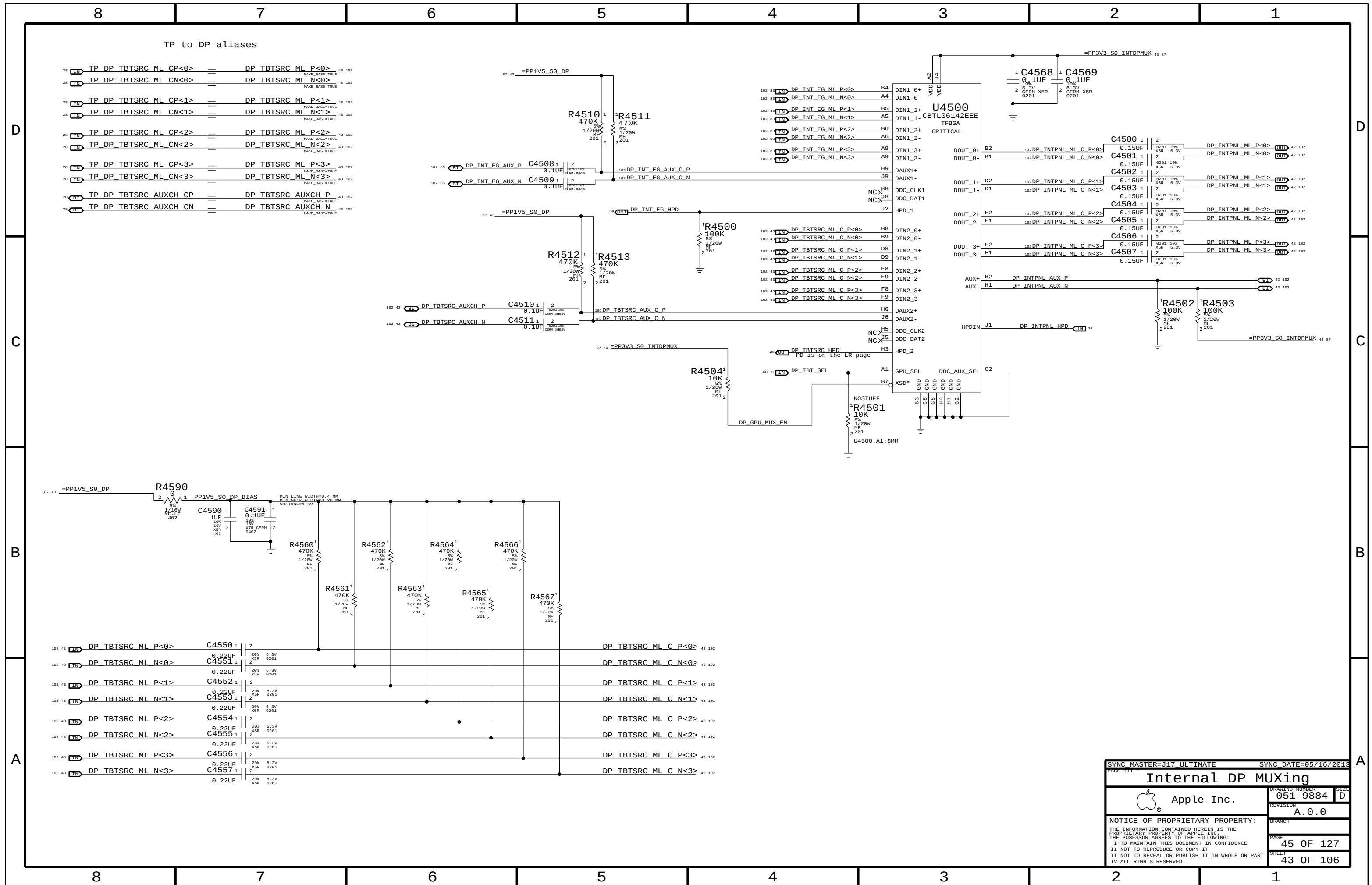


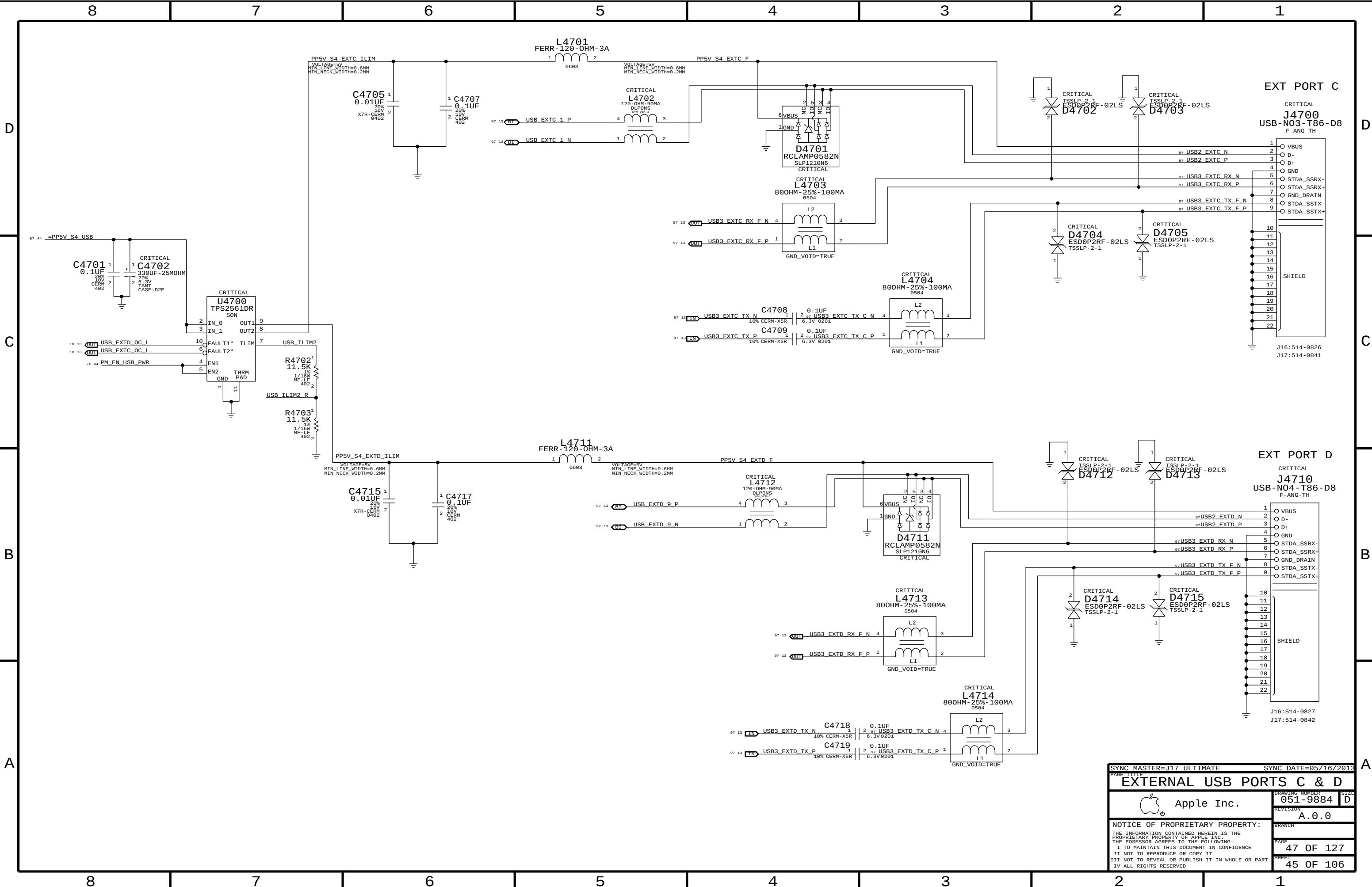
BACKLIGHT CONTROL SUPPORT

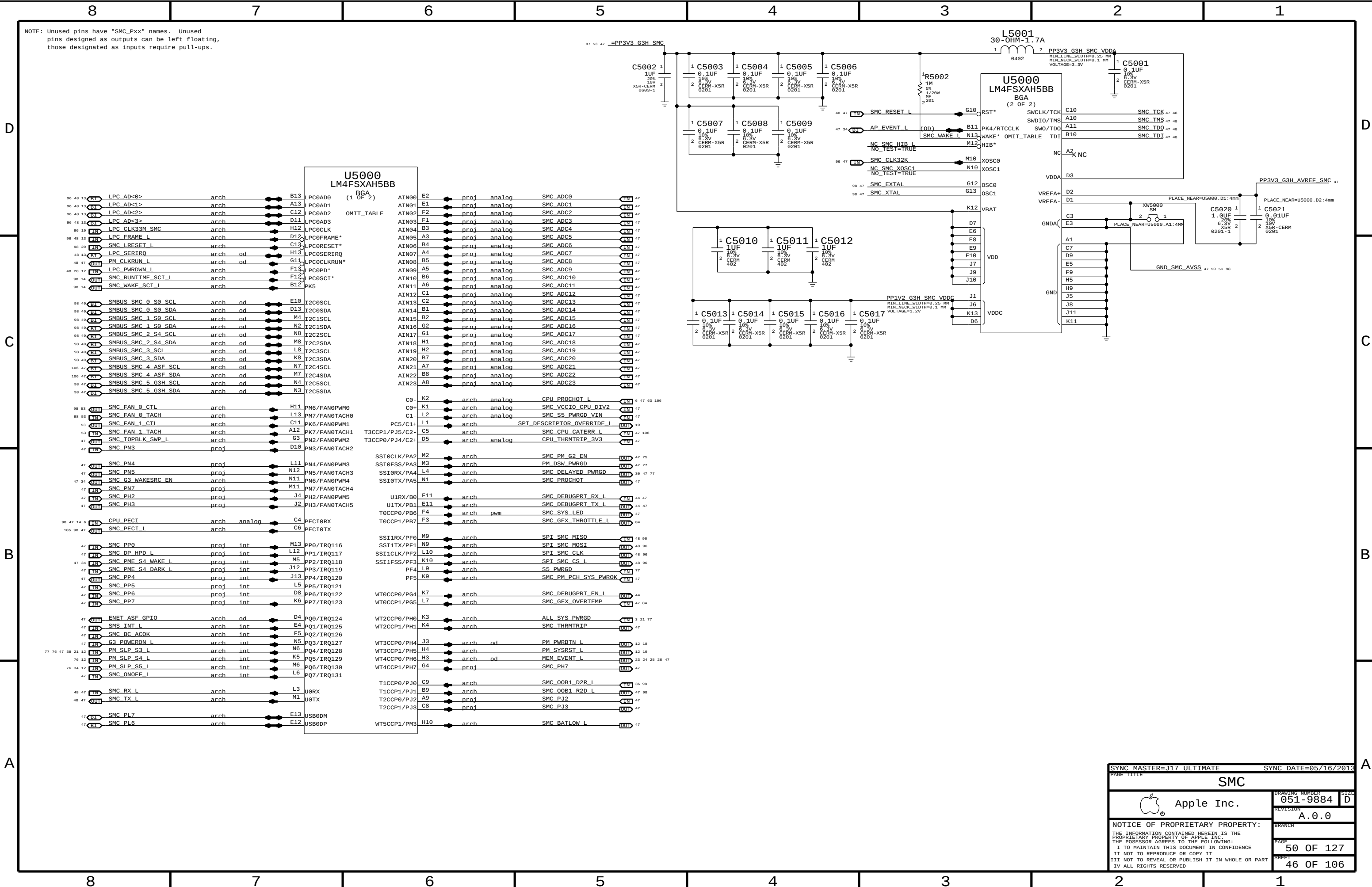
guarantee backlight is
only on when Panel has valid video

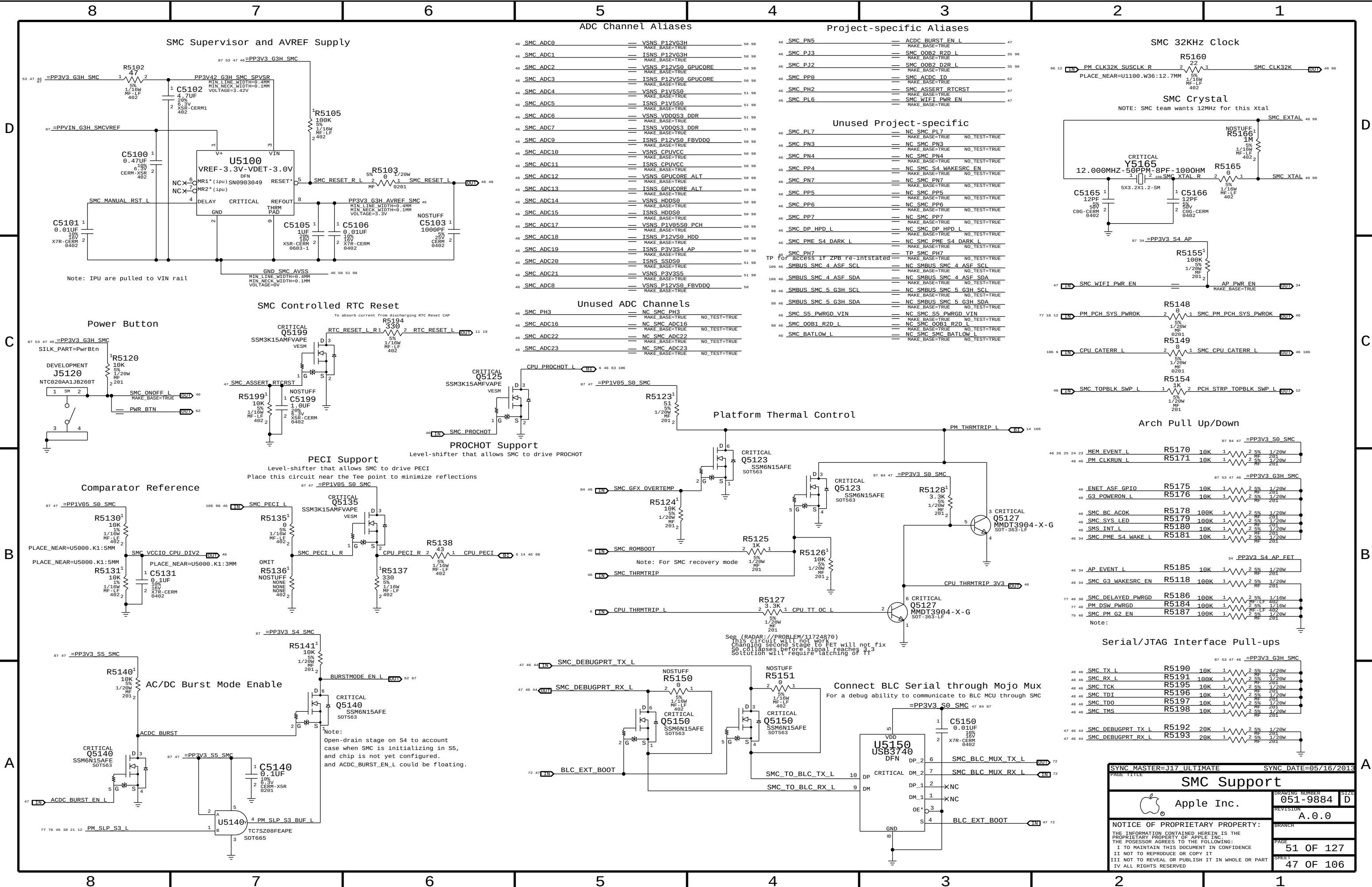


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Internal DP Support		DRAWING NUMBER	051-9884
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SMC Supervisor and AVREF Supply

ADC Channel Aliases

Project-specific Aliases

SMC 32KHz Clock

SMC Crystal

Unused Project-specific

Unused ADC Channels

Platform Thermal Control

PECI Support

Comparator Reference

Level-shifter that allows SMC to drive PCHOT
Place this circuit near the Tee point to minimize reflections

Arch Pull Up/Down

Serial/JTAG Interface Pull-ups

Connect BLC Serial through Mojo Mux
For a debug ability to communicate to BLC MCU through SMC

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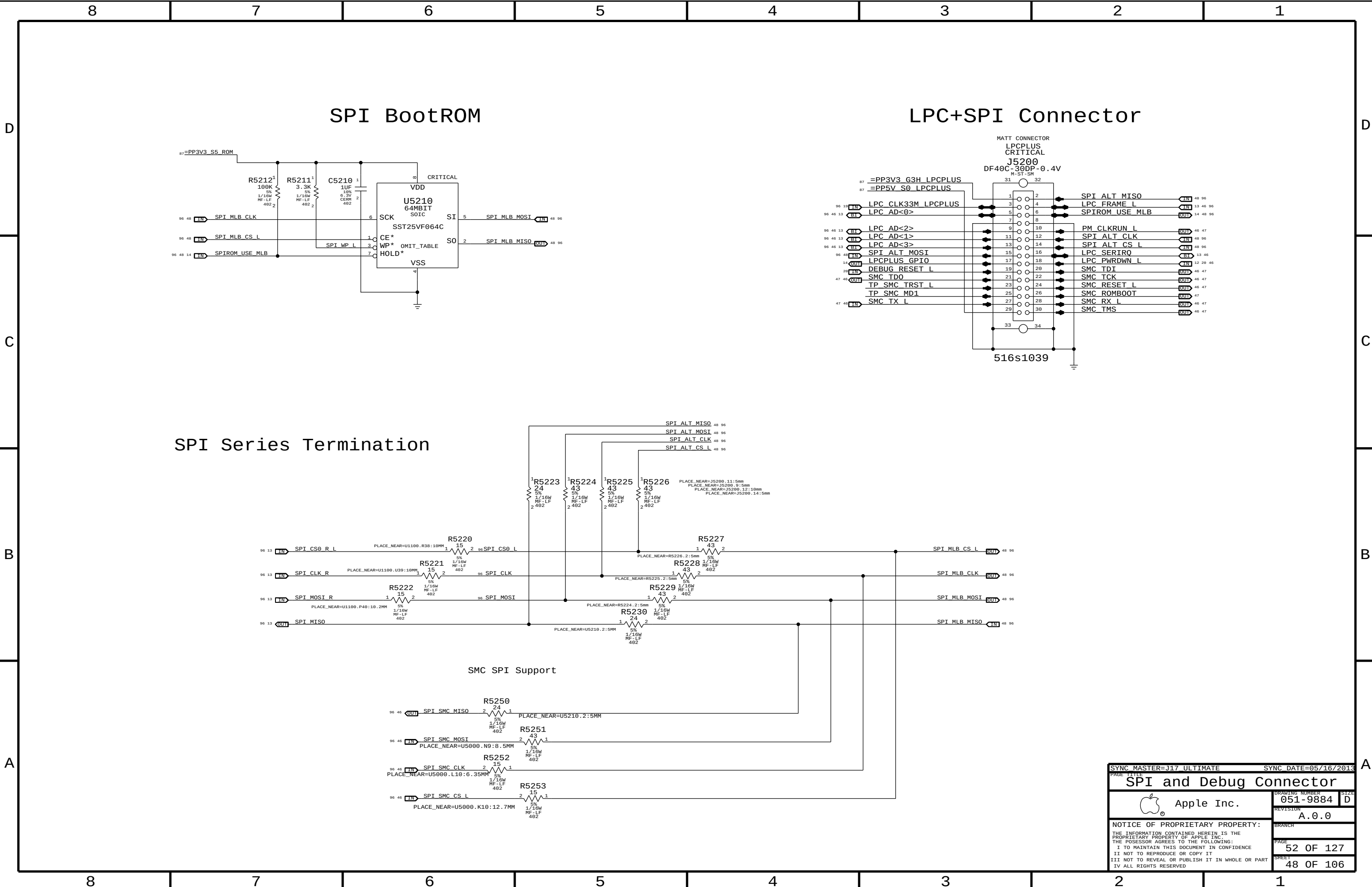
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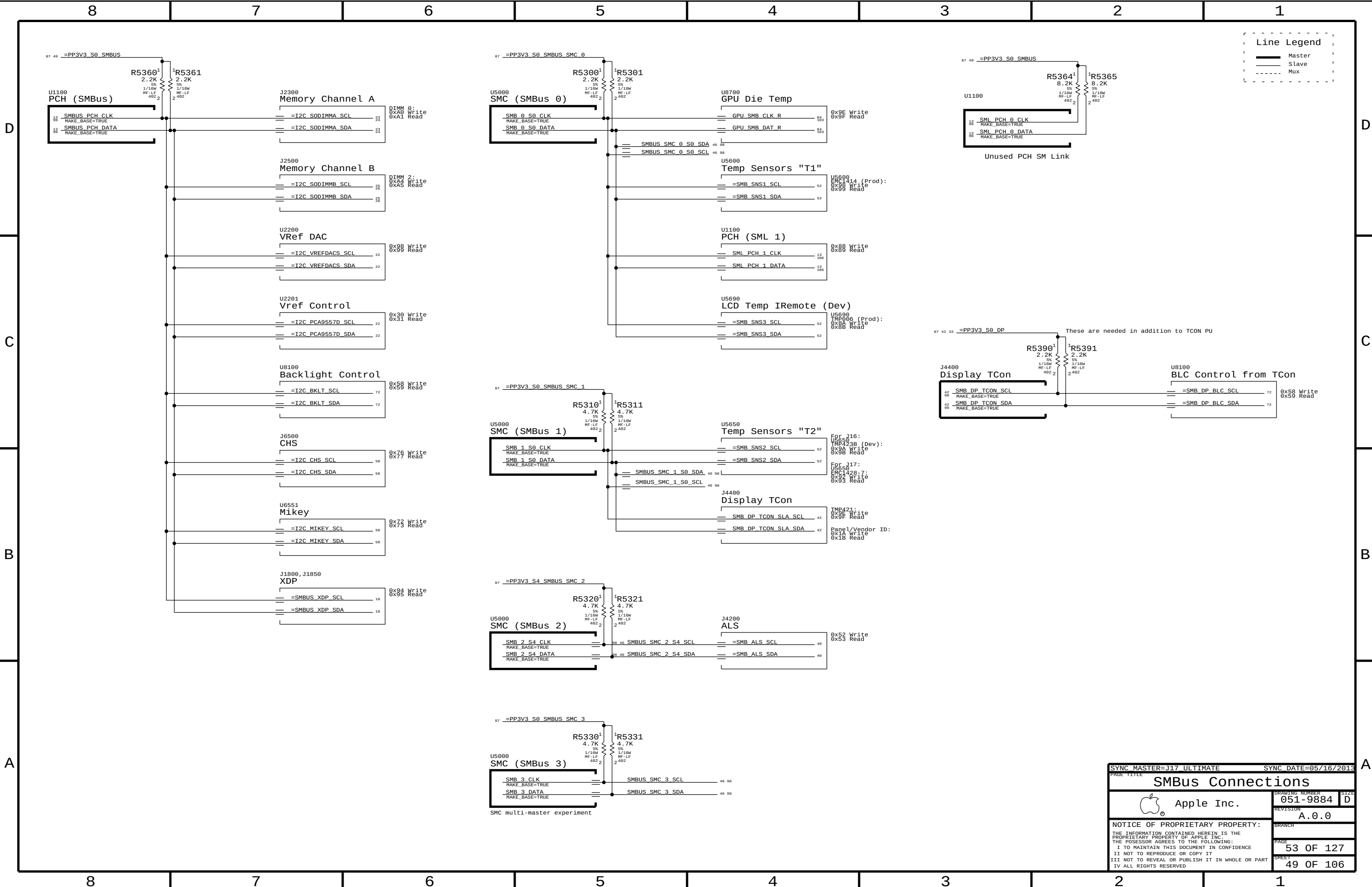
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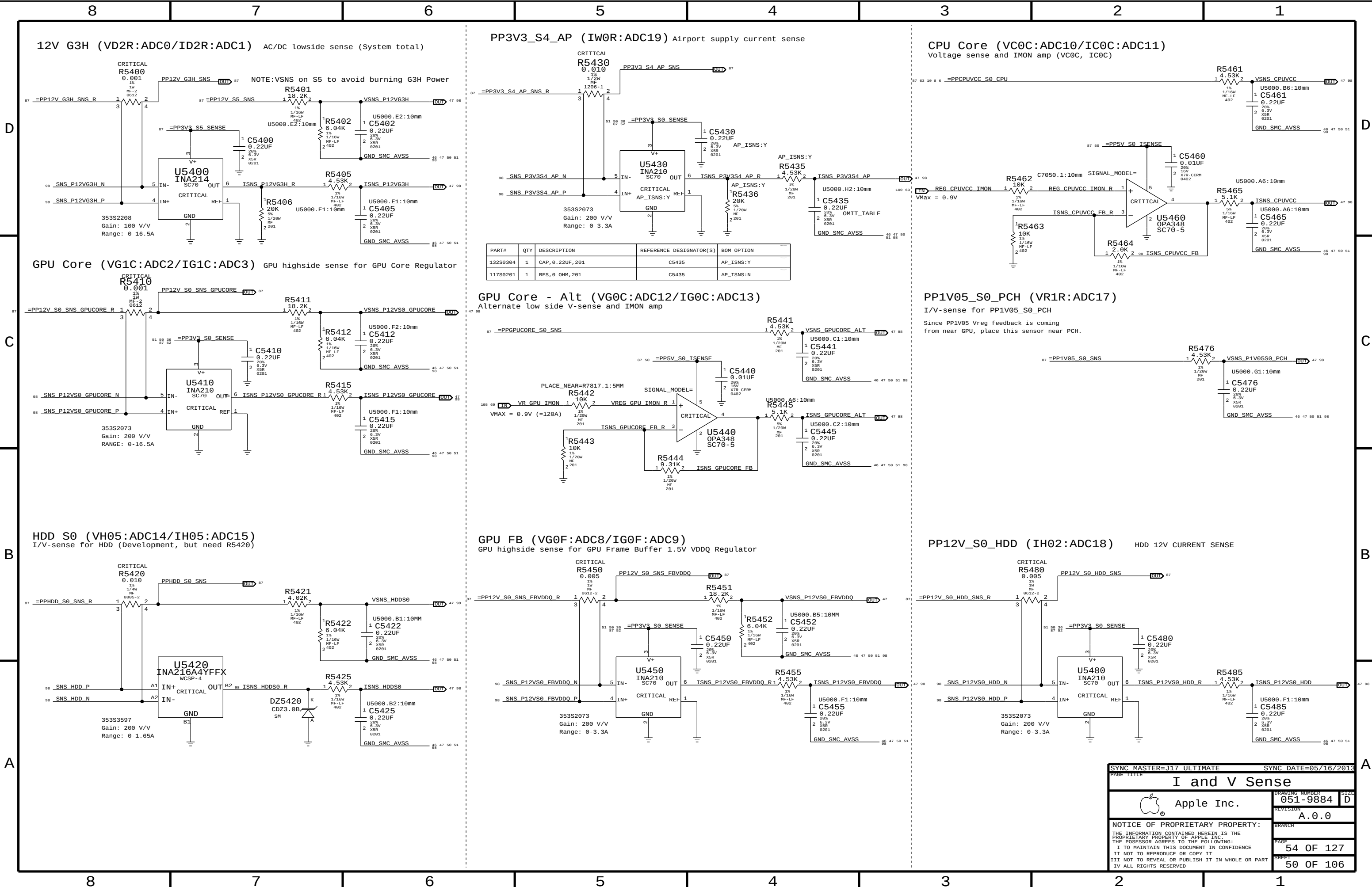
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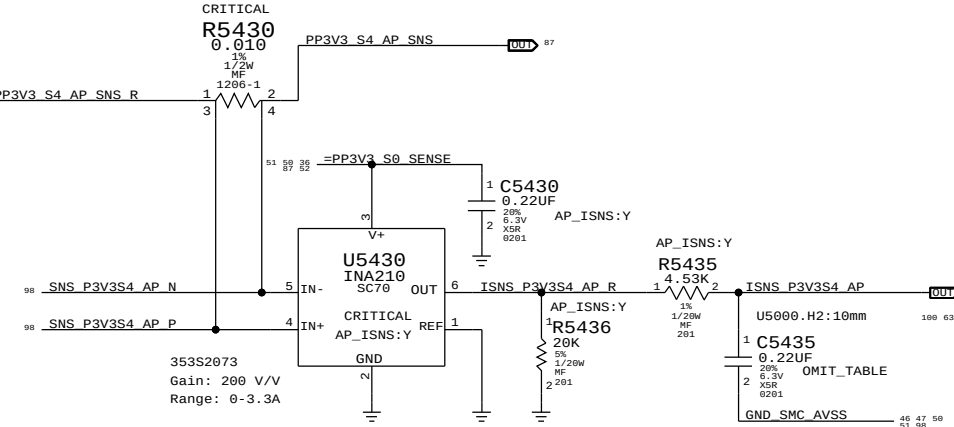
47 OF 106





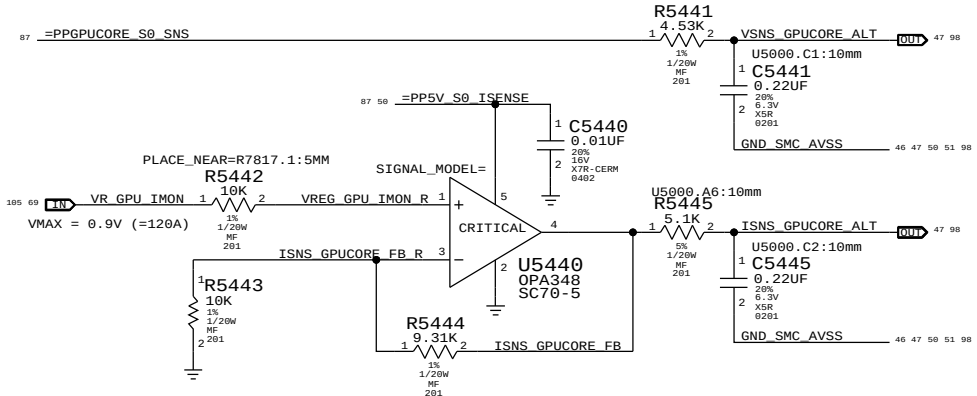


PP3V3_S4_AP (IW0R:ADC19) Airport supply current sense

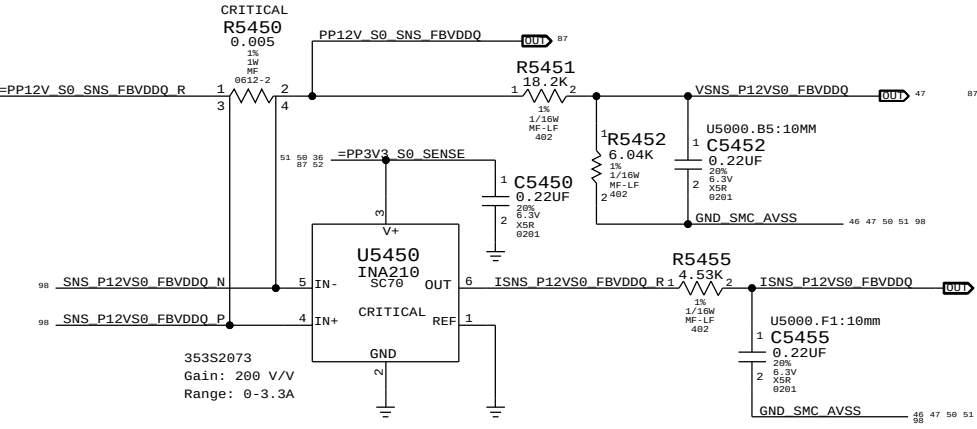


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
13250304	1	CAP, 0.22UF, 201	C5435	AP_ISNS:Y
11750201	1	RES, 0 OHM, 201	C5435	AP_ISNS:N

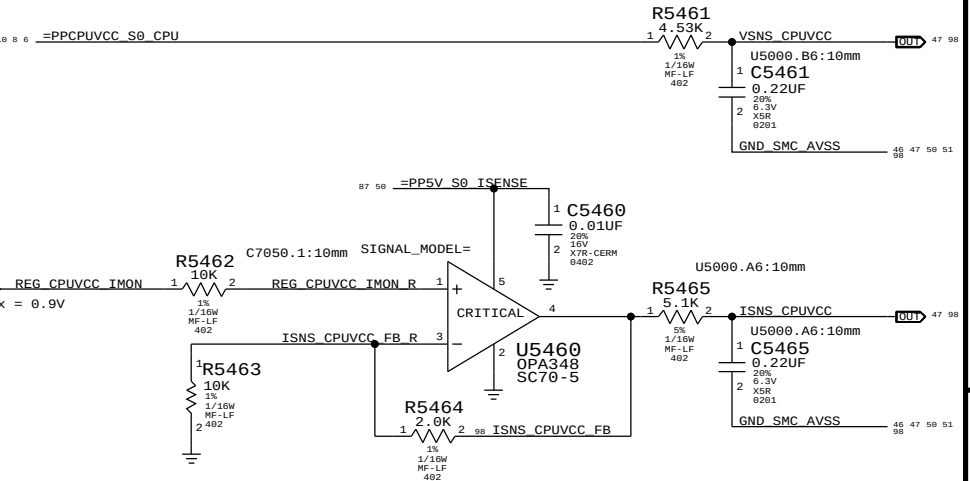
GPU Core - Alt (VG0C:ADC12/IG0C:ADC13)
Alternate low side V-sense and IMON amp



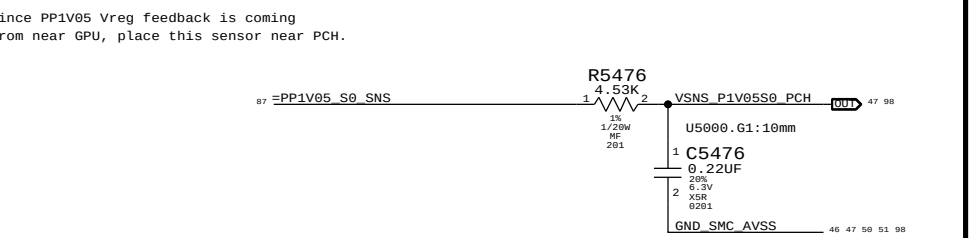
GPU FB (VG0F:ADC8/IG0F:ADC9)
GPU highside sense for GPU Frame Buffer 1.5V VDDQ Regulator



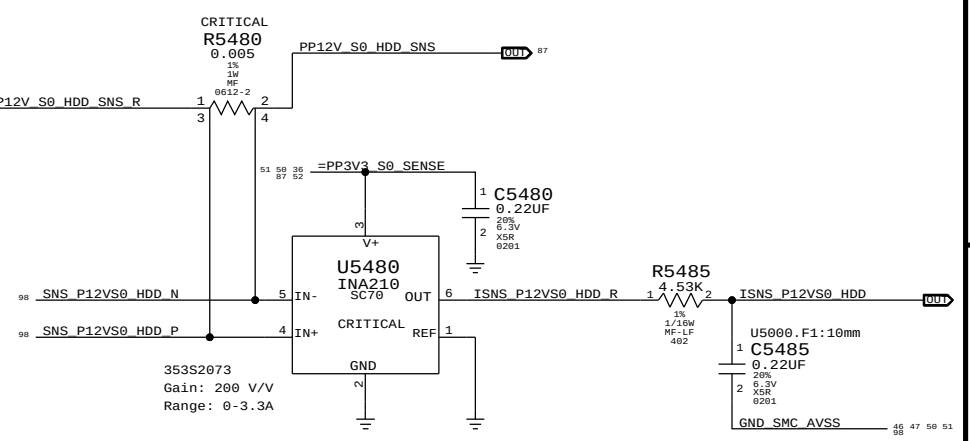
CPU Core (VC0C:ADC10/IC0C:ADC11)
Voltage sense and IMON amp (VC0C, IC0C)



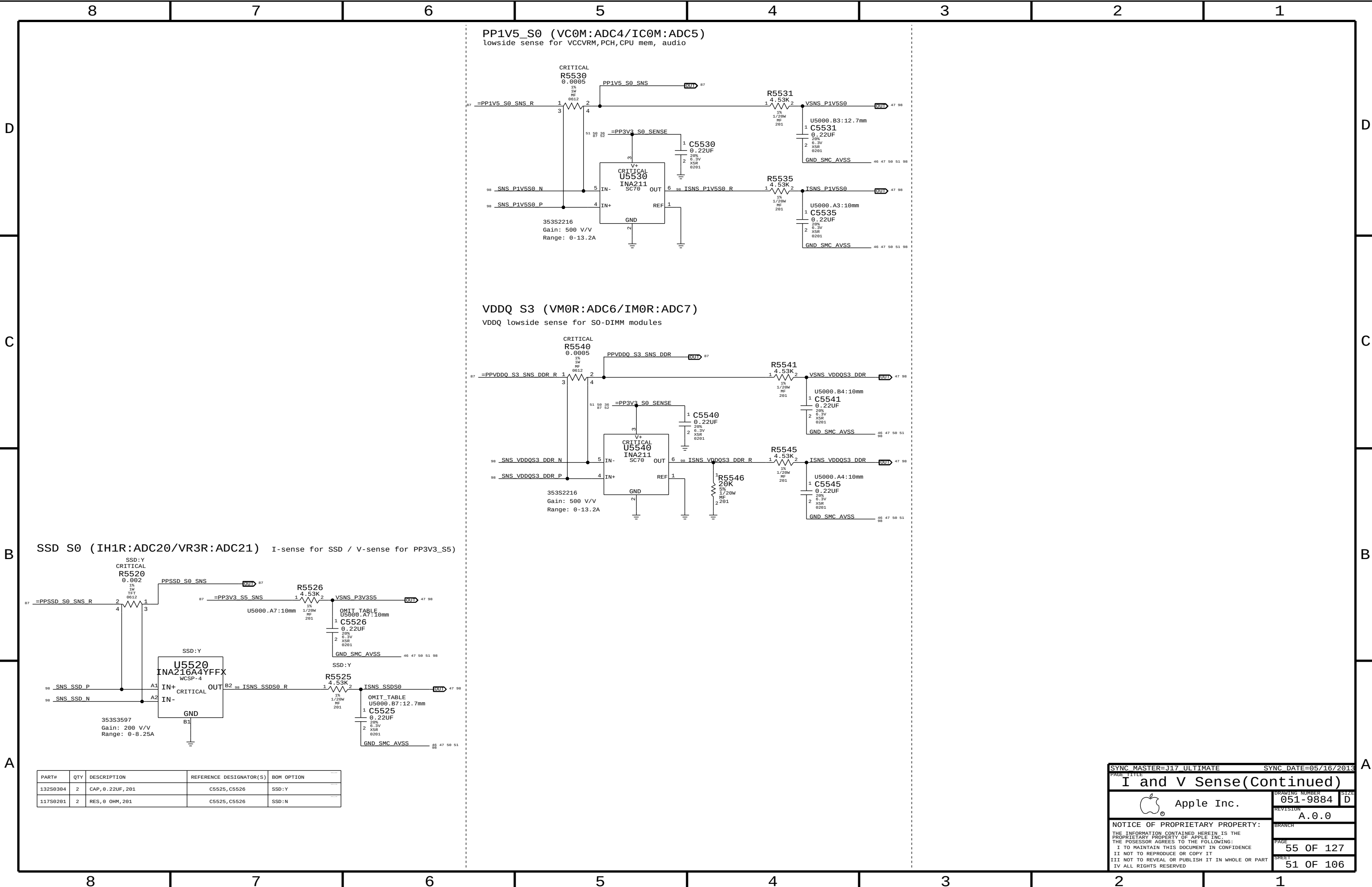
PP1V05_S0_PCH (VR1R:ADC17)
I/V-sense for PP1V05_S0_PCH



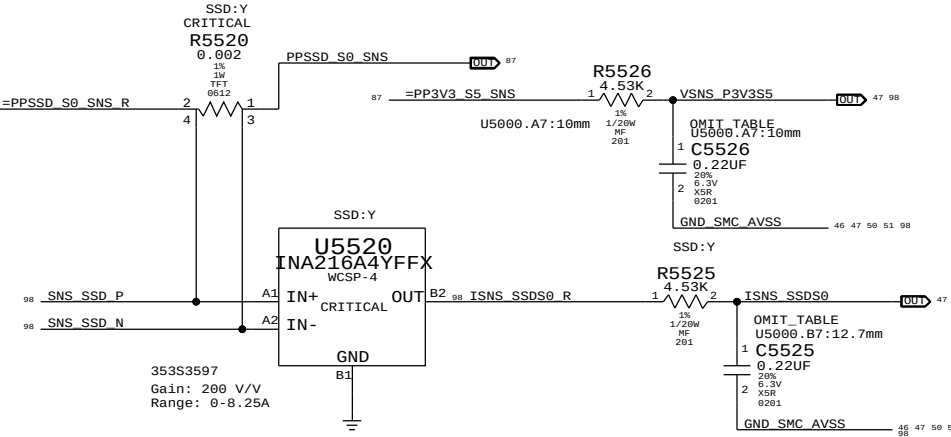
PP12V_S0_HDD (IH02:ADC18)
HDD 12V CURRENT SENSE



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		I and V Sense	
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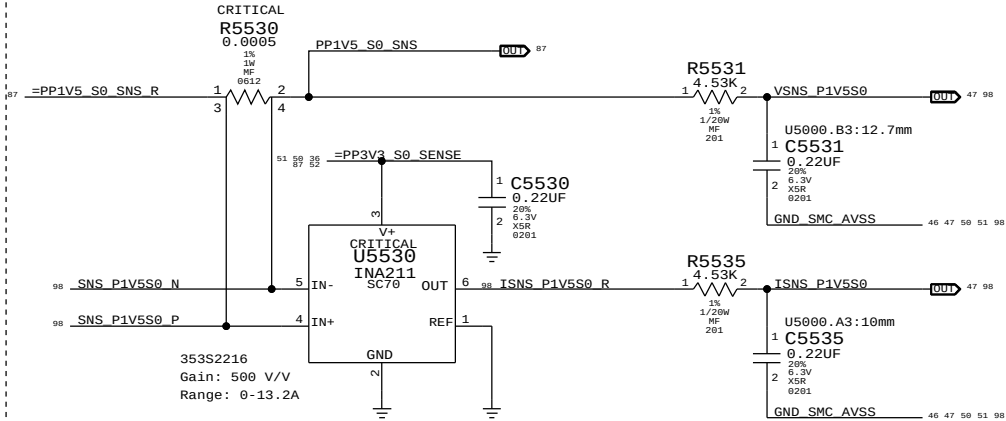


SSD_S0 (IH1R:ADC20/VR3R:ADC21) I-sense for SSD / V-sense for PP3V3_S5)

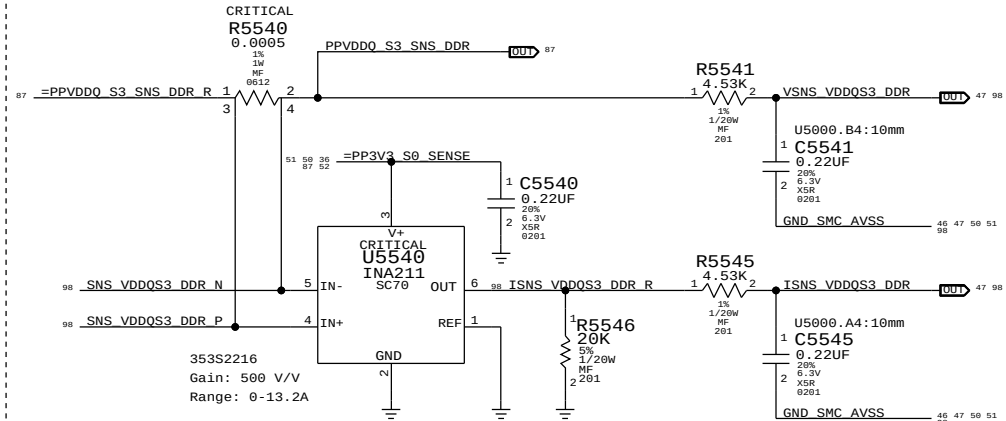


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0304	2	CAP, 0.22UF, 201	C5525, C5526	SSD:Y
117S0201	2	RES, 0 OHM, 201	C5525, C5526	SSD:N

PP1V5_S0 (VC0M:ADC4/IC0M:ADC5)
lowside sense for VCCVRM, PCH, CPU mem, audio



VDDQ_S3 (VM0R:ADC6/IM0R:ADC7)
VDDQ lowside sense for S0-DIMM modules



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SYNC DATE=05/16/2013

I and V Sense(Continued)

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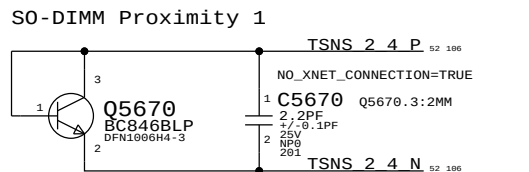
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PAGE
55 OF 127

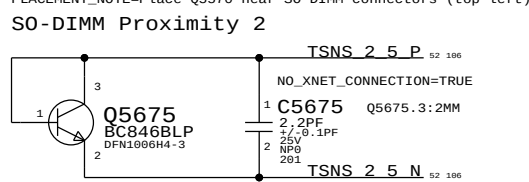
SHEET
51 OF 106

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

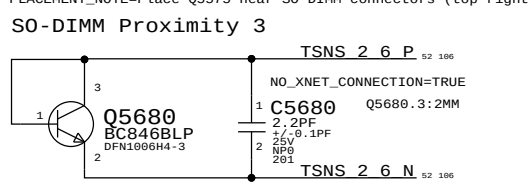
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87 52 51 50 36 =PP3V3 S0 SENSE
```



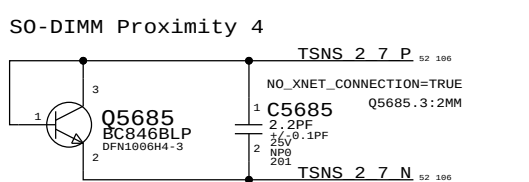
PLACEMENT_NOTE=Place Q5570 near S0-DIMM connectors (top left)



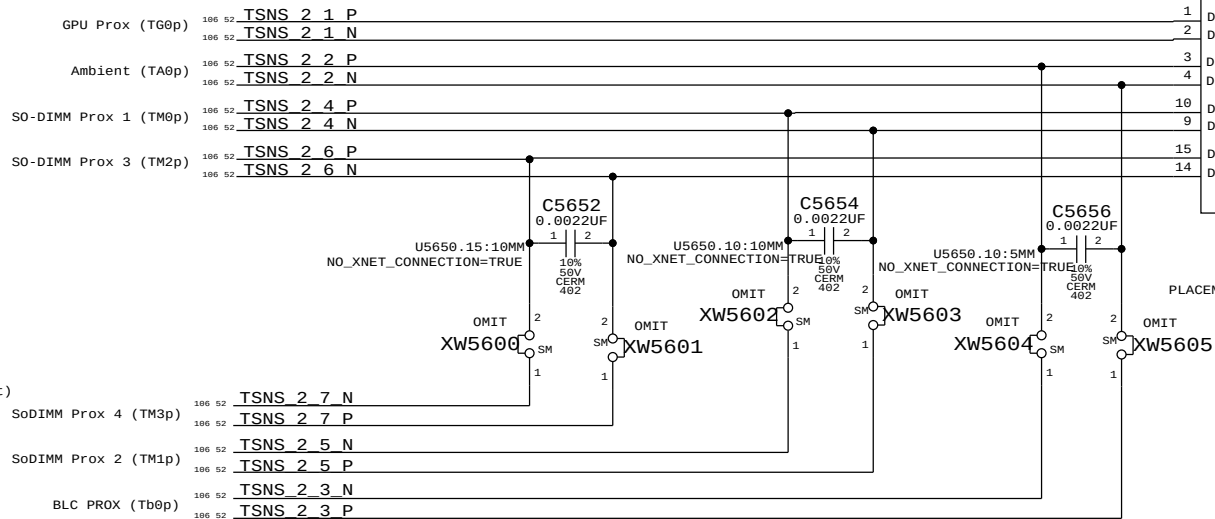
PLACEMENT_NOTE=Place Q5575 near S0-DIMM connectors (top right



PLACEMENT_NOTE=Place Q5580 near SO-DIMM connectors (bottom left)

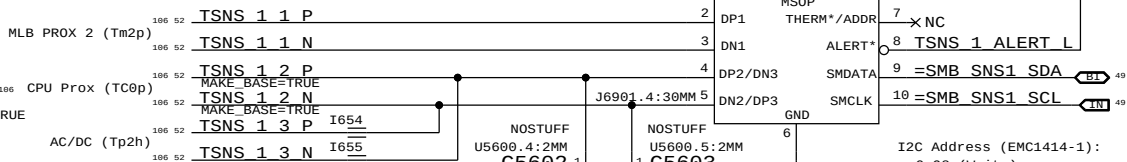


PLACEMENT_NOTE=Place Q5585 near SO-DIMM connectors (bottom ri



EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

```
87 52 51 50 36 =PP3V3_S0_SENSE
```



Pin	Symbol	Function
1	V _{CC}	Power supply
2	Q ₁	Output 1
3	Q ₂	Output 2
4	Q ₃	Output 3
5	Q ₄	Output 4
6	Q ₅	Output 5
7	Q ₆	Output 6
8	Q ₇	Output 7
9	Q ₈	Output 8
10	Q ₉	Output 9
11	Q ₁₀	Output 10
12	Q ₁₁	Output 11
13	Q ₁₂	Output 12
14	Q ₁₃	Output 13
15	Q ₁₄	Output 14
16	Q ₁₅	Output 15
17	Q ₁₆	Output 16
18	Q ₁₇	Output 17
19	Q ₁₈	Output 18
20	Q ₁₉	Output 19
21	Q ₂₀	Output 20
22	Q ₂₁	Output 21
23	Q ₂₂	Output 22
24	Q ₂₃	Output 23
25	Q ₂₄	Output 24
26	Q ₂₅	Output 25
27	Q ₂₆	Output 26
28	Q ₂₇	Output 27
29	Q ₂₈	Output 28
30	Q ₂₉	Output 29
31	Q ₃₀	Output 30
32	Q ₃₁	Output 31
33	Q ₃₂	Output 32
34	Q ₃₃	Output 33
35	Q ₃₄	Output 34
36	Q ₃₅	Output 35
37	Q ₃₆	Output 36
38	Q ₃₇	Output 37
39	Q ₃₈	Output 38
40	Q ₃₉	Output 39
41	Q ₄₀	Output 40
42	Q ₄₁	Output 41
43	Q ₄₂	Output 42
44	Q ₄₃	Output 43
45	Q ₄₄	Output 44
46	Q ₄₅	Output 45
47	Q ₄₆	Output 46
48	Q ₄₇	Output 47
49	Q ₄₈	Output 48
50	Q ₄₉	Output 49
51	Q ₅₀	Output 50
52	Q ₅₁	Output 51
53	Q ₅₂	Output 52
54	Q ₅₃	Output 53
55	Q ₅₄	Output 54
56	Q ₅₅	Output 55
57	Q ₅₆	Output 56
58	Q ₅₇	Output 57
59	Q ₅₈	Output 58
60	Q ₅₉	Output 59
61	Q ₆₀	Output 60
62	Q ₆₁	Output 61
63	Q ₆₂	Output 62
64	Q ₆₃	Output 63
65	Q ₆₄	Output 64
66	Q ₆₅	Output 65
67	Q ₆₆	Output 66
68	Q ₆₇	Output 67
69	Q ₆₈	Output 68
70	Q ₆₉	Output 69
71	Q ₇₀	Output 70
72	Q ₇₁	Output 71
73	Q ₇₂	Output 72
74	Q ₇₃	Output 73
75	Q ₇₄	Output 74
76	Q ₇₅	Output 75
77	Q ₇₆	Output 76
78	Q ₇₇	Output 77
79	Q ₇₈	Output 78
80	Q ₇₉	Output 79
81	Q ₈₀	Output 80
82	Q ₈₁	Output 81
83	Q ₈₂	Output 82
84	Q ₈₃	Output 83
85	Q ₈₄	Output 84
86	Q ₈₅	Output 85
87	Q ₈₆	Output 86
88	Q ₈₇	Output 87
89	Q ₈₈	Output 88
90	Q ₈₉	Output 89
91	Q ₉₀	Output 90
92	Q ₉₁	Output 91
93	Q ₉₂	Output 92
94	Q ₉₃	Output 93
95	Q ₉₄	Output 94
96	Q ₉₅	Output 95
97	Q ₉₆	Output 96
98	Q ₉₇	Output 97
99	Q ₉₈	Output 98
100	Q ₉₉	Output 99

AC/DC Via connector to diode inside PSU

J6901.5:30MM
L5610
FERR-220-OHM

1 2

0482

TSNS 1 3 P 52 186

J6901.4:30MM
L5611
FERR-220-OHM

1 2

0482

TSNS 1 3 N 52 186

L5610.2:10MM
L5610
0.0022UF
10%
50V
CERM
482

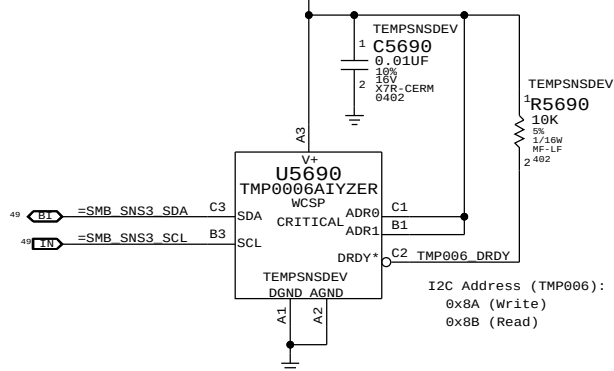
52 186

52 186

Make sure these caps are OK with U5600 vendor!	Internal sensor of the EMC 1414 will be used as MLB sensor.
PLACEMENT NOTE=PLACE U5600 NEAR PSU CONNECTOR	MLB PROX 0 (TmOp)

PLACEMENT_NOTE=PLACE U5600 NEAR PSU CONNECTOR

```
87 52 51 50 36 =PP3V3 S0 SENSE
```



NOTE - Follow TI layout guide (SB0U108.pdf) for this part!!!

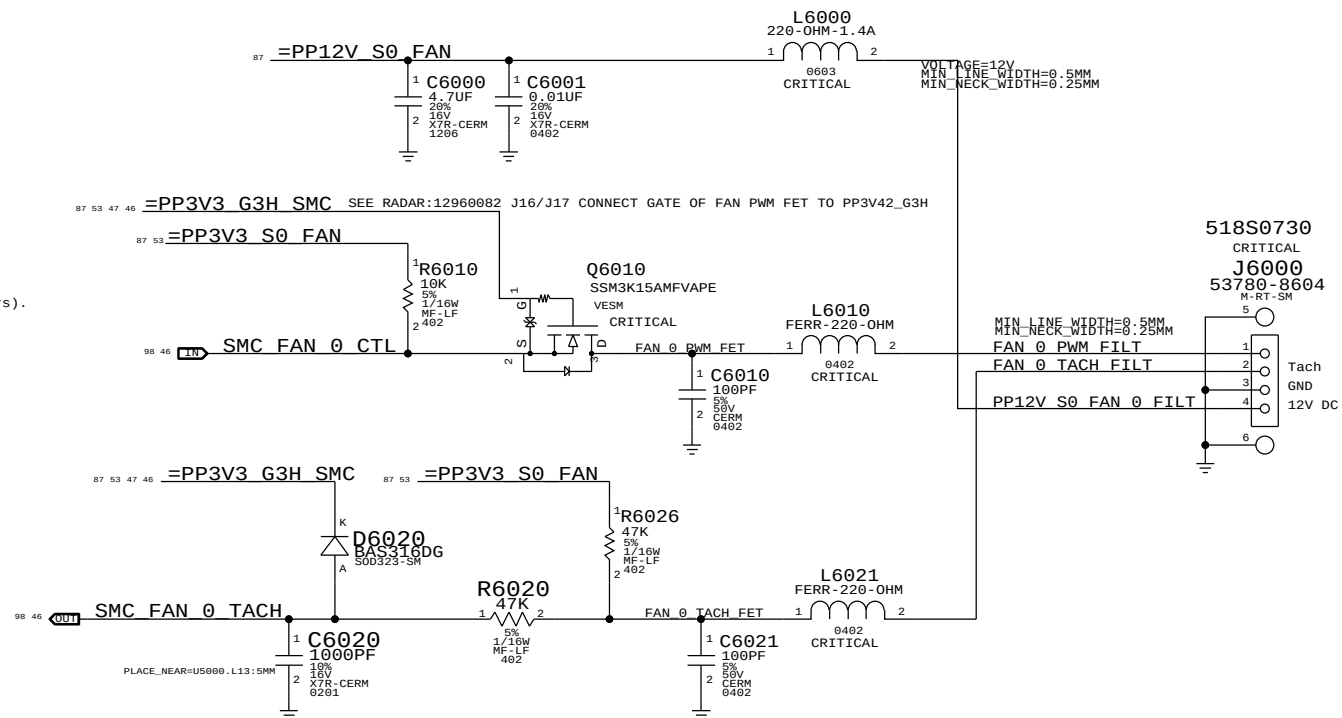
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Temperature Sensors			
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Note:

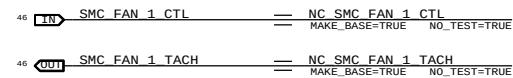
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q6010 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q6010 is at common and the SMC sinks current when Q6010 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

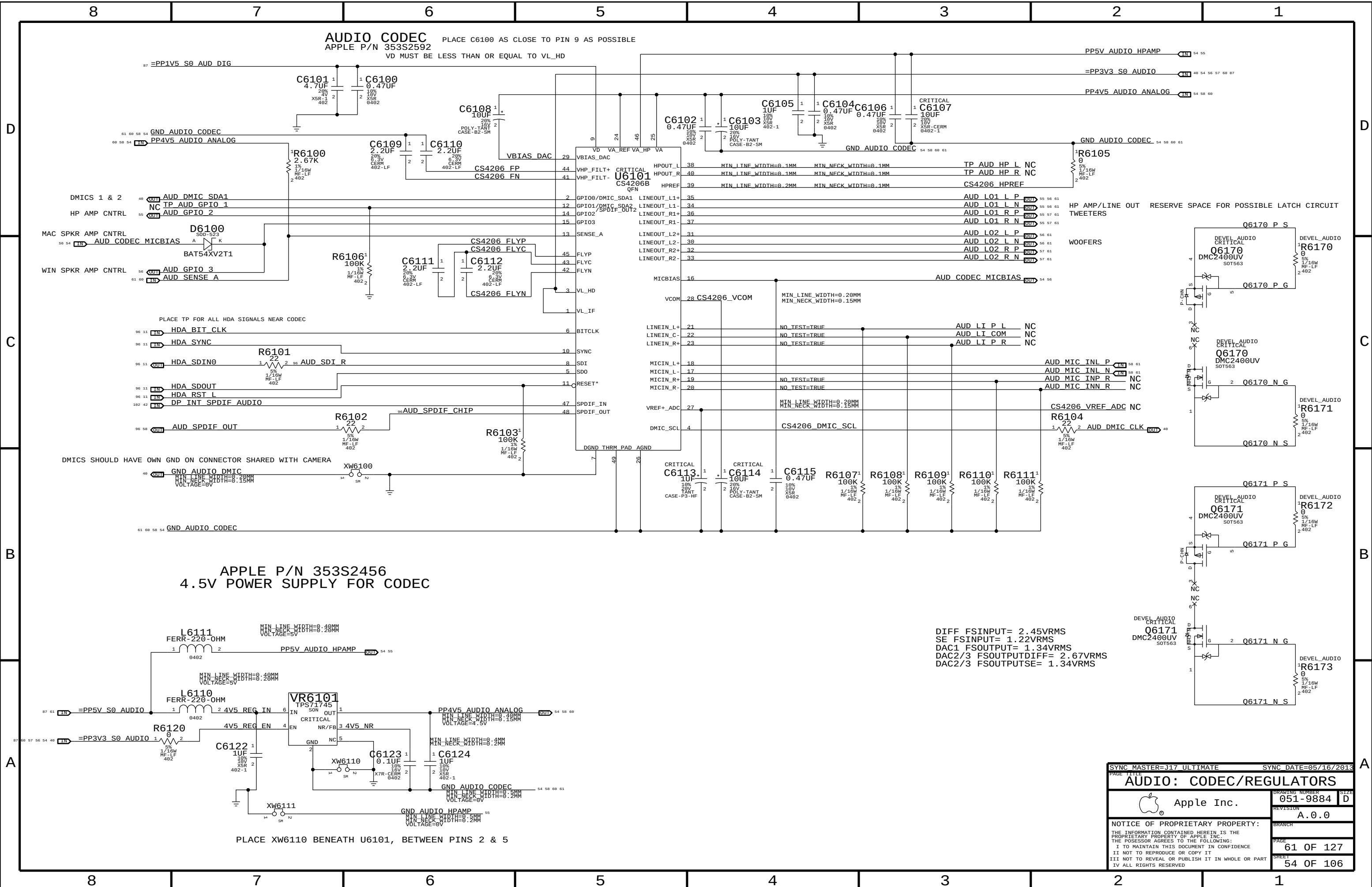
Otherwise, this is simply a pass-FET. See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



SMC Fan 1 (Unused)



SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
System Fan			
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		PAGE	60 OF 127
		SHEET	53 OF 106



PAGE TITLE		SYNC DATE=05/16/2013	
AUDIO: CODEC/REGULATORS		DRAWING NUMBER	
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LEFT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

C

B

B

A

A

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

R6304
0 OHM
NOSTUFF

R6305
0 OHM
NOSTUFF

AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6306
NOSTUFF
NOSTUFF
47 KOHM
0 OHM

R6307
0 OHM
NOSTUFF
NOSTUFF
NOSTUFF

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE		AUDIO: LEFT SPKR AMP	
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		SHEET	56 OF 106

RIGHT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +12 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPFB, TWEETERS = ~847 HZ (4700 PF)
FC_HPFB, WOOFERS = ~4 HZ (1.0 UF)

D

D

C

C

B

B

A

A

INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

R6404
0 OHM
NOSTUFF

R6405
NOSTUFF
0 OHM

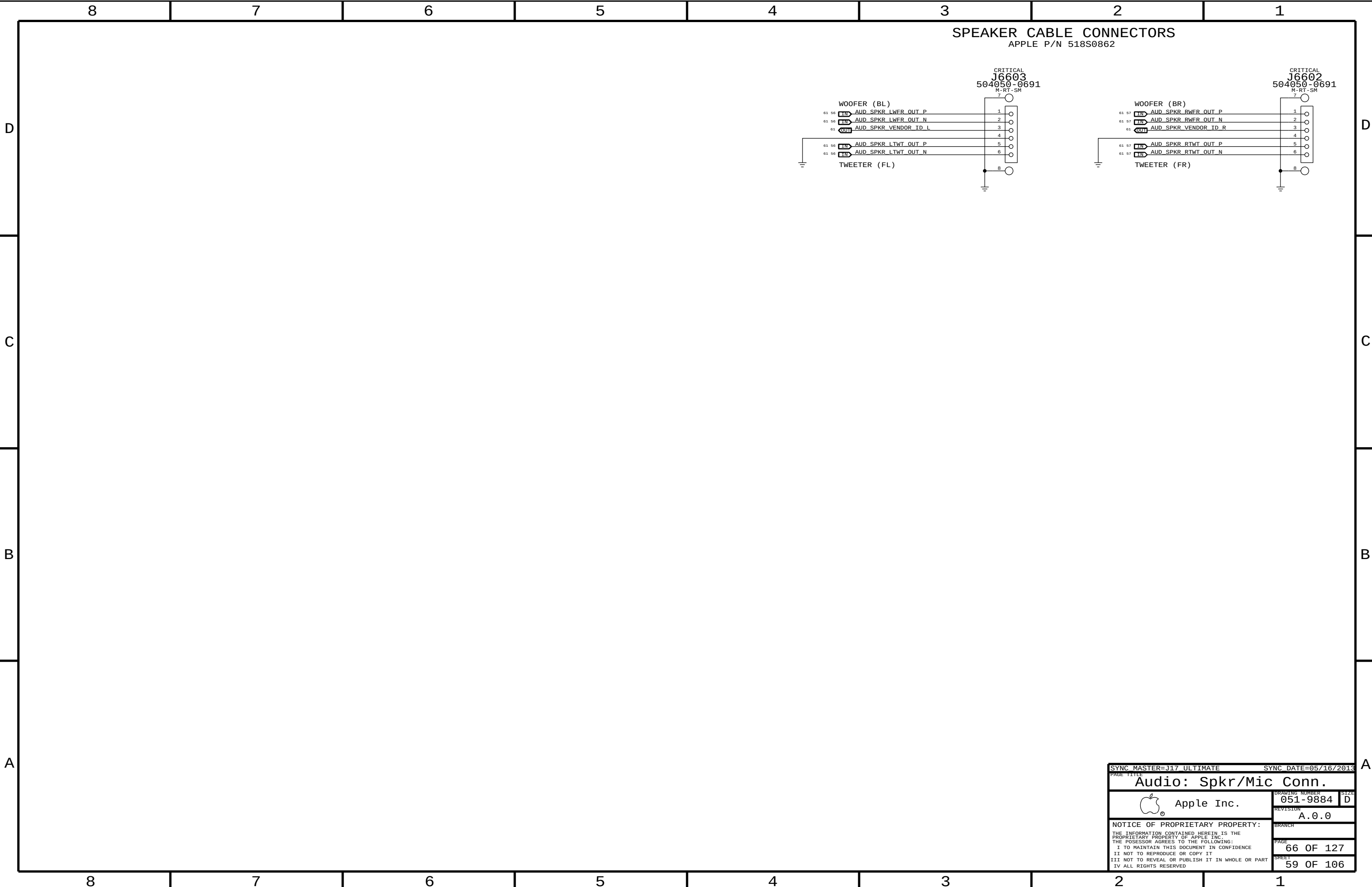
AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6406
NOSTUFF
NOSTUFF
47 KOHM
0 OHM

R6407
0 OHM
NOSTUFF
NOSTUFF
NOSTUFF
NOSTUFF

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AUDIO: RIGHT SPKR AMP			
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8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

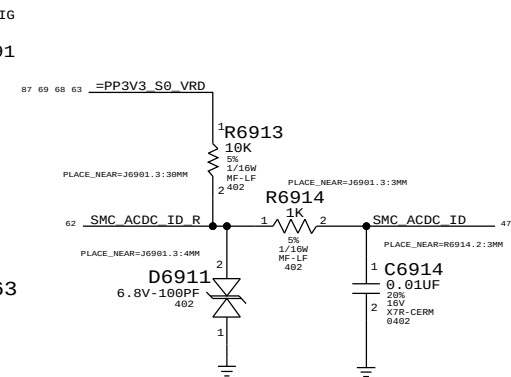
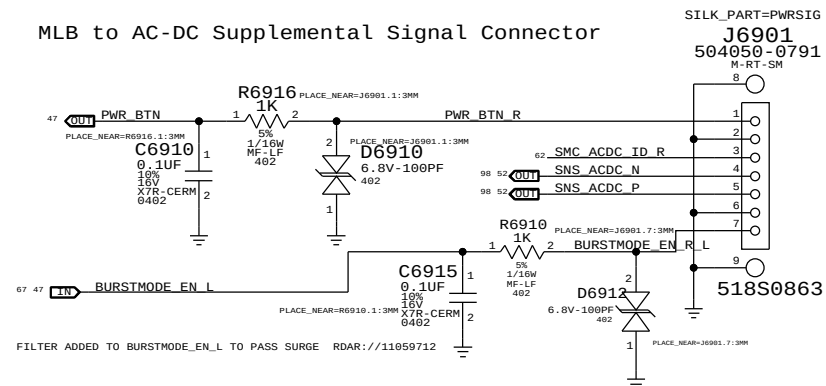
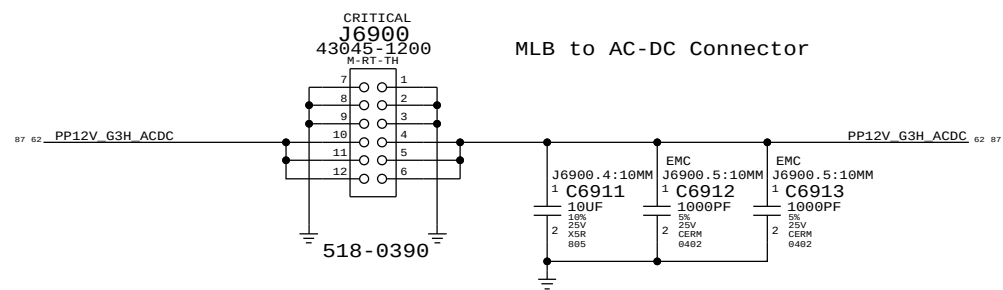


AUDIO CONNECTOR DETECT STATES			
	NOTHING	SPDIF	HEADPHONE
AUD_TYDEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_INA	1	20K/2.67K RDIV	5.11K/2.67K RDIV

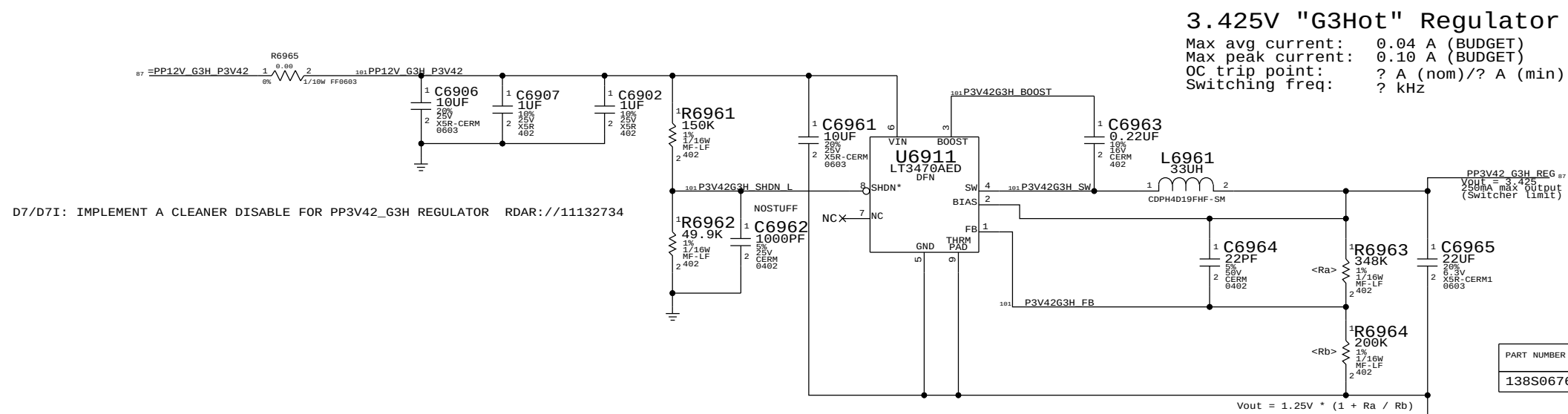


C





D8:CONTROLLER CHANGE FOR 3.42V SMC SUPPLY RDAR://11003901



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0676	138S0691		C6965	

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---	---	---	---	---	---	---	---

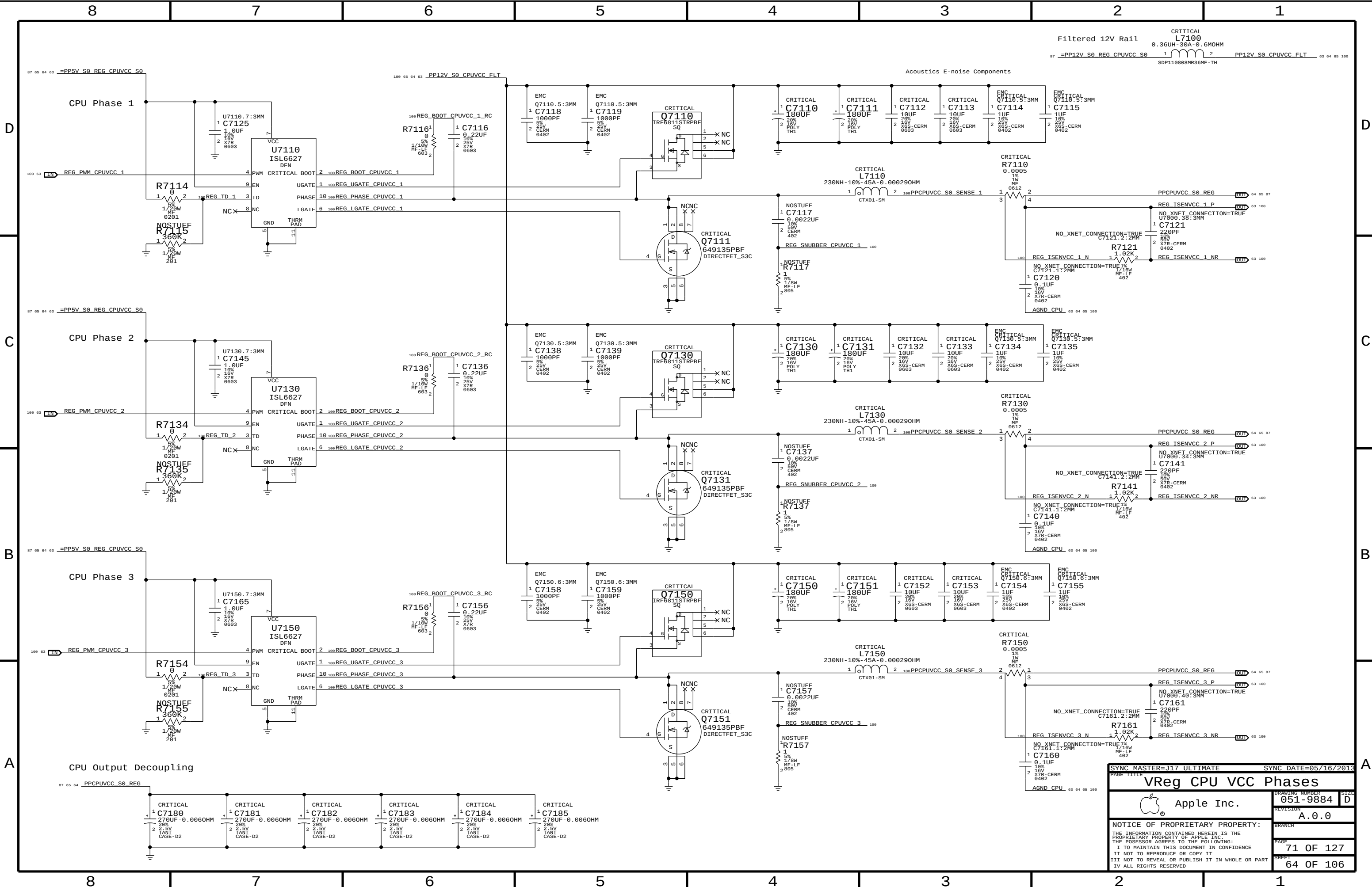
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SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

VReg CPU VCC Phases

Apple Inc.

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DRAWING NUMBER

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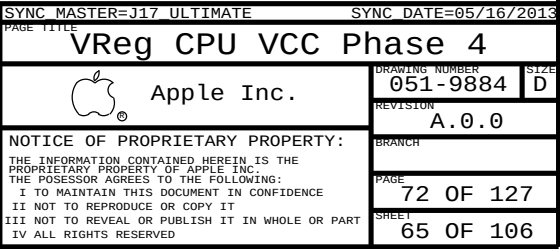
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PAGE

71 OF 127

SHEET

64 OF 106

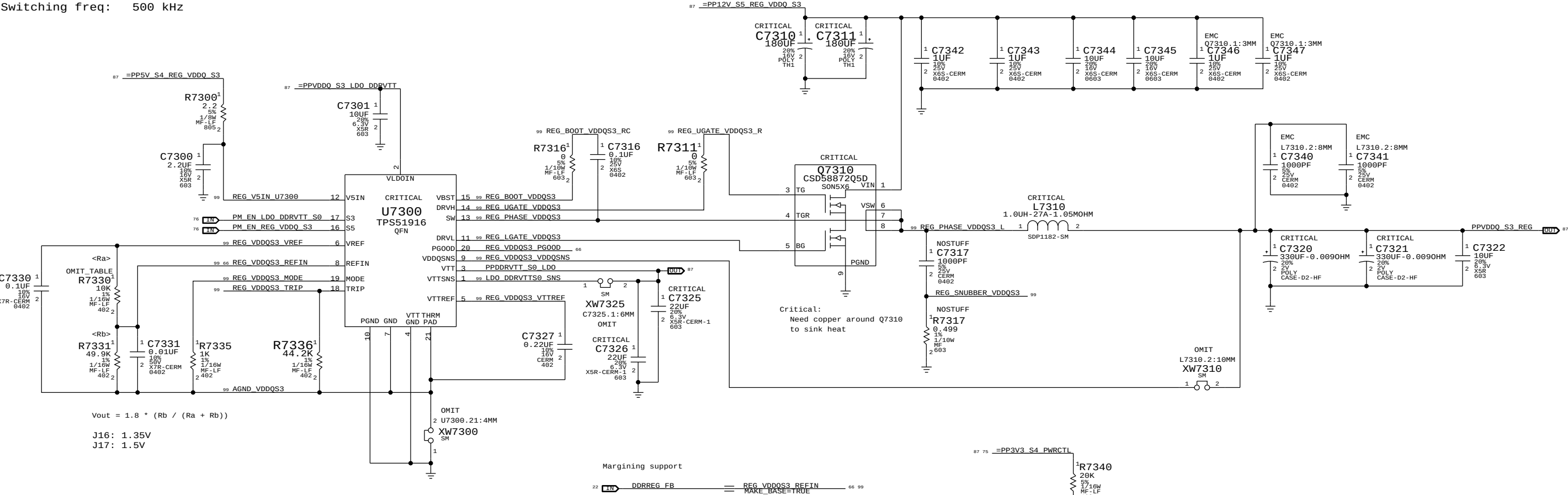


VDDQ (1.5V / 1.35V) S3 Regulator

OC trip point: $30.4 \text{ A VDDQ} = \frac{R7336}{8 \text{ E5} * \text{Rds}(Q7310)} + \frac{0.65625}{L7310 * f(\text{switch})}$

3 A VTT (FIXED)
10 mA VTTREF (FIXED)

Switching freq: 500 kHz



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
11450335	1	RES, 16.5K, 402	R7330	VDDQ:P1V35
11450315	1	RES, 10K, 402	R7330	VDDQ:P1V5

3.3V S5 Regulator

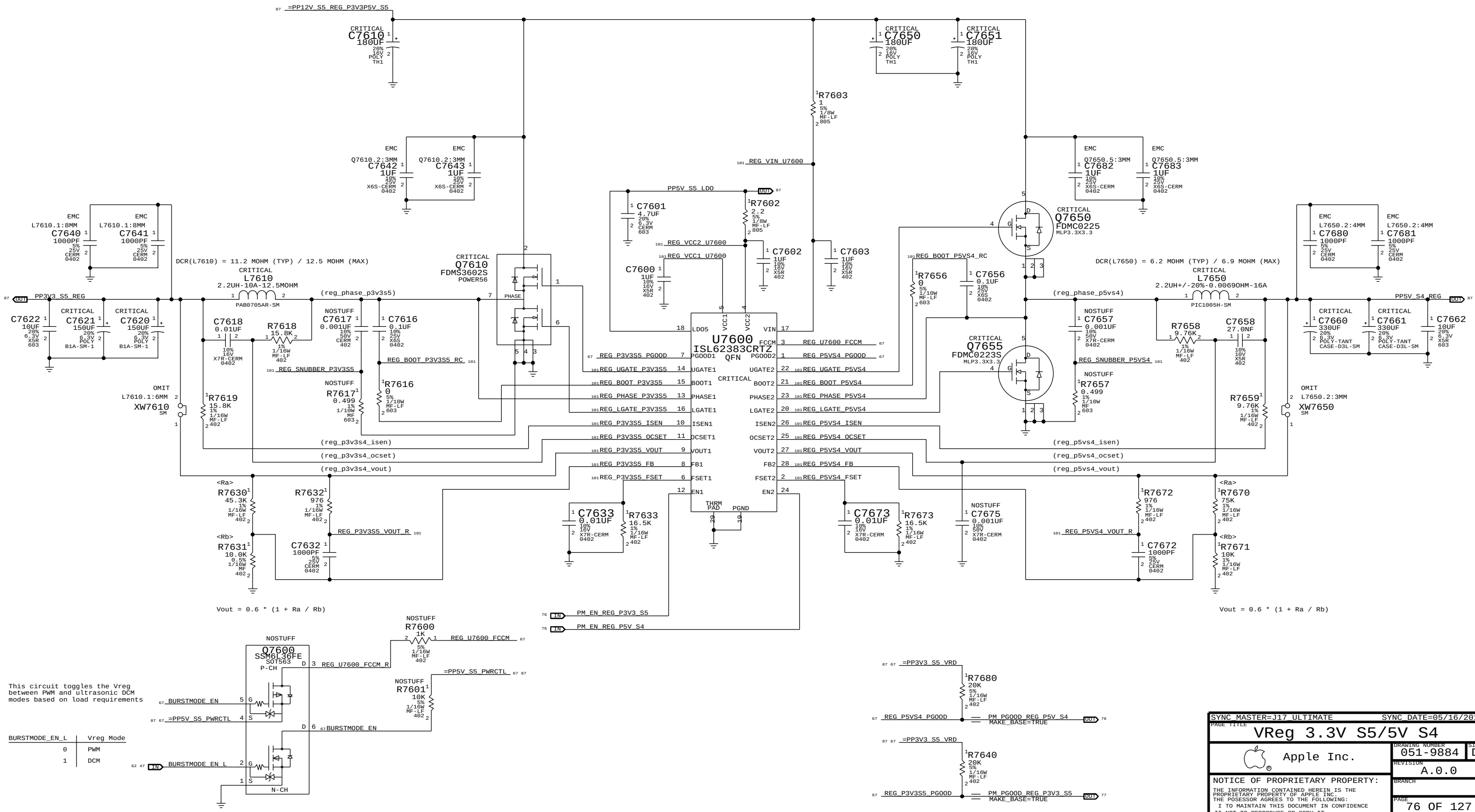
OC trip point: $12.5 \text{ A} = \frac{R7618 * 10 \text{ E-6}}{\text{DCR}(\text{L7610})}$

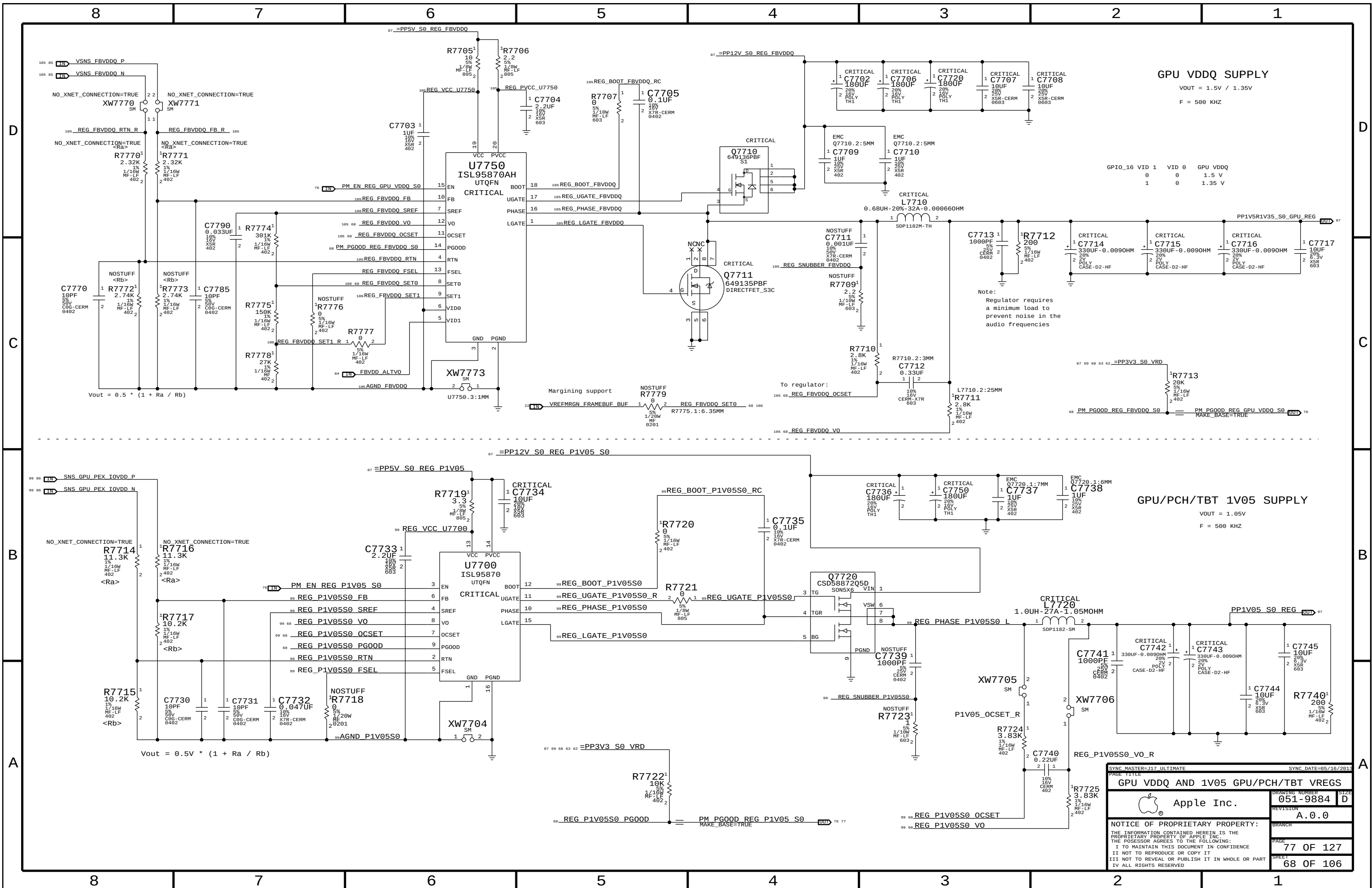
Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7633}$

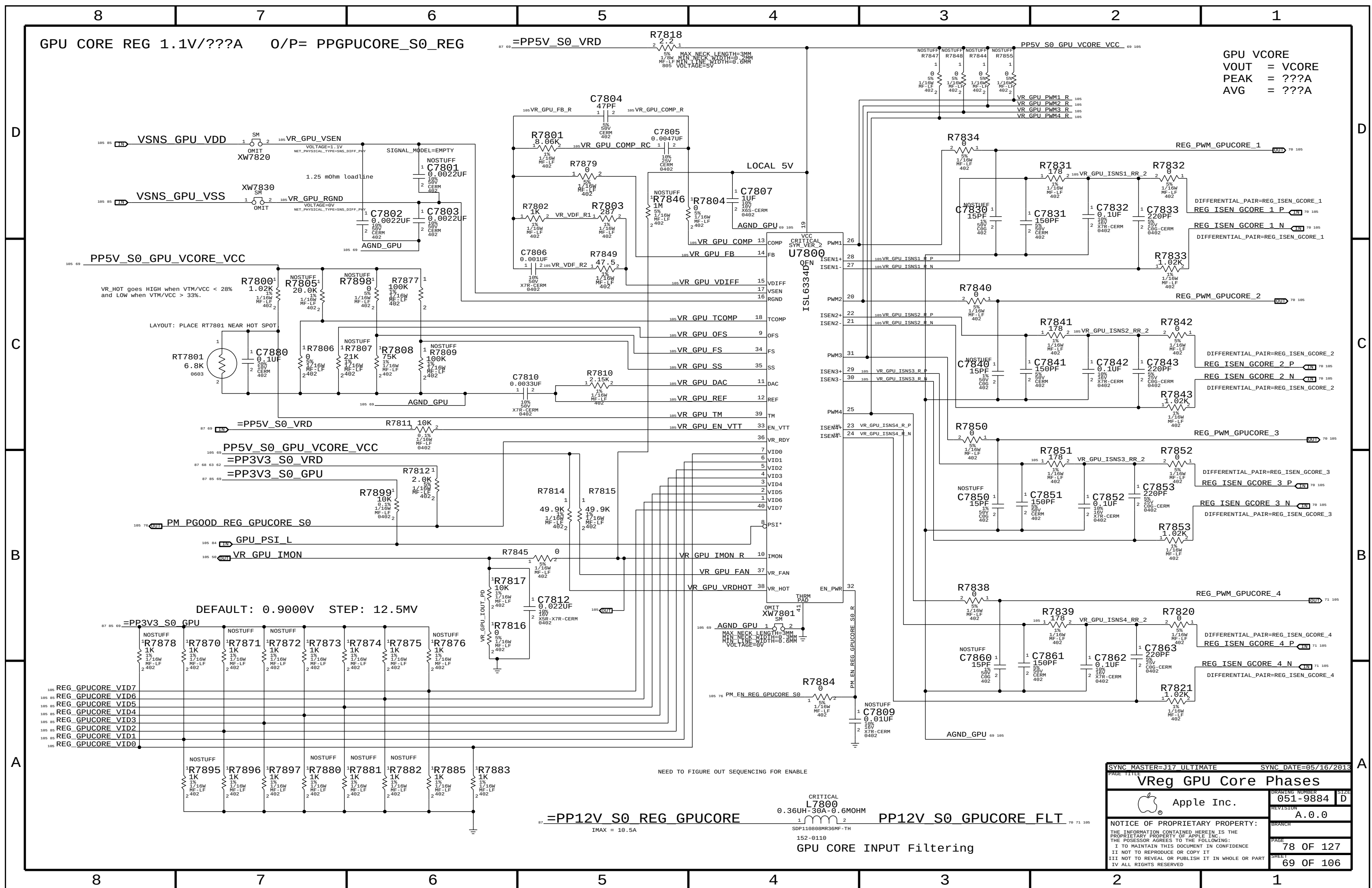
5V S4 Regulator

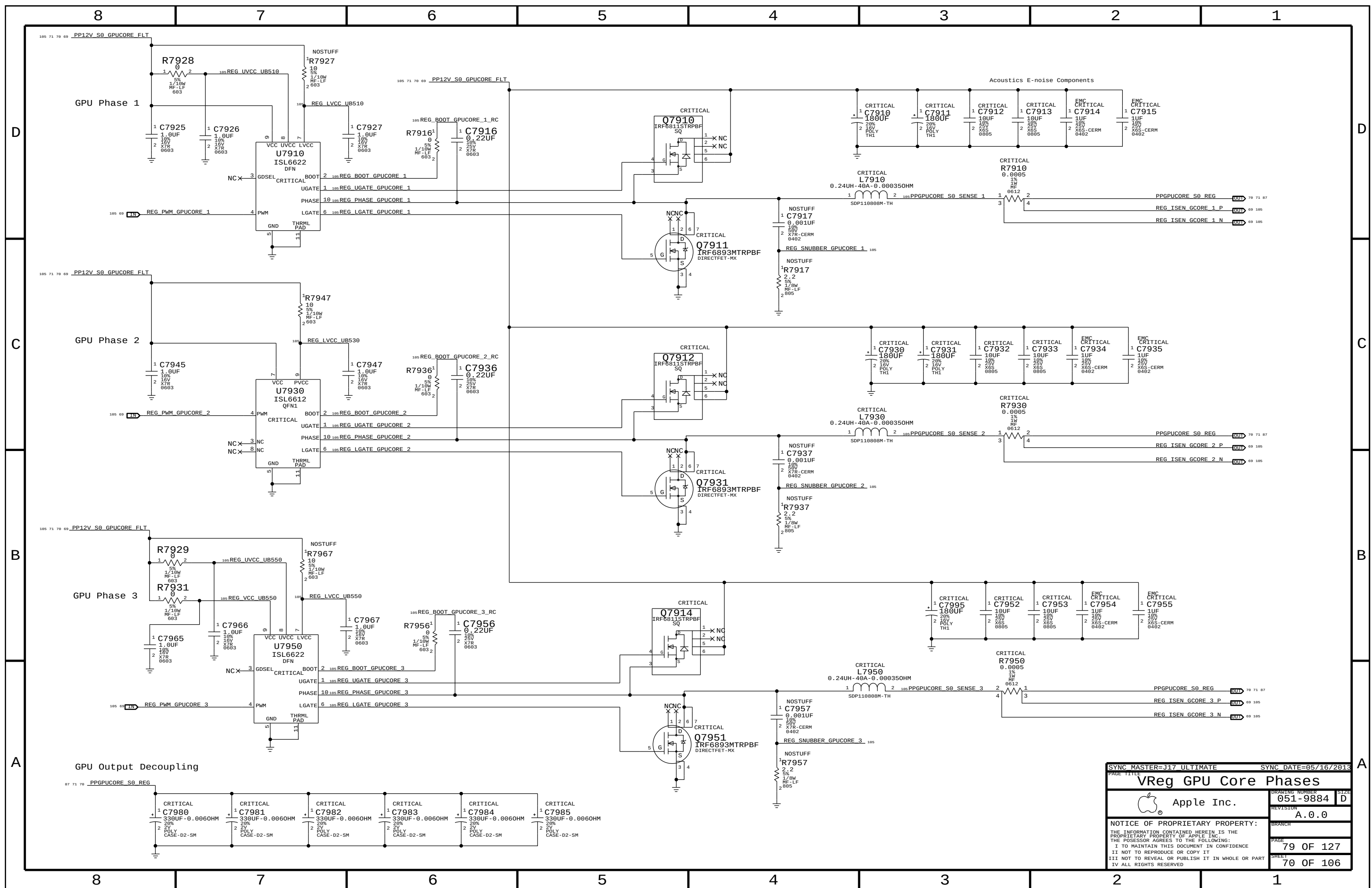
OC trip point: $14.1 \text{ A} = \frac{R7658 * 10 \text{ E-6}}{\text{DCR}(\text{L7650})}$

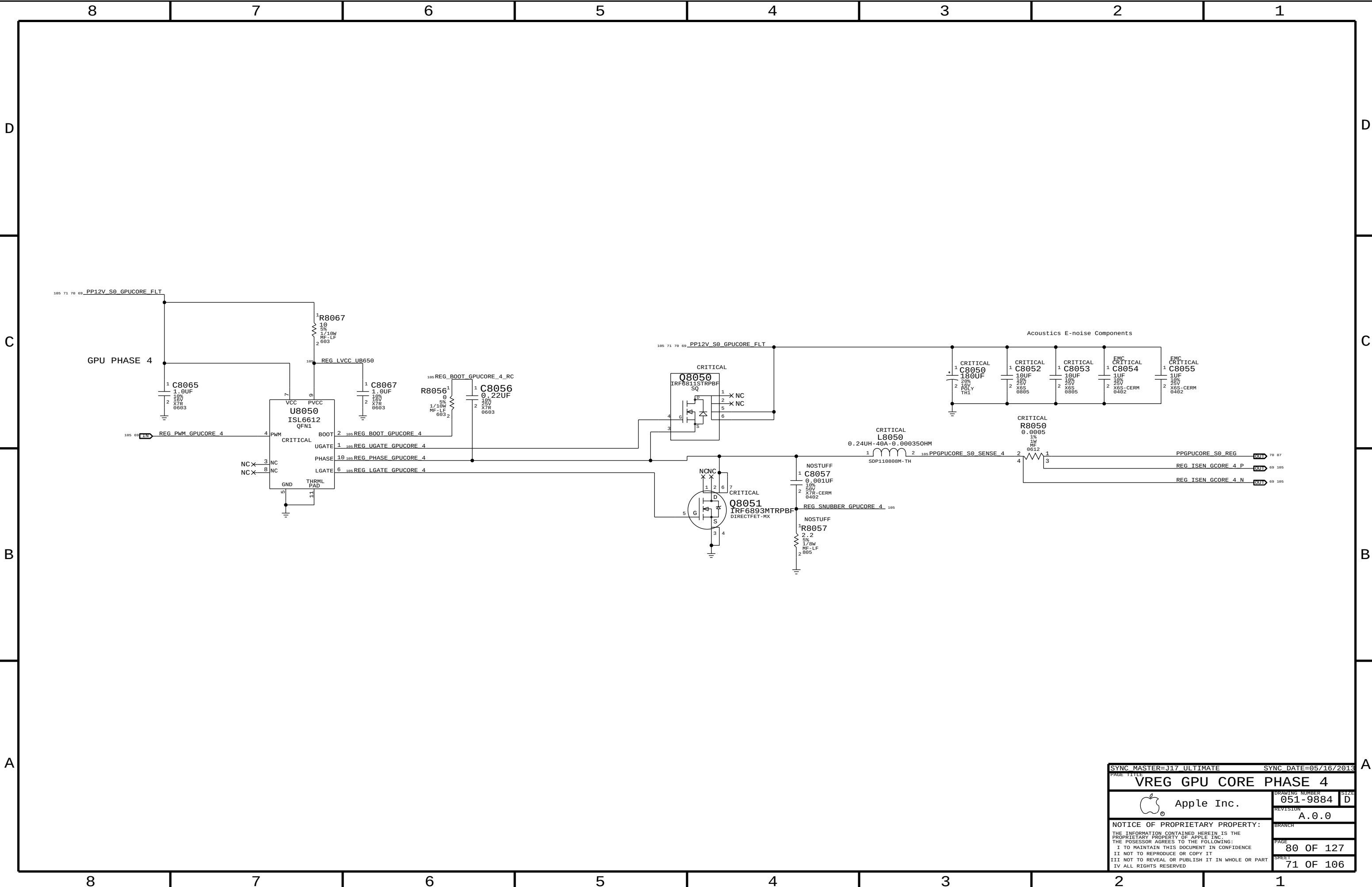
Switching freq: $356 \text{ kHz} = \frac{1}{170 \text{ E-12} * R7673}$

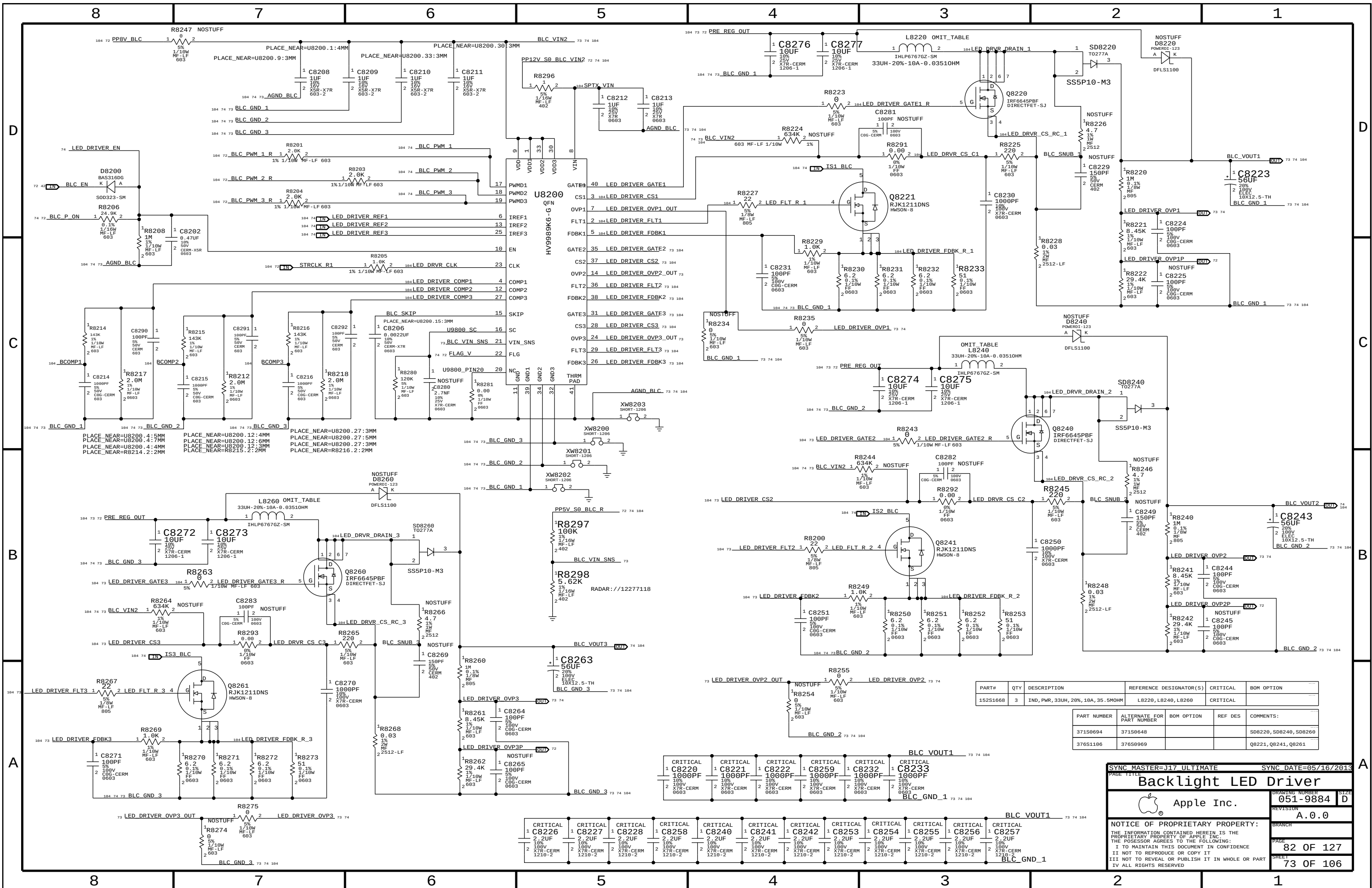












PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1668	3	IND, PWR, 33UH, 20%, 10A, 35.5MOHM	L8220, L8240, L8260	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
371S0694	371S0648			SD8220, SD8240, SD8260
376S1106	376S0969			Q8221, Q8241, Q8261

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SYNC DATE=05/16/2013

Backlight LED Driver

Apple Inc.

051-9884

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82 OF 127

73 OF 106

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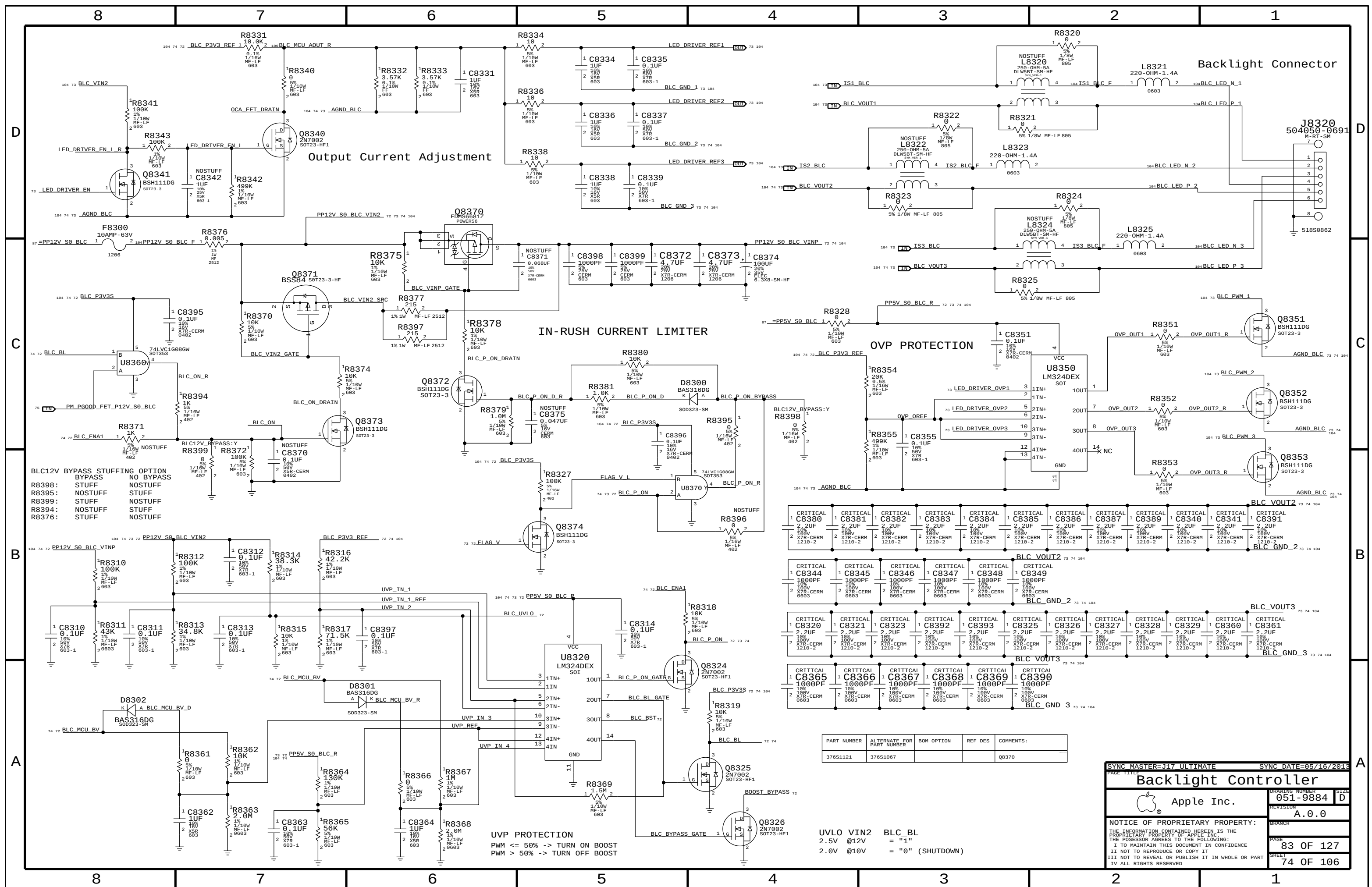
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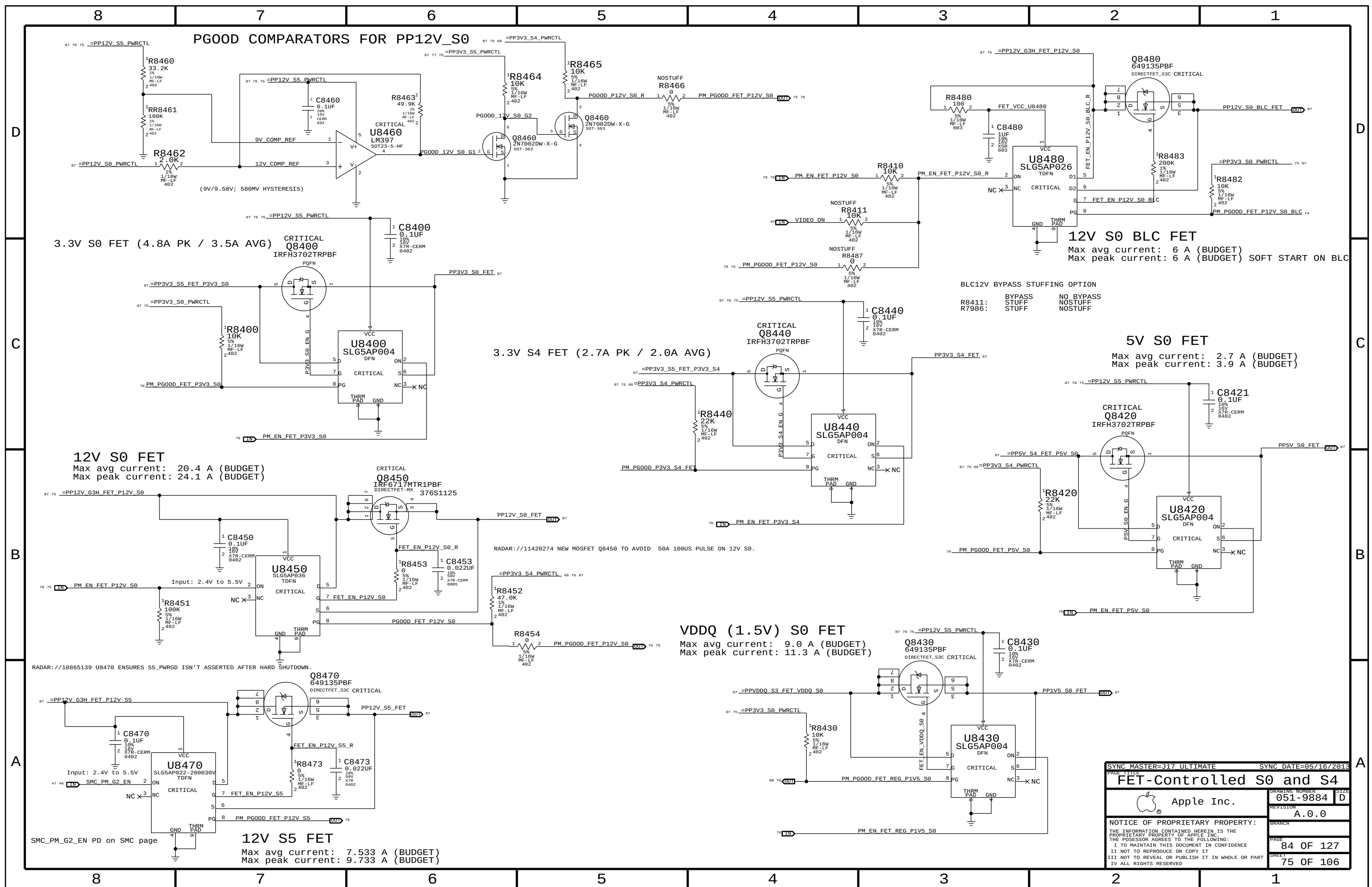
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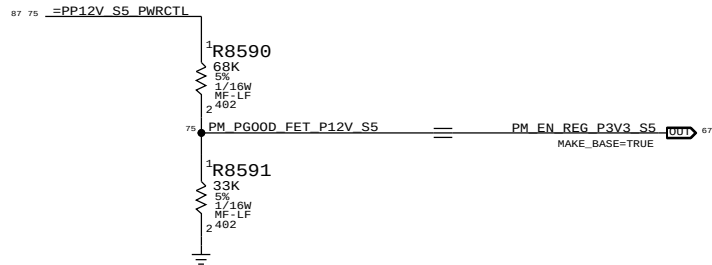
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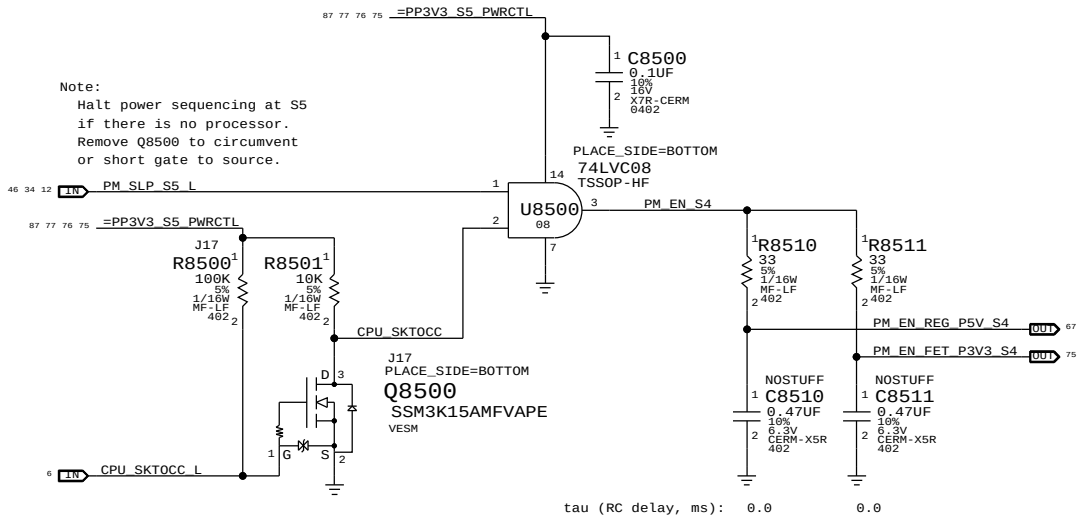




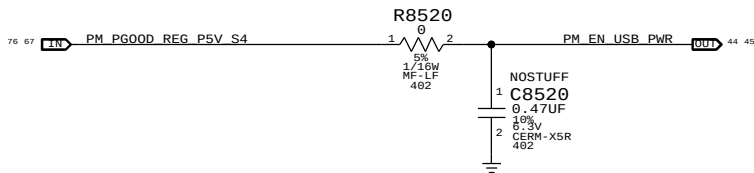
S5 Enable



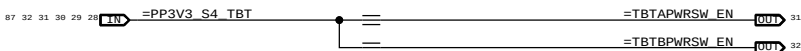
S4 Enables



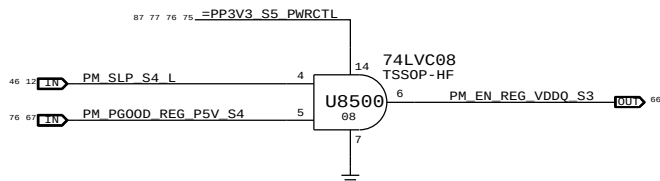
S4 USB Enable



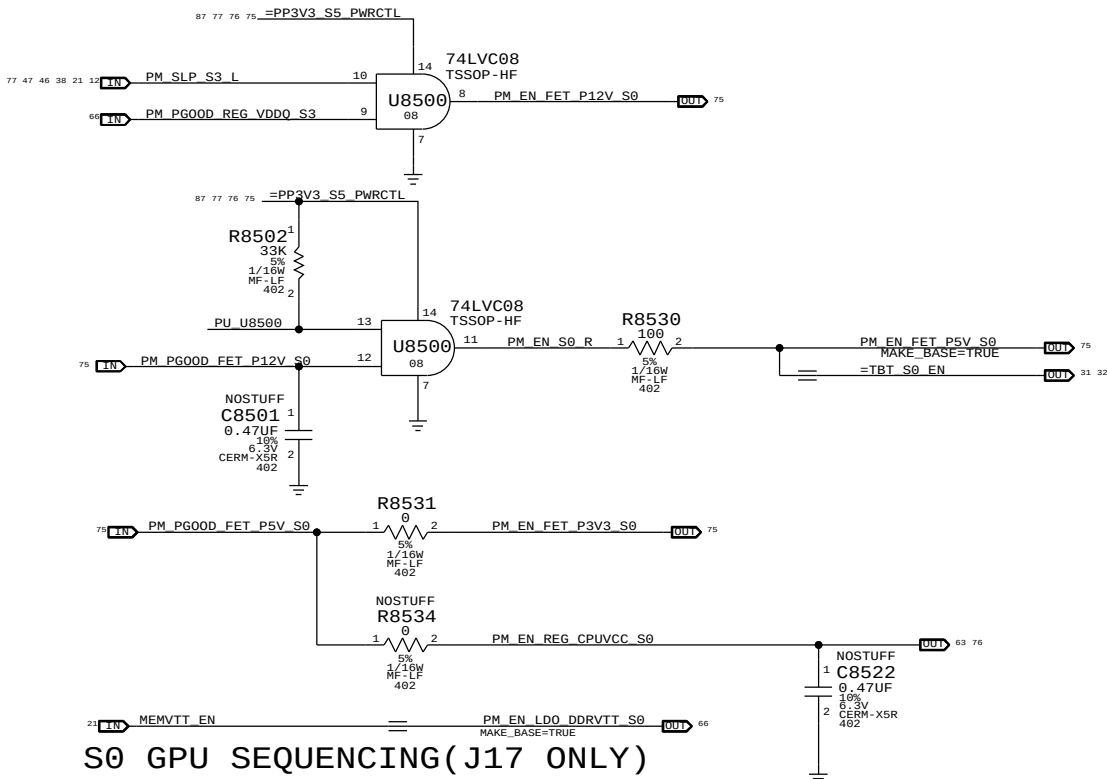
S4 TBT S4 Port Enable



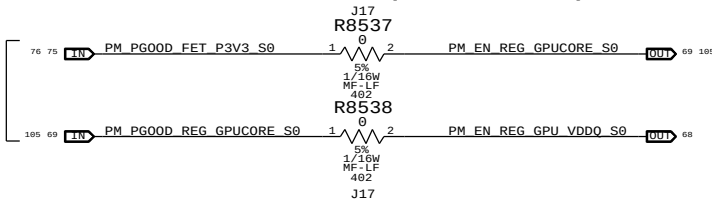
S3 VDDQ Enable



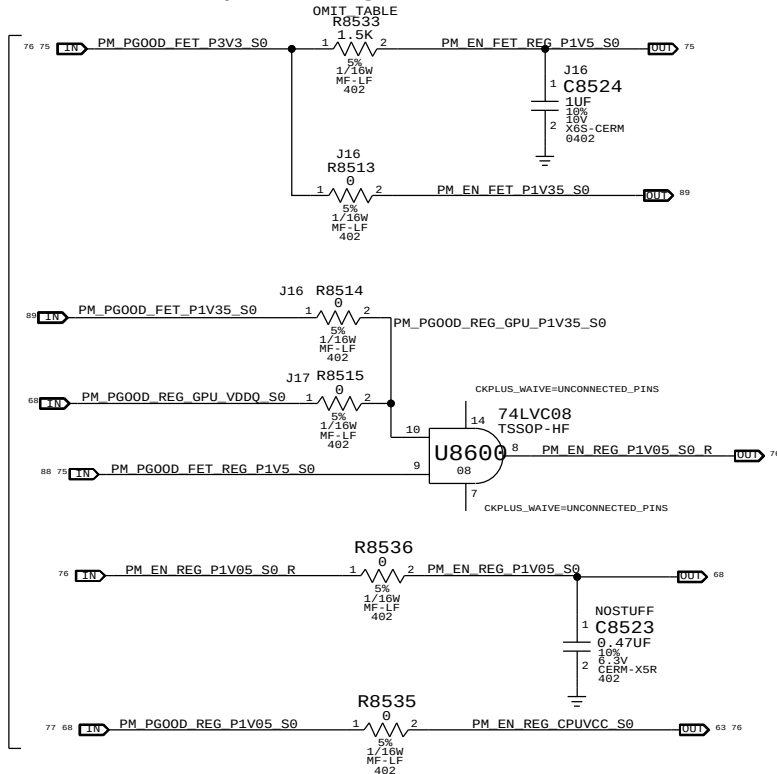
S0 Enables



S0 GPU SEQUENCING(J17 ONLY)



S0 PCH Sequencing



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
116S0070	1	RES,1.5K,0402,5%	R8533	J16
116S0094	1	RES,00HM,0402,5%	R8533	J17

Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.5 V, 1.05V for PCH/TBT/GPU)
Uncore: VDDQ

Notes on sequencing requirements

Intel:

1. No hard specification on platform rails
2. SMC guarantees timing on PCH DPWROK and PWROK
3. VCC3_3 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC3V3 ramping to 2.6V
4. VCC1_5 may power up before VCC, VCC must ramp to 0.6V within 25ms of VCC1V5 ramping to 1.35V
5. VCC may power down before VCC3_3, VCC3_3 must ramp down to 2.6V within 35ms
6. VCC may power down before VCC1_5, VCC1_5 must ramp down to 1.35V within 35ms

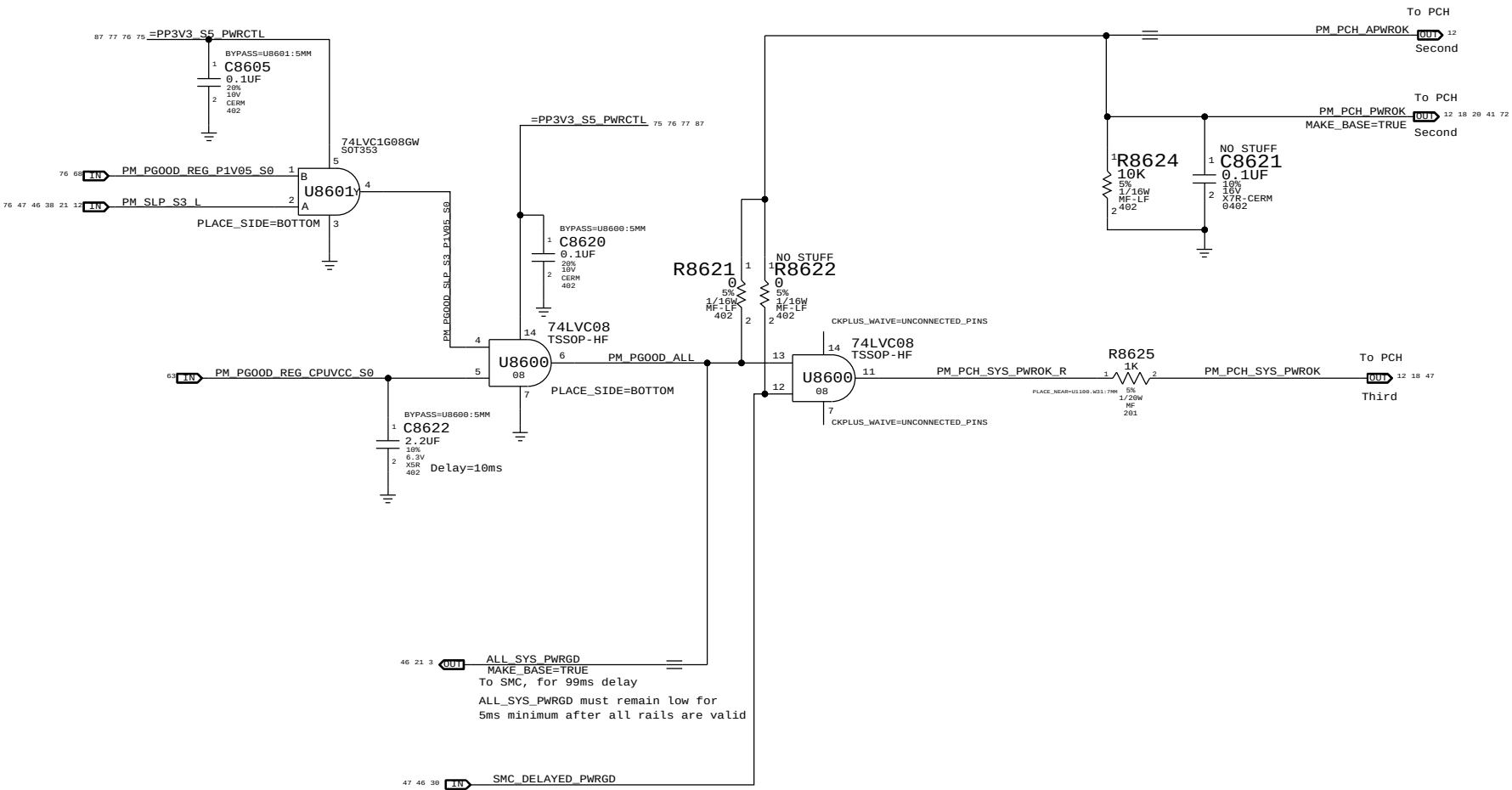
NVIDIA:

1. 3V3_S0 must ramp first
2. VDDQ MUST RAMP AFTER GPUOP-HF
3. PEX_VDD with IFPC/D/E/F_IOVDD (1.05V) must ramp after VDDQ
4. All rails must reach their target voltages in more than 40 uS

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PAGE TITLE		PM Regulator Enables	
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ALL_SYS_PWRGD, PCH_PWROK & SYS_PWROK Generation

PCH Power Goods



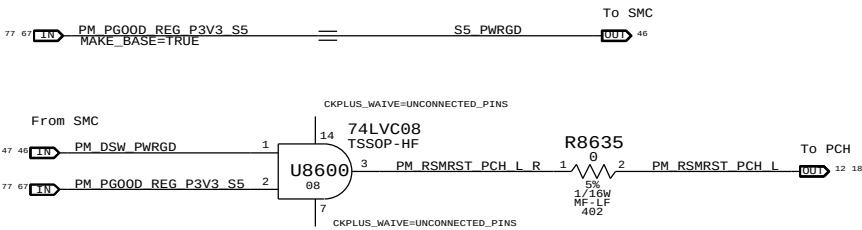
Resume Reset

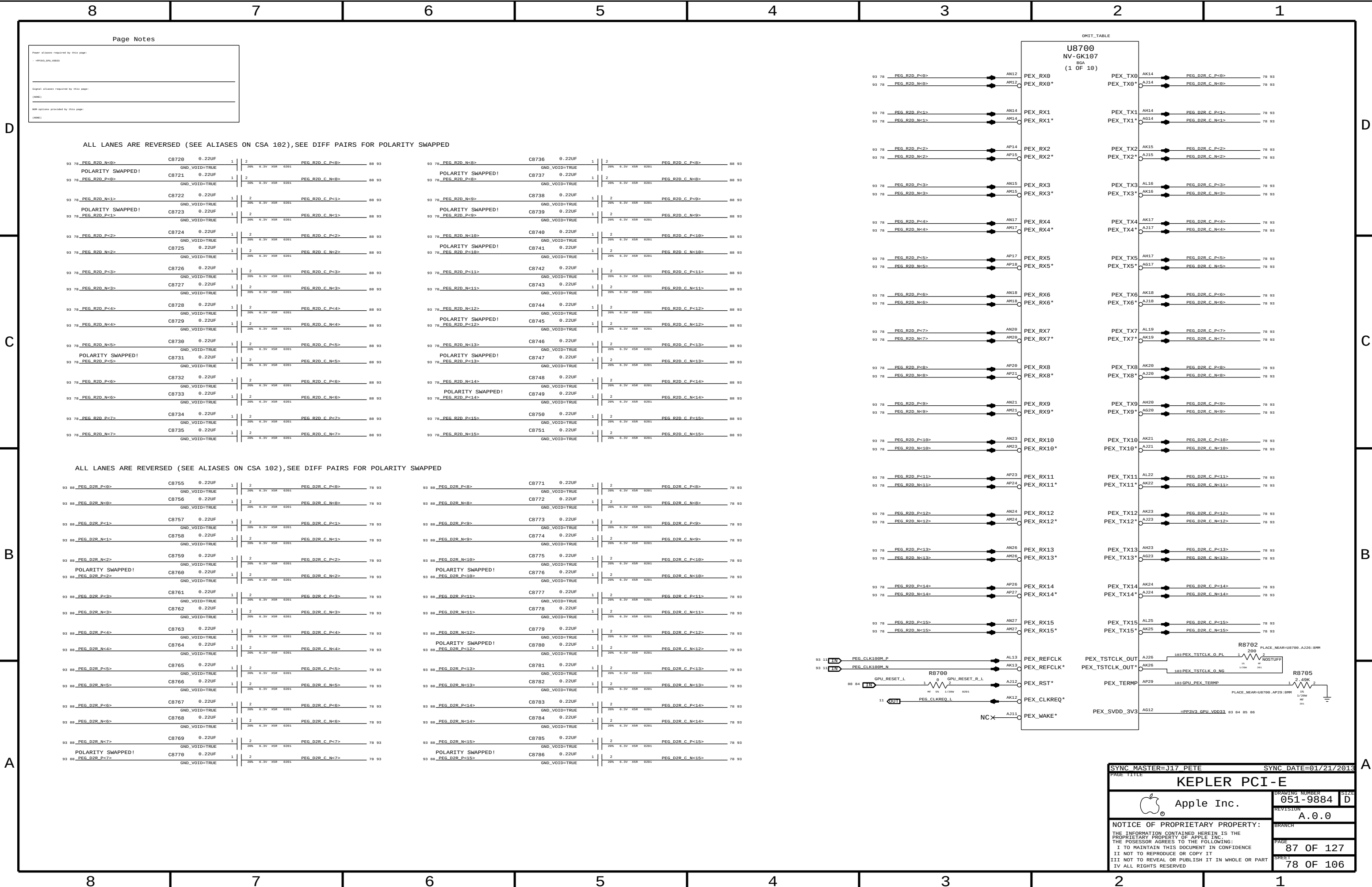
Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

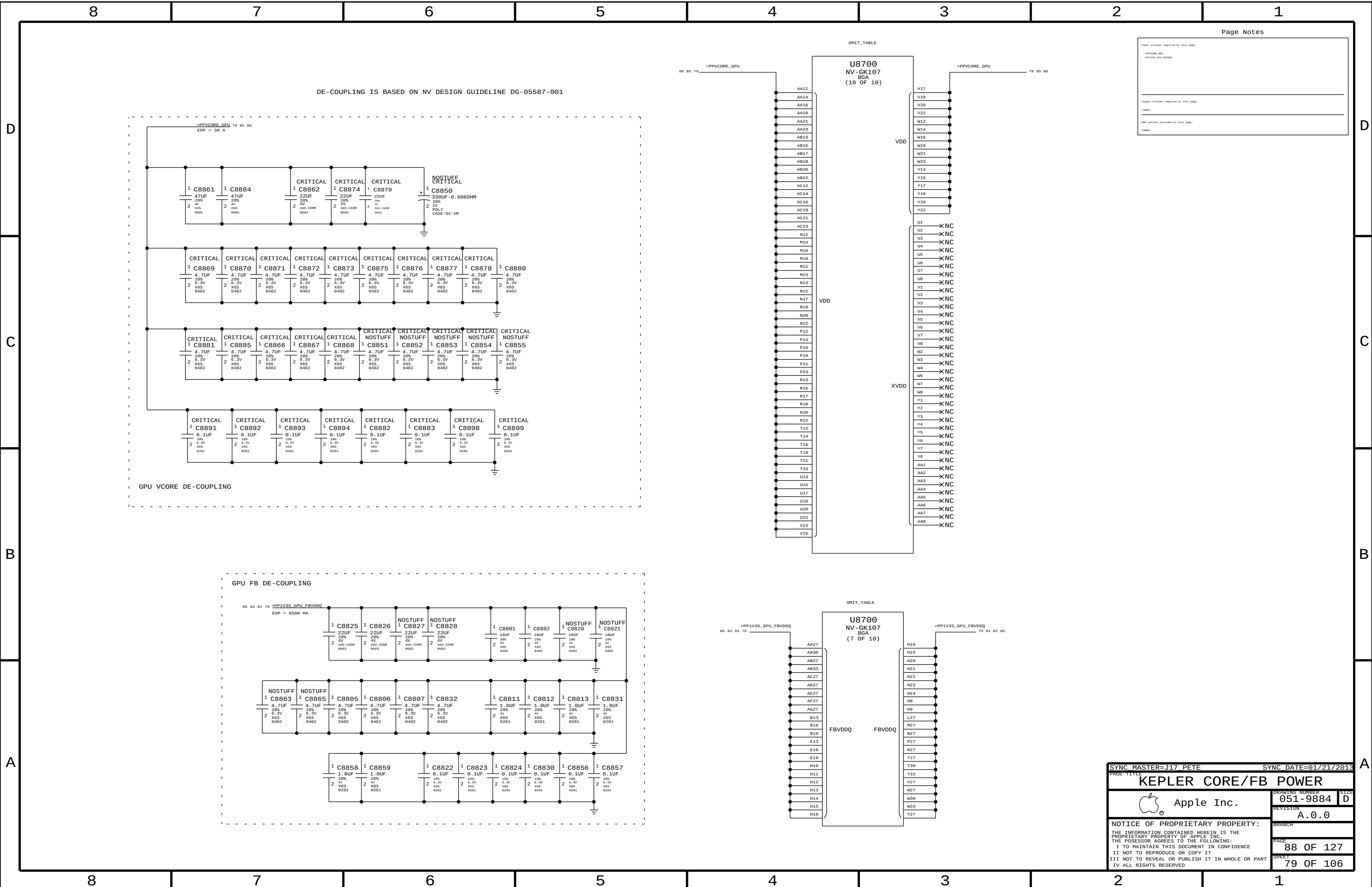
Note:
The iMac J16/J17 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.







D

C

B

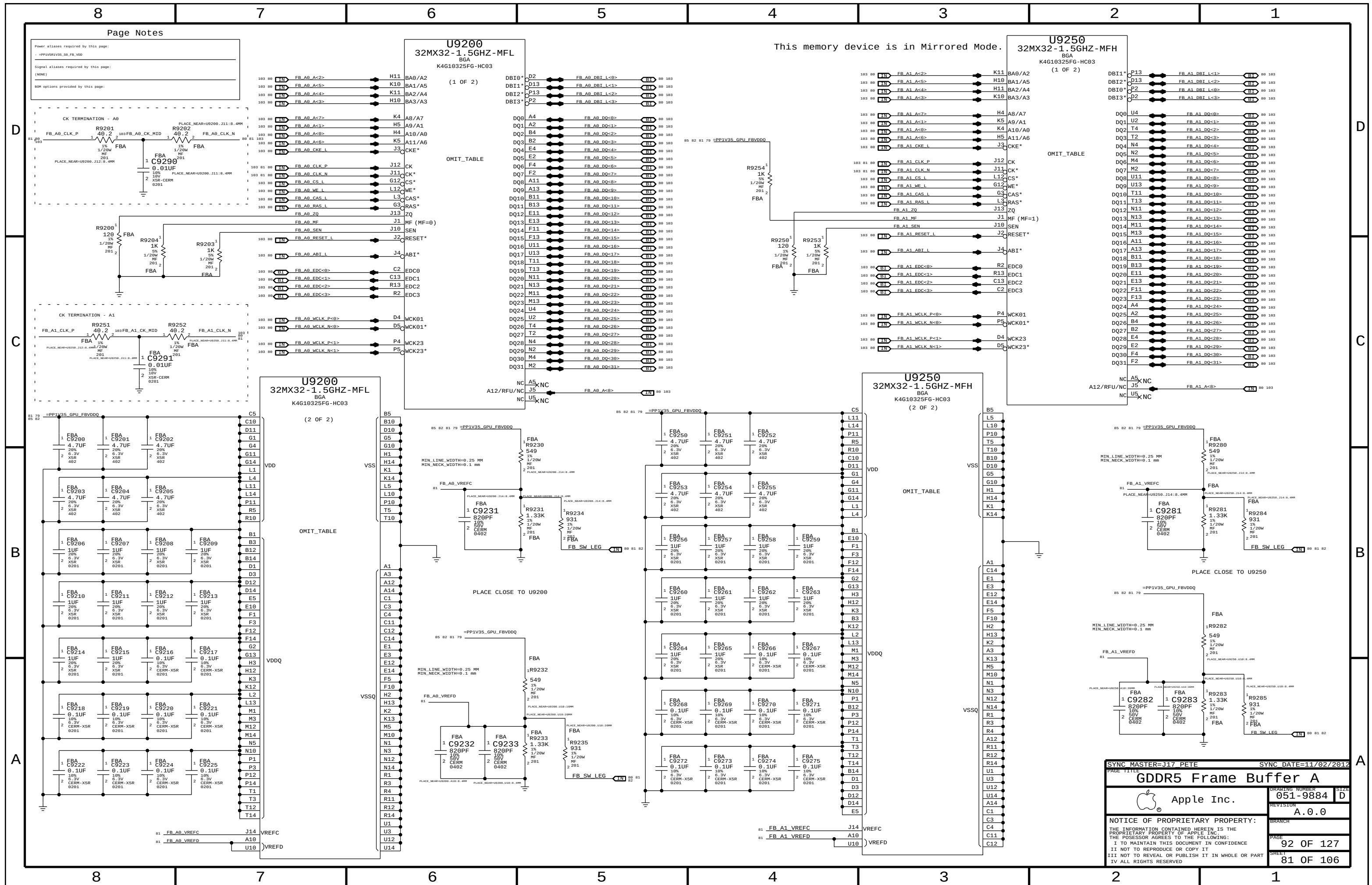
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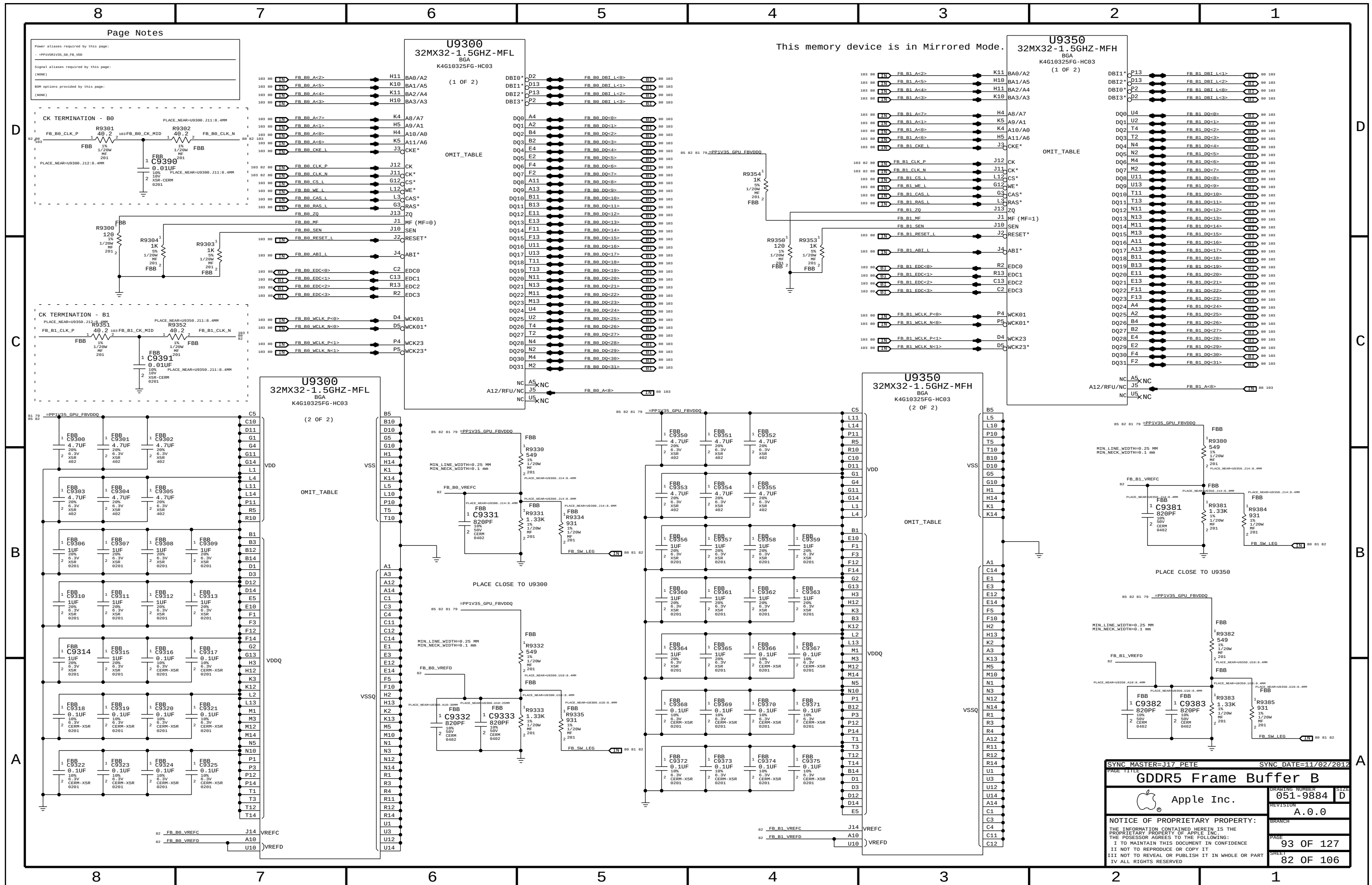
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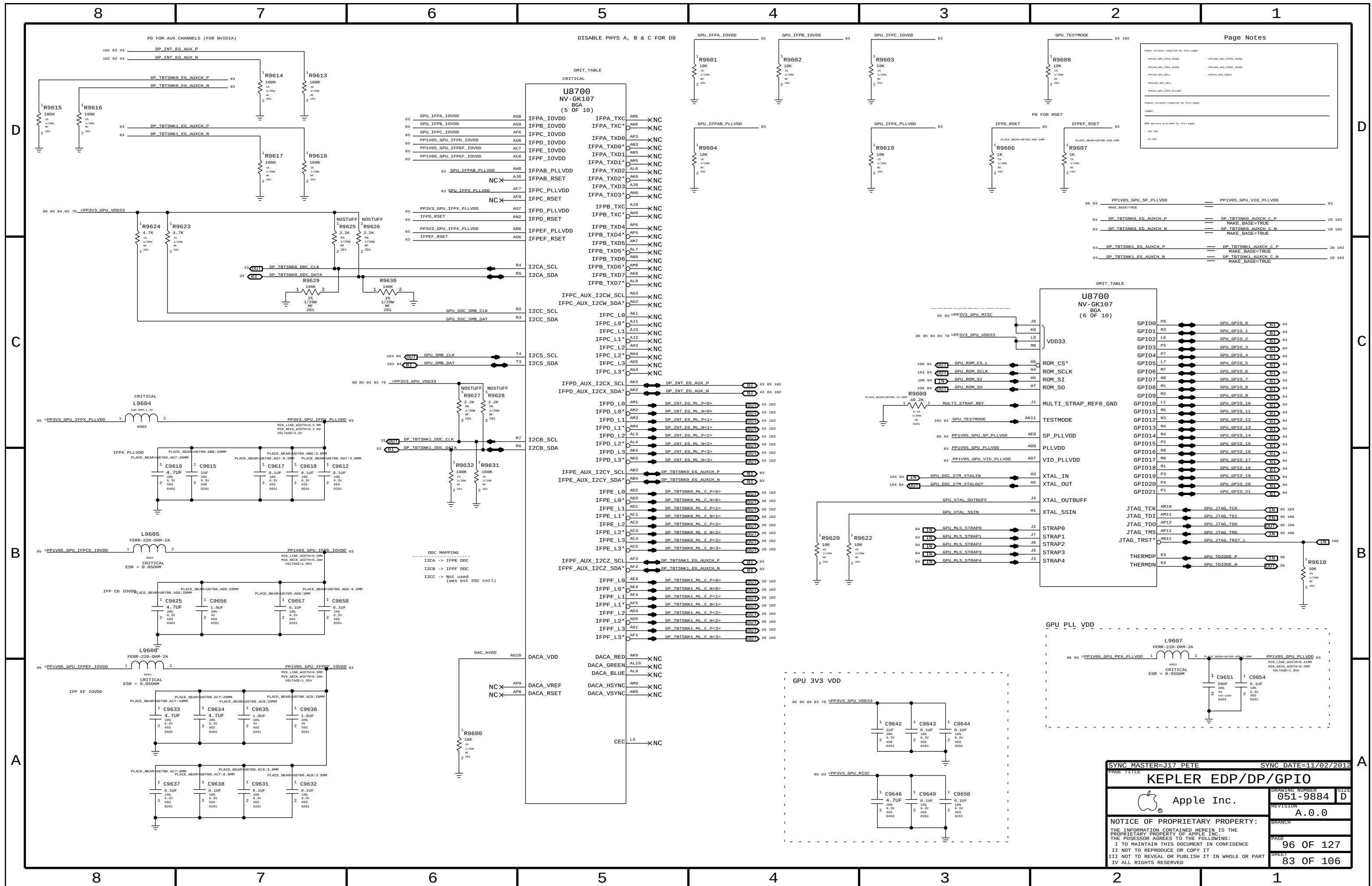
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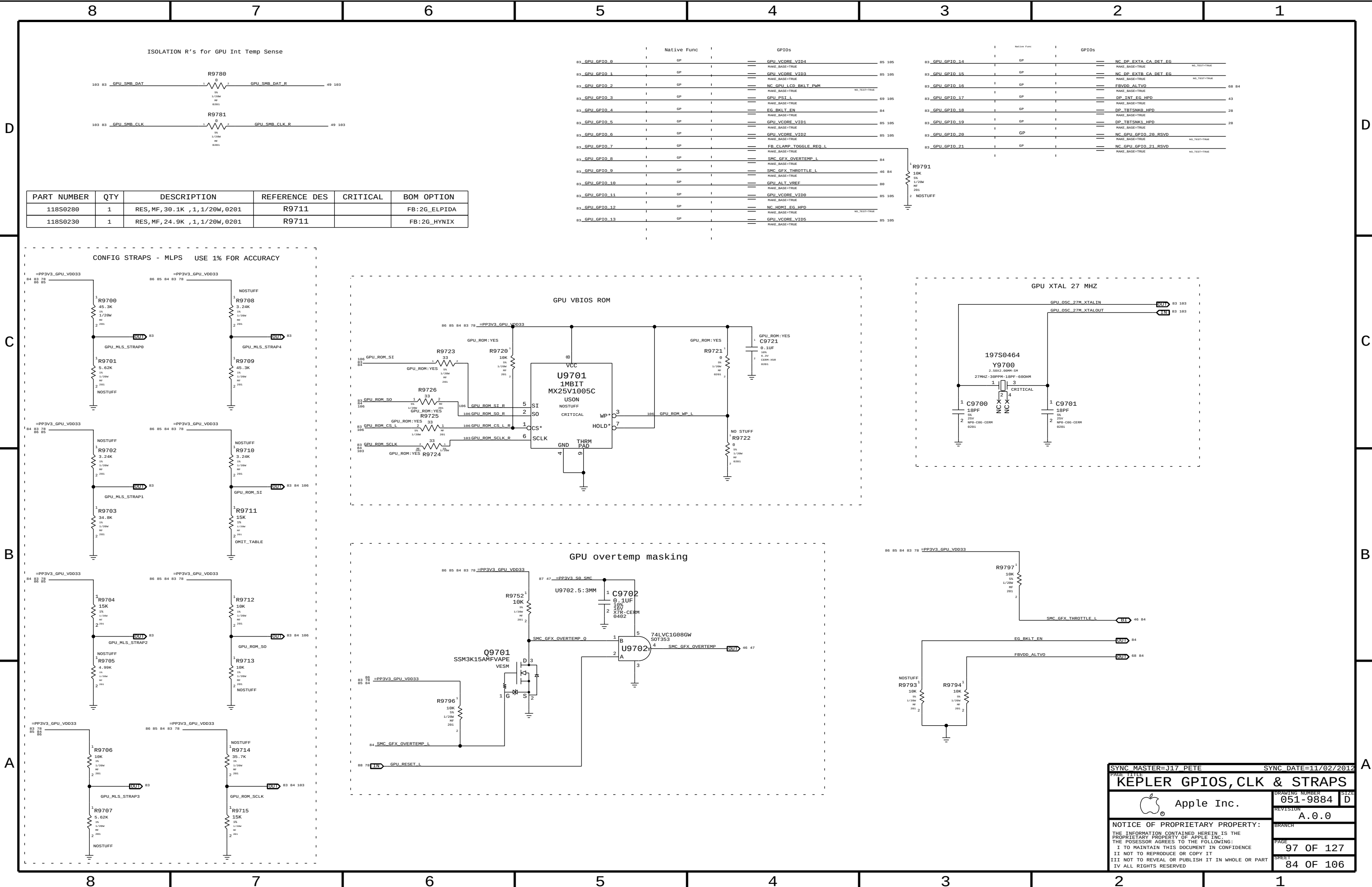
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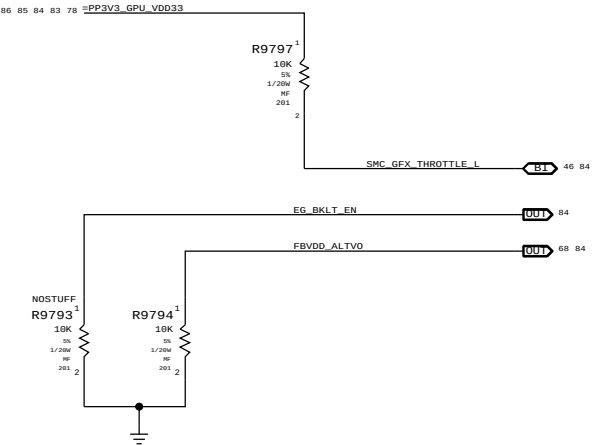
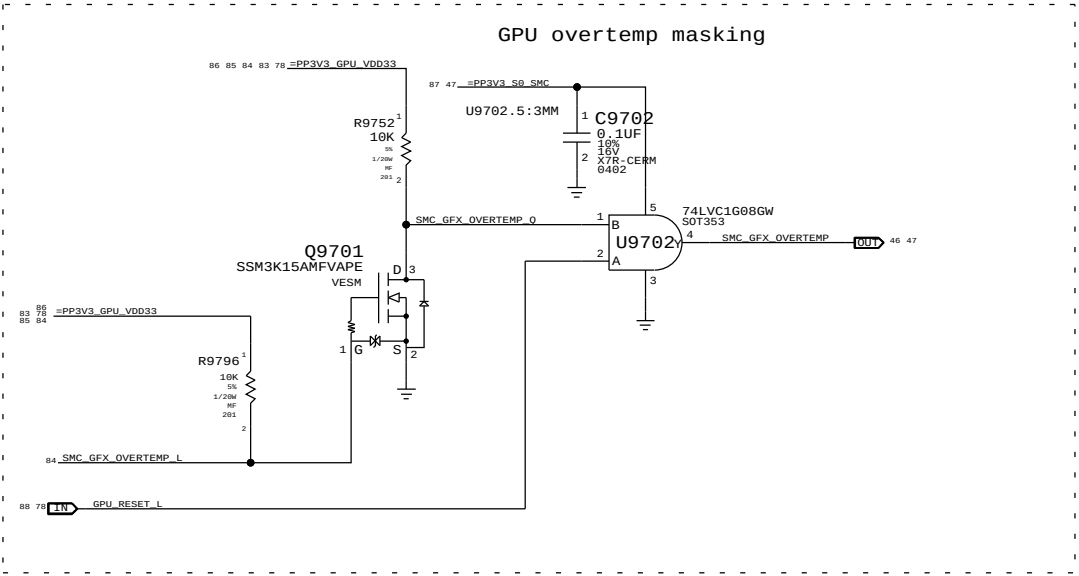
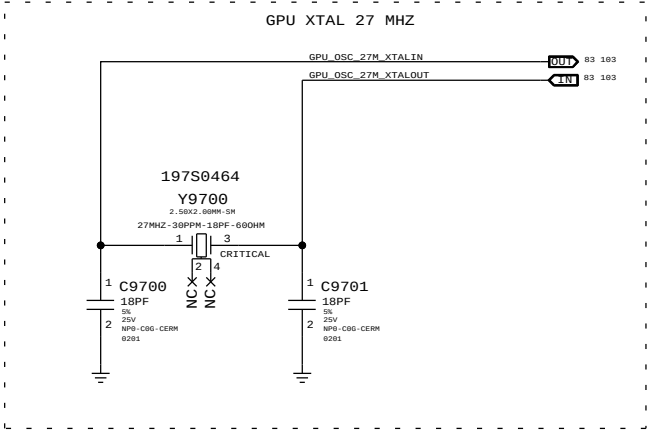
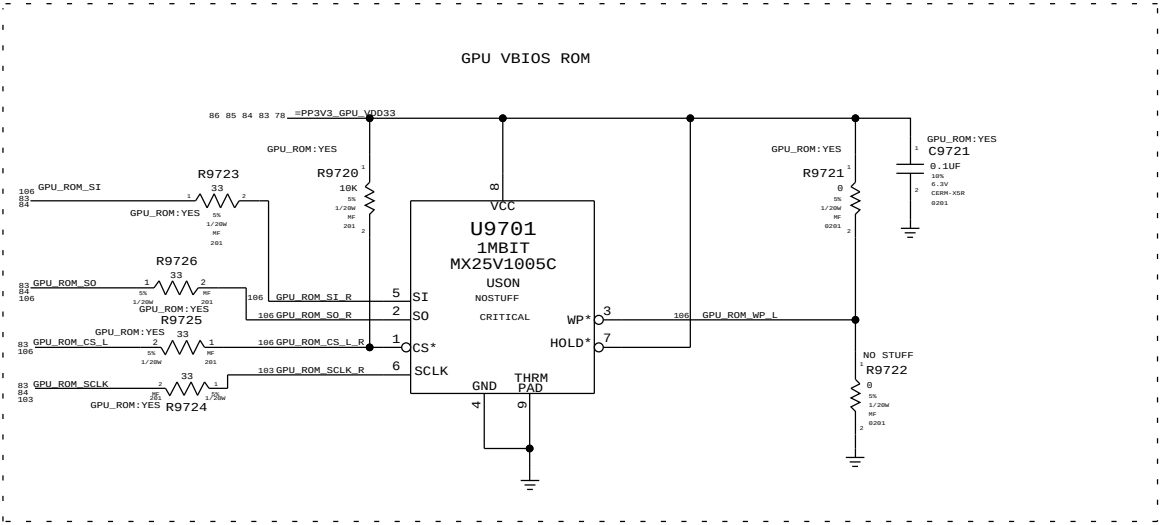
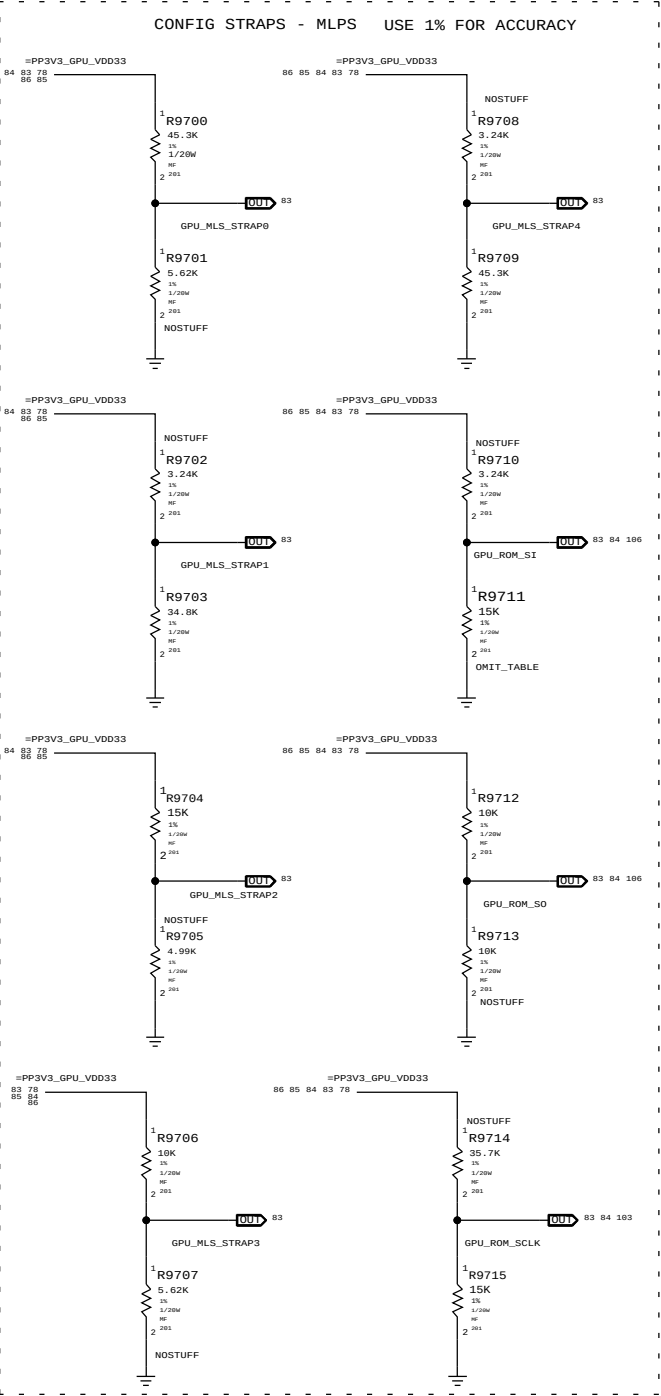




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0280	1	RES,MF,30.1K ,1,1/20W,0201	R9711		FB:2G_ELPIDA
118S0230	1	RES,MF,24.9K ,1,1/20W,0201	R9711		FB:2G_HYNIX

	Native Func		GPIOs	
83_GPU_GPIO_0	GP		GPU_VCORE_VID4	85 105
83_GPU_GPIO_1	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_2	GP		GPU_VCORE_VID3	85 105
83_GPU_GPIO_3	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_4	GP		NC_GPU_LCD_BKLT_PWM	NO_TEST+TRUE
83_GPU_GPIO_5	GP		GPU_PSI_L	69 105
83_GPU_GPIO_6	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_7	GP		EG_BKLT_EN	84
83_GPU_GPIO_8	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_9	GP		GPU_VCORE_VID1	85 105
83_GPU_GPIO_10	GP		GPU_VCORE_VID2	85 105
83_GPU_GPIO_11	GP		FB_CLAMP_TOGGLE_REQ_L	
83_GPU_GPIO_12	GP		SMC_GFX_OVERTEMP_L	84
83_GPU_GPIO_13	GP		SMC_GFX_THROTTLE_L	46 84
			GPU_ALT_VREF	80
			GPU_VCORE_VID0	85 105
			MAKE_BASE=TRUE	
			NC_HDMI_EG_HPD	NO_TEST+TRUE
			GPU_VCORE_VID5	85 105
			MAKE_BASE=TRUE	

	Native Func		GPIOs	
83_GPU_GPIO_14	GP		NC_DP_EXTN_CA_DET_EG	
83_GPU_GPIO_15	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_16	GP		NC_DP_EXTN_CA_DET_EG	
83_GPU_GPIO_17	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_18	GP		FBVDD_ALTVDD	68 84
83_GPU_GPIO_19	GP		MAKE_BASE=TRUE	
83_GPU_GPIO_20	GP		DP_INT_EG_HPD	43
83_GPU_GPIO_21	GP		MAKE_BASE=TRUE	
			DP_TBTSNK0_HPD	28
			MAKE_BASE=TRUE	
			DP_TBTSNK1_HPD	28
			MAKE_BASE=TRUE	
			NC_GPU_GPIO_28_RSVD	NO_TEST+TRUE
			MAKE_BASE=TRUE	
			NC_GPU_GPIO_21_RSVD	NO_TEST+TRUE
			MAKE_BASE=TRUE	



SYNC MASTER=J17 PETE

SYNC DATE=11/02/2012

KEPLER GPIOs,CLK & STRAPS

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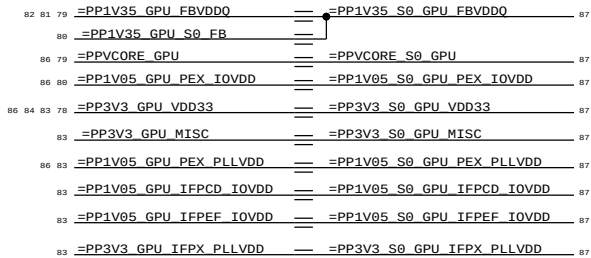
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PAGE
97 OF 127

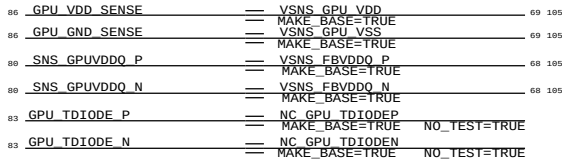
SHEET
84 OF 106

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GPU POWER ALIAS

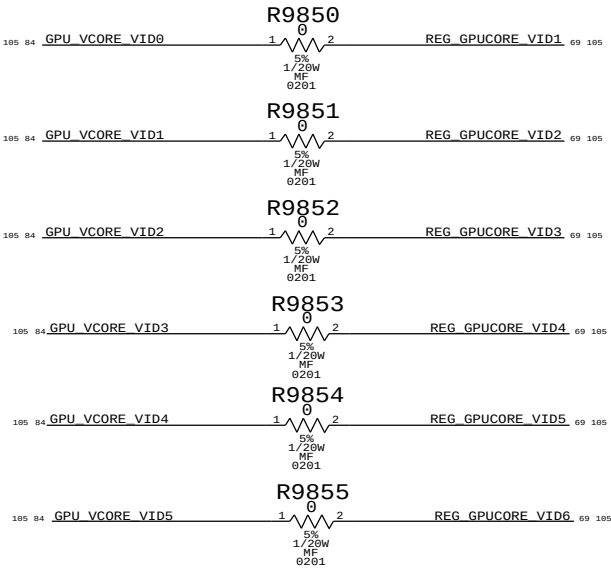


GPU SIGNAL & SENSE ALIAS



C

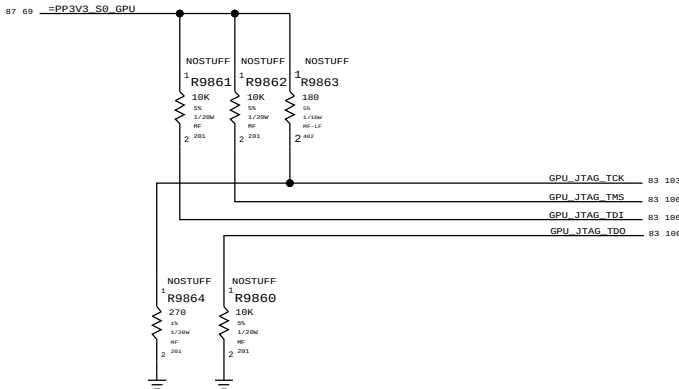
GPU VIDS ALIAS (VR VID0 IS TIED LOW)



D

B

PU/PD IS BASED ON RECOMMENDATION FROM NVIDIA FOR THEIR GPU JTAG DEBUGGER

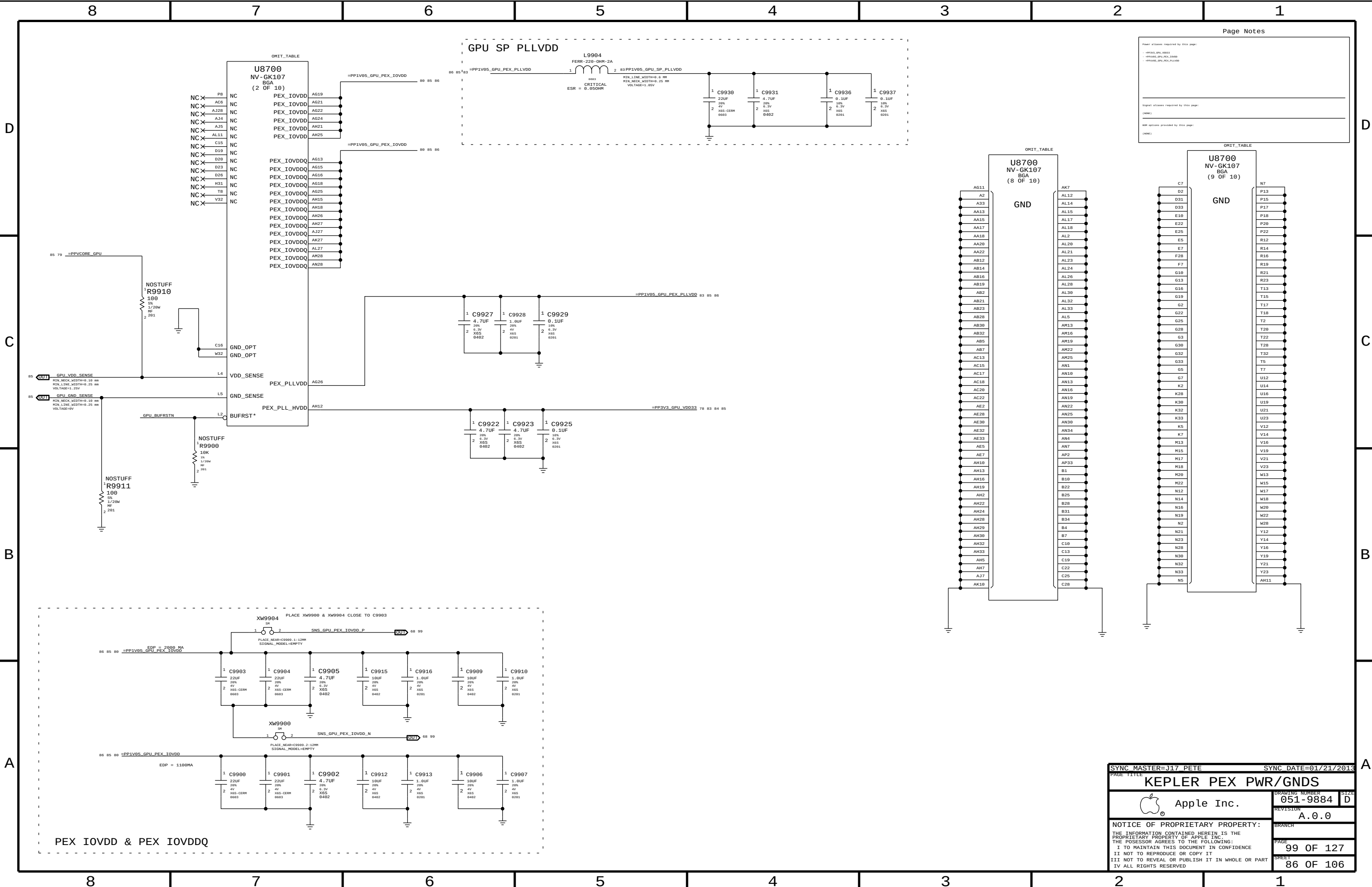


B

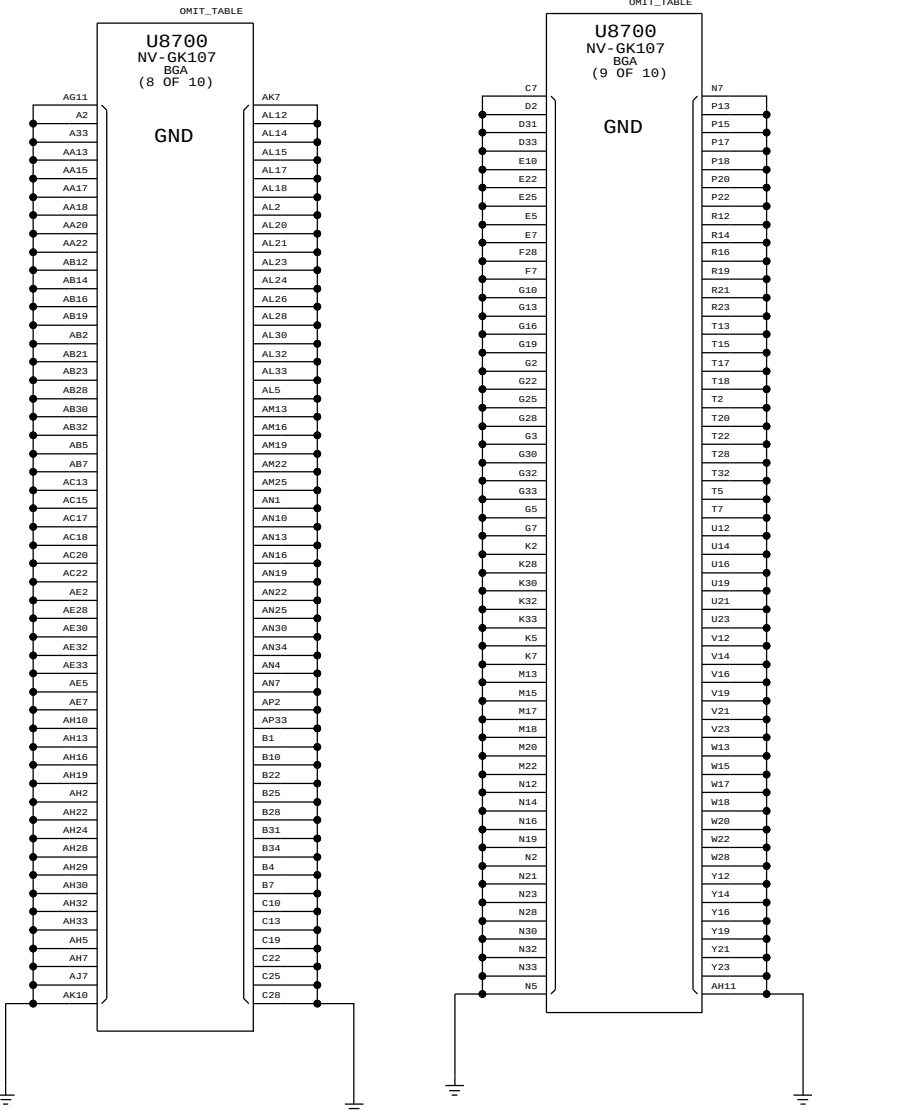
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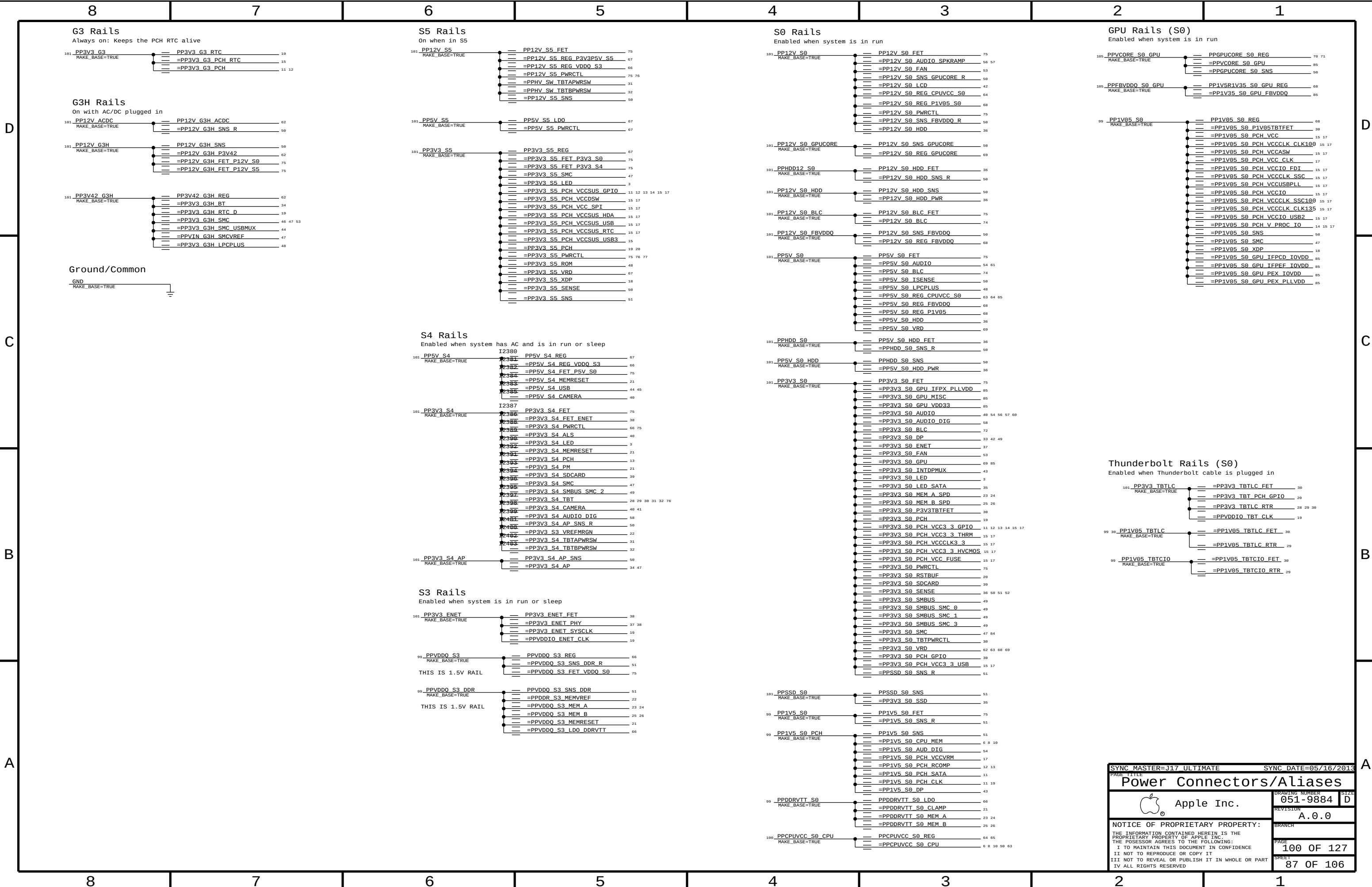
SYNC MASTER=J17 PETE		SYNC DATE=01/21/2013	
PAGE TITLE		GPU SIGNAL & POWER ALIASES	
 Apple Inc.		DRAWING NUMBER	051-9884
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		PAGE	98 OF 127
		SHEET	85 OF 106

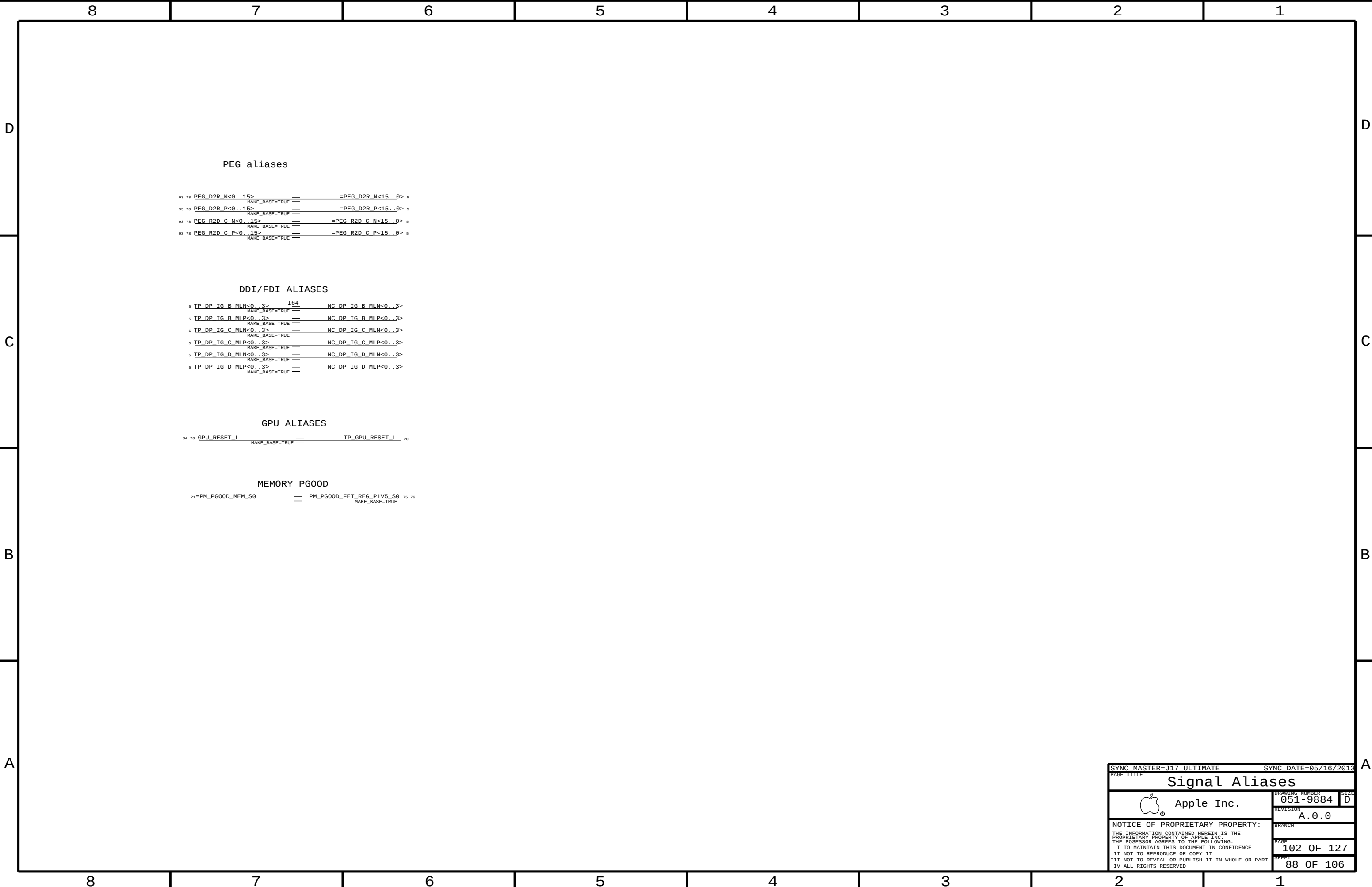


Page Notes	
Power aliases required by this page:	
- PP1V05_GPU_VDDQ1	
- PP1V05_GPU_PEX_PLLVDD	
- PP1V05_GPU_PEX_PLLVDD	
Signal aliases required by this page:	
(NONE)	
SOP options provided by this page:	
(NONE)	



PAGE TITLE		SYNC DATE=01/21/2013	
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PEG aliases

93 78 PEG_D2R_N<0..15> == =PEG_D2R_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_D2R_P<0..15> == =PEG_D2R_P<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_N<0..15> == =PEG_R2D_C_N<15..0> 5
MAKE_BASE=TRUE ==
93 78 PEG_R2D_C_P<0..15> == =PEG_R2D_C_P<15..0> 5
MAKE_BASE=TRUE ==

DDI/FDI ALIASES

5 TP_DP_IG_B_MLN<0..3> I64 == NC_DP_IG_B_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_B_MLP<0..3> == NC_DP_IG_B_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLN<0..3> == NC_DP_IG_C_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_C_MLP<0..3> == NC_DP_IG_C_MLP<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLN<0..3> == NC_DP_IG_D_MLN<0..3>
MAKE_BASE=TRUE ==
5 TP_DP_IG_D_MLP<0..3> == NC_DP_IG_D_MLP<0..3>
MAKE_BASE=TRUE ==

GPU ALIASES

84 78 GPU_RESET_L == TP_GPU_RESET_L 20
MAKE_BASE=TRUE ==

MEMORY PG00D

21 ==PM_PG00D_MEM_S0 == PM_PG00D_FET_REG_P1V5_S0 75 76
MAKE_BASE=TRUE ==

SYNC_MASTER=J17_ULTIMATE

SYNC_DATE=05/16/2013

PAGE TITLE

Signal Aliases

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051-9884

SIZE

D

REVISION

A.0.0

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PAGE

102 OF 127

SHEET

88 OF 106

8	7	6	5	4	3	2	1
D							D
C							C
B							B
A							A
8	7	6	5	4	3	2	1

SYNC MASTER=J17 ULTIMATE

SYNC DATE=05/16/2013

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Functional / ICT Test

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051-9884

REVISION

A.0.0

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PAGE

105 OF 127

SHEET

90 OF 106

J17 BOARD SPECIFIC PHYSICAL AND SPACING CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA, BGA_TBT	MM	16.2

General Physical Rule Definitions

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	0.1 MM	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	*	Y	0.185 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	ISL5, ISL8	Y	0.205 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
34_OHM_SE	TOP, BOTTOM	Y	0.220 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	*	Y	0.150 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	ISL5, ISL8	Y	0.165 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
39_OHM_SE	TOP, BOTTOM	Y	0.175 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	*	Y	0.130 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	ISL5, ISL8	Y	0.145 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
42_OHM_SE	TOP, BOTTOM	Y	0.155 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	*	Y	0.115 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	ISL5, ISL8	Y	0.126 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
45_OHM_SE	TOP, BOTTOM	Y	0.135 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	*	Y	0.090 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	ISL5, ISL8	Y	0.100 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD
50_OHM_SE	TOP, BOTTOM	Y	0.105 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_0HM_SE	*	Y	0.075 MM	0.075 MM	=STANDARD	=STANDARD	=STANDARD
55_0HM_SE	ISL5, ISL8	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD
55_0HM_SE	TOP, BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	Y	0.171 MM	0.085 MM	=STANDARD	0.130 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.185 MM	0.085 MM	=STANDARD	0.150 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	Y	0.136 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.141 MM	0.085 MM	=STANDARD	0.185 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	Y	0.121 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.190 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	Y	0.109 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.111 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	Y	0.086 MM	0.085 MM	=STANDARD	0.200 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.090 MM	0.085 MM	=STANDARD	0.230 MM	0.1 MM

General Spacing Definitions

Default

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

Fixed and Dielectric

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?
1X_DIELECTRIC	TOP,BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3,ISL10	0.101 MM	?
1X_DIELECTRIC	*	0.076 MM	?

Board Stack-up

FINISHED BOARD THICKNESS: 1.94 MM

Top	Signal	0.5 oz (Cu plated)
	Prepreg	0.071 MM
2	Plane	1 oz
	Core	0.101 MM
3	Signal	0.5 oz
	Prepreg	0.115 MM
4	Plane	1 oz
	Core	0.076 MM
5	Signal	0.5 oz
	Prepreg	0.300 MM
6	Plane	1 oz
	Core	0.076 MM
7	Plane	1 oz
	Prepreg	0.300 MM
8	Signal	0.5 oz
	Core	0.076 MM
9	Plane	1 oz
	Prepreg	0.115 MM
10	Signal	0.5 oz
	Core	0.101 MM
11	Plane	1 oz
	Prepreg	0.071 MM
Btm	Signal	0.5 oz (Cu plated)

BGA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=STANDARD	?

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_ISO	*	=STANDARD	8000
GND_P2MM	*	=2:1_SPACING	1000
PWR_P2MM	*	=2:1_SPACING	1100

GENERIC

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GENERIC_ISO	*	=1:1_SPACING	?

BGA Area Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1MM

SYNC MASTER-J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
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		BRANCH	
		PAGE	110 OF 127
		SHEET	91 OF 106

DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
DDR_COMP	*	Y	0.395 MM	0.195 MM	=STANDARD	=STANDARD	=STANDARD

Minimum diff spacing is 4 mil
Table 4-5, Intel Doc# 486712

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_68D
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_QDQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D
DDR_COMP_PHY	*	DDR_COMP

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?
DDR_COMP_ISO	*	0.381 MM	?

Main Segment Min Spacing Rules (mils) (Shark Bay PDG, Intel Doc# 486712)

Table	Trace	Design	Iso	Design	Comments
4-2	4	(diff)	15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
4-3	8	9.84	12	13.78	
4-4	6	7.87	12	13.78	
4-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_COMP	*	*	DDR_COMP_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_DATA_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_DATA_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

See Note (3)

See Note (1)

See Note (3)

See Note (1)

See Note (2)

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggested 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):

In order for the constraints $DDR_*_DQ_BYTE*$ to =SAME to win out over $DDR\{A,B\}_DQ_BYTE*$ to $DDR\{A,B\}_DQ_BYTE*$ so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL .

DDR3

Electrical Constraint Set		Physical	Spacing	
Channel A				
H177	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<1..0> 7 23
H178	DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<1..0> 7 23
H255	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_P<3..2> 7 24
H256	DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_A_CLK_N<3..2> 7 24
H189	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<1..0> 7 23
H189	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<1..0> 7 23
H189	DDR_A_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<1..0> 7 23
H240	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CKE<3..2> 7 24
H240	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_CS_L<3..2> 7 24
H240	DDR_A_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_A_ODT<3..2> 7 24
H189	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_A<15..0> 7 23 24
H216	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_BA<2..0> 7 23 24
H216	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_RAS_L 7 23 24
H216	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_CAS_L 7 23 24
H189	DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM_A_WE_L 7 23 24
H189	DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM_A_DQ<7..0> 7 27
H189	DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM_A_DQ<15..8> 7 27
H189	DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM_A_DQ<23..16> 7 27
H189	DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM_A_DQ<31..24> 7 27
H189	DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM_A_DQ<39..32> 7 27
H189	DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM_A_DQ<47..40> 7 27
H189	DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM_A_DQ<55..48> 7 27
H189	DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM_A_DQ<63..56> 7 27
H189	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_P<0> 7 27
H189	DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM_A_DQS_N<0> 7 27
H189	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_P<1> 7 27
H189	DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM_A_DQS_N<1> 7 27
H189	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_P<2> 7 27
H189	DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM_A_DQS_N<2> 7 27
H189	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_P<3> 7 27
H189	DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM_A_DQS_N<3> 7 27
H189	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_P<4> 7 27
H189	DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM_A_DQS_N<4> 7 27
H189	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_P<5> 7 27
H189	DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM_A_DQS_N<5> 7 27
H189	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_P<6> 7 27
H189	DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM_A_DQS_N<6> 7 27
H189	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_P<7> 7 27
H189	DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM_A_DQS_N<7> 7 27
Channel B				
H229	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<1..0> 7 25
H229	DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<1..0> 7 25
H255	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_P<3..2> 7 26
H256	DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM_B_CLK_N<3..2> 7 26
H216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<1..0> 7 25
H216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<1..0> 7 25
H216	DDR_B_CTRL0	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<1..0> 7 25
H240	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CKE<3..2> 7 26
H240	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_CS_L<3..2> 7 26
H240	DDR_B_CTRL1	DDR_CTRL_PHY	DDR_CTRL	MEM_B_ODT<3..2> 7 26
H216	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_A<15..0> 7 25 26
H216	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_BA<2..0> 7 25 26
H216	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_RAS_L 7 25 26
H216	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_CAS_L 7 25 26
H216	DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM_B_WE_L 7 25 26
H229	DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM_B_DQ<7..0> 7 27
H229	DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM_B_DQ<15..8> 7 27
H229	DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM_B_DQ<23..16> 7 27
H229	DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM_B_DQ<31..24> 7 27
H229	DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM_B_DQ<39..32> 7 27
H229	DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM_B_DQ<47..40> 7 27
H229	DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM_B_DQ<55..48> 7 27
H229	DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM_B_DQ<63..56> 7 27
H229	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_P<0> 7 27
H229	DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM_B_DQS_N<0> 7 27
H229	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_P<1> 7 27
H229	DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM_B_DQS_N<1> 7 27
H229	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_P<2> 7 27
H229	DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM_B_DQS_N<2> 7 27
H229	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_P<3> 7 27
H229	DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM_B_DQS_N<3> 7 27
H229	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_P<4> 7 27
H229	DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM_B_DQS_N<4> 7 27
H229	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_P<5> 7 27
H229	DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM_B_DQS_N<5> 7 27
H229	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_P<6> 7 27
H229	DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM_B_DQS_N<6> 7 27
H229	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_P<7> 7 27
H229	DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM_B_DQS_N<7> 7 27
Reset				
H240		DDR_50S		MEM_RESET_L 21 23 24
SM COMP				
H240		DDR_COMP_PHY	DDR_COMP	CPU_SM_RCOMP<0..2> 6

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
DDR3 Constraints			
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		REVISION A.0.0	
		BRANCH	
		PAGE 111 OF 127	
		SHEET 92 OF 106	

PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALL ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE3_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
PEG_R2D	PEG_R2D	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_SAME_DIR
PEG_D2R	PEG_R2D	*	PEG_ALT_DIR
PEG_D2R	*	*	PEG_ISO
PEG_R2D	*	*	PEG_ISO

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PEG_SAME_DIR	*	=3.5X_DIELECTRIC	?
PEG_ALT_DIR	*	=7X_DIELECTRIC	?
PEG_ISO	*	=4:1_SPACING	?

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

PCIe (CPU)

Electrical Constraint Set	Physical	Spacing
x16 Graphics		
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D P<15>
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D N<15>
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D C P<15>
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D C N<15>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R P<15>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R N<15>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C P<15>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C N<15>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<14>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<14>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<14>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C N<14>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<14>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<14>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<14>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<14>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<13>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<13>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<13>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C N<13>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<13>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<13>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<13>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<13>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<12>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<12>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<12>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C N<12>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R P<12>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R N<12>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C P<12>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C N<12>
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D P<11>
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D N<11>
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D C P<11>
PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D C N<11>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<11>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<11>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<11>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<11>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<10>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<10>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<10>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C N<10>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R P<10>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R N<10>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C P<10>
PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R C N<10>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<9>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<9>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<9>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C N<9>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<9>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<9>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<9>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<9>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D P<8>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D N<8>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C P<8>
PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D C N<8>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R P<8>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R N<8>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C P<8>
PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R C N<8>

PCIe (CPU)

Electrical Constraint Set		Physical	Spacing	
x16 Graphics				
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<7> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<7> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<7> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<7> 78 88
16000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R P<7> 78 88
16000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R N<7> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<7> 78
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<7> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<6> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<6> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<6> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<6> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<6> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<6> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<6> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<6> 78
16000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D P<5> 78
16000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D N<5> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<5> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<5> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<5> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<5> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<5> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<5> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<4> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<4> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<4> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<4> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<4> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<4> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<4> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<4> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<3> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<3> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<3> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<3> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<3> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<3> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<3> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<3> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D P<2> 78
16000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG_R2D N<2> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<2> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<2> 78 88
16000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R P<2> 78 88
16000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG_D2R N<2> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<2> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<2> 78
16000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D P<1> 78
16000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D N<1> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<1> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<1> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<1> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<1> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<1> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<1> 78
16000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D P<0> 78
16000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG_R2D N<0> 78
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C P<0> 78 88
16000		PCIE3_PHY	PEG_R2D	PEG_R2D C N<0> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R P<0> 78 88
16000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG_D2R N<0> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C P<0> 78 88
16000		PCIE3_PHY	PEG_D2R	PEG_D2R C N<0> 78
CPU PCIe Clocks				
16000	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M P 11 78
16000	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG_CLK100M N 11 78
CPU PCIe Compensation				
16000		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_RCOMP 5
CPU eDP Compensation				
16000		COMP_PCIE_PHY	COMP_PCIE	CPU_EDP_RCOMP 5

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	DMI_COMP

PCIE-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_COMP	*	Y	0.2032 MM	0.2032 MM	3 MM	=STANDARD	=STANDARD

PCie (PCH)

Electrical Constraint Set	Physical	Spacing
x4 Thunderbolt		
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x1 AirPort		
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
x1 Caesar IV		
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x2 SSD		
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCH PCIE Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_PCIE_RCOMP

CPU DP REF CLK

Electrical Constraint Set	Physical	Spacing
CPU DP REF CLK		
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE
CPU_CLK135_PL1	PCIE_PHY	CLK_PCIE

DMI

Electrical Constraint Set	Physical	Spacing
DMI		
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_S2N	PCIE_PHY	DMI_S2N
DMI_S2N	PCIE_PHY	DMI_S2N
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
DMI Compensation		
COMP_DMI_PHY	COMP_PCIE	PCH_DMI_RCOMP

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Constraint Set	Physical	Spacing
PCI Clock		
100M	CLK_PCT_55S	CLK_PCT
100M	CLK_PCT_55S	CLK_PCT

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	LPC_55S	LPC	LPC_AD<3..0> 13 46 48
	LPC_55S	LPC	LPC_AD_R<3..0> 13
	LPC_55S	LPC	LPC_FRAME_L 13 46 48
	LPC_55S	LPC	LPC_FRAME_R_L 13
LPC Clocks			
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS 19 48
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R 11 19
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC 19 46
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R 11 19

PCH Clocks

Electrical Constraint Set	Physical	Spacing		
PCH Reference Clock				
	CLK_PCH_55S	CLK_PCH	SYSLCK_CLK25M_SB	11 19
	CLK_PCH_55S	CLK_PCH	SYSLCK_CLK25M_SB_R	11
PCH RTC 32K				
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	11 19
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	11 19
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	19
SMC 32K				
	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R	12 47
	CLK_PCH_55S	CLK_PCH	SMC_CLK32K	46 47

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
REQD	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1 19
REQD	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2 19
REQD	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R 19
25M Reference Clocks			
REQD	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET 19 37
REQD	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R 19
REQD	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT 19 28
REQD	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R 28

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
H35P	HDA_55S	HDA	HDA_BIT_CLK 11 54
H35P	HDA_55S	HDA	HDA_BIT_CLK_R 11
H35P	HDA_55S	HDA	HDA_RST_L 11 54
H35P	HDA_55S	HDA	HDA_RST_R_L 11
H35P	HDA_55S	HDA	HDA_SDOUT 11 54
H35P	HDA_55S	HDA	HDA_SDOUT_R 11 19
H35P	HDA_55S	HDA	HDA_SYNC 11 54
H35P	HDA_55S	HDA	HDA_SYNC_R 11
H35P	HDA_55S	HDA	HDA_SDIN0 11 54
H35P	HDA_55S	HDA	AUD_SDI_R 54
SPDIF			
H37P		HDA	AUD_SPDIF_CHIP 54
H37P		HDA	AUD_SPDIF_OUT 54 58

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H09D	SPI 56S	SPT	SPI CLK R 13 48
H09E	SPI 56S	SPT	SPI CLK 48
H09F	SPI 56S	SPT	SPI ALT CLK 48
H09G	SPI 56S	SPT	SPI SMC CLK 46 48
H09H	SPI 56S	SPT	SPI MLB CLK 48
H09I	SPI 56S	SPT	SPI CS0 R L 13 48
H09J	SPI 56S	SPT	SPI CS0 L 48
H09K	SPI 56S	SPT	SPI ALT CS L 48
H09L	SPI 56S	SPT	SPI SMC CS L 46 48
H09M	SPI 56S	SPT	SPI MLB CS L 48
H09P	SPI 56S	SPT	SPI MOSI R 13 48
H09Q	SPI 56S	SPT	SPI MOSI 48
H09R	SPI 56S	SPT	SPI ALT MOSI 48
H09S	SPI 56S	SPT	SPI SMC MOSI 46 48
H09T	SPI 56S	SPT	SPI MLB MOSI 48
H09U	SPI 56S	SPT	SPI MISO 13 48
H09V	SPI 56S	SPT	SPI ALT MISO 48
H09W	SPI 56S	SPT	SPI SMC MISO 46 48
H09X	SPI 56S	SPT	SPI MLB MISO 48
H09Y	SPI 56S	SPT	SPIROM USE MLB 14 48

SYNC MASTER=J17 ULTIMATE		SYNC DATE=05/16/2013	
PAGE TITLE			
PCH and BR Constraints			
 Apple Inc.		DRAWING NUMBER 051-9884	SIZE D
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		SHEET 96 OF 106	

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<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>D1D1</th><th>NO_TEST</th></tr><tr><td colspan="6">Input Bus</td></tr><tr><td>PP12V_S0_CPUVCC_FLT</td><td>POWER</td><td>POWER</td><td>12V</td><td></td><td></td></tr><tr><td>REG_VCC_U7000</td><td>POWER</td><td>POWER</td><td>5V</td><td></td><td></td></tr><tr><td colspan="6">Local Ground</td></tr><tr><td>AGND_CPU</td><td>GND</td><td>GND</td><td>0V</td><td></td><td></td></tr><tr><td colspan="6">Phase 1</td></tr><tr><td>REG_TD_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_1_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PHASE_CPUVCC_1</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_1</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_1_RC</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td></tr><tr><td>REG_UGATE_CPUVCC_1</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_LGATE_CPUVCC_1</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_SNUBBER_CPUVCC_1</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>PPCPUVCC_S0_SENSE_1</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr><tr><td>REG_ISENVCC_1_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_1_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_1_NR</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td 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2</td></tr><tr><td>REG_TD_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_2_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PHASE_CPUVCC_2</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_2</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_2_RC</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td></tr><tr><td>REG_UGATE_CPUVCC_2</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_LGATE_CPUVCC_2</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_SNUBBER_CPUVCC_2</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>PPCPUVCC_S0_SENSE_2</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr><tr><td>REG_ISENVCC_2_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_2_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_2_NR</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td 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3</td></tr><tr><td>REG_TD_3</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_3</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PWM_CPUVCC_3_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_PHASE_CPUVCC_3</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_3</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_BOOT_CPUVCC_3_RC</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td></tr><tr><td>REG_UGATE_CPUVCC_3</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_LGATE_CPUVCC_3</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>REG_SNUBBER_CPUVCC_3</td><td>VR_D1D1_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td></tr><tr><td>PPCPUVCC_S0_SENSE_3</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr><tr><td>REG_ISENVCC_3_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_3_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_ISENVCC_3_NR</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td 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Constraint Set	Physical	Spacing	Voltage	D1D1	NO_TEST	Input Bus						PP12V_S0_CPUVCC_FLT	POWER	POWER	12V			REG_VCC_U7000	POWER	POWER	5V			Local Ground						AGND_CPU	GND	GND	0V			Phase 1						REG_TD_1	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_1_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_1_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_1	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_1	POWER	POWER	1.8V			REG_ISENVCC_1_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_1_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_1_NR	SNS_D1FF_PHY	SENSE				Phase 2						REG_TD_2	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_2_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_2_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_2	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_2	POWER	POWER	1.8V			REG_ISENVCC_2_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_2_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_2_NR	SNS_D1FF_PHY	SENSE				Phase 3						REG_TD_3	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_3_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_3_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_3	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_3	POWER	POWER	1.8V			REG_ISENVCC_3_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_3_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_3_NR	SNS_D1FF_PHY	SENSE				Phase 4						REG_TD_4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4	VR_CTL_PHY	VR_CTL				REG_PWM_CPUVCC_4_R	VR_CTL_PHY	VR_CTL				REG_PHASE_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUVCC_4_RC	VR_D1D1_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_UGATE_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_CPUVCC_4	VR_D1D1_PHY	VR_SWITCH	12V	TRUE		PPCPUVCC_S0_SENSE_4	POWER	POWER	1.8V			REG_ISENVCC_4_P	SNS_D1FF_PHY	SENSE				REG_ISENVCC_4_N	SNS_D1FF_PHY	SENSE				REG_ISENVCC_4_NR	SNS_D1FF_PHY	SENSE				<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>D1D1</th><th>NO_TEST</th></tr><tr><td colspan="6">ISL6372</td></tr><tr><td>REG_CPUVCC_DVC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_DVC_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_RC_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_COMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_COMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_FB</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_R_1</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_FB_R_2</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_PSICOMP_RC</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_PSICOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_HFCOMP</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_R_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>CPU_VCCSENSE_R_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>SNS_VCC_XW_P</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td>1.8V</td><td></td><td></td></tr><tr><td>SNS_VCC_XW_N</td><td>SNS_D1FF_PHY</td><td>SENSE</td><td>0V</td><td></td><td></td></tr><tr><td>REG_CPUVCC_VSEN</td><td></td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_RGND</td><td></td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_VIN</td><td></td><td>SENSE</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_IMON</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPUVCC_IMON_R</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_TM</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_IMX</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_NPSI</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_FDVID</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_TMX</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_MEMVSEL</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>REG_CPUVCC_RSET</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSCLK</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDCLK_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDALERT_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDALERT_R_L</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSOUT</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td>CPU_VIDSOUT_R</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td></td></tr><tr><td colspan="6">Output Bus</td></tr><tr><td>PPCPUVCC_S0_CPU</td><td>POWER</td><td>POWER</td><td>1.8V</td><td></td><td></td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	D1D1	NO_TEST	ISL6372						REG_CPUVCC_DVC	VR_CTL_PHY	VR_CTL				CPUVCC_DVC_RC	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC_2	VR_CTL_PHY	VR_CTL				REG_CPUVCC_COMP	VR_CTL_PHY	VR_CTL				CPUVCC_COMP_RC	VR_CTL_PHY	VR_CTL				REG_CPUVCC_FB	VR_CTL_PHY	VR_CTL				CPUVCC_FB_RC	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_1	VR_CTL_PHY	VR_CTL				CPUVCC_FB_R_2	VR_CTL_PHY	VR_CTL				CPUVCC_PSICOMP_RC	VR_CTL_PHY	VR_CTL				REG_CPUVCC_PSICOMP	VR_CTL_PHY	VR_CTL				REG_CPUVCC_HFCOMP	VR_CTL_PHY	VR_CTL				CPU_VCCSENSE_P	SNS_D1FF_PHY	SENSE				CPU_VCCSENSE_N	SNS_D1FF_PHY	SENSE				CPU_VCCSENSE_R_P	SNS_D1FF_PHY	SENSE				CPU_VCCSENSE_R_N	SNS_D1FF_PHY	SENSE				SNS_VCC_XW_P	SNS_D1FF_PHY	SENSE	1.8V			SNS_VCC_XW_N	SNS_D1FF_PHY	SENSE	0V			REG_CPUVCC_VSEN		SENSE				REG_CPUVCC_RGND		SENSE				REG_CPUVCC_VIN		SENSE				REG_CPUVCC_IMON	VR_CTL_PHY	VR_CTL				CPUVCC_IMON_R	VR_CTL_PHY	VR_CTL				REG_CPUVCC_TM	VR_CTL_PHY	VR_CTL				REG_CPUVCC_IMX	VR_CTL_PHY	VR_CTL				REG_CPUVCC_NPSI	VR_CTL_PHY	VR_CTL				REG_CPUVCC_FDVID	VR_CTL_PHY	VR_CTL				REG_CPUVCC_TMX	VR_CTL_PHY	VR_CTL				REG_CPUVCC_MEMVSEL	VR_CTL_PHY	VR_CTL				REG_CPUVCC_RSET	VR_CTL_PHY	VR_CTL				CPU_VIDSCLK	VR_VID_PHY	VR_VID				CPU_VIDCLK_R	VR_VID_PHY	VR_VID				CPU_VIDALERT_L	VR_VID_PHY	VR_VID				CPU_VIDALERT_R_L	VR_VID_PHY	VR_VID				CPU_VIDSOUT	VR_VID_PHY	VR_VID				CPU_VIDSOUT_R	VR_VID_PHY	VR_VID				Output 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PPCPUVCC_S0_SENSE_3	POWER	POWER	1.8V																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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SNS_VCC_XW_P	SNS_D1FF_PHY	SENSE	1.8V																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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REG_CPUVCC_IMON	VR_CTL_PHY	VR_CTL																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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3.3V S5/5V S4

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U624 POWER	POWER	12V			REG VIN U7600 67
U626 POWER	POWER	5V			REG VCC1 U7600 67
U628 POWER	POWER	5V			REG VCC2 U7600 67
3.3V S5					
U646 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P3V3S5 67
U647 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T P3V3S5 67
U651 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T P3V3S5_RC 67
U648 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P3V3S5 67
U650 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P3V3S5 67
U652 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P3V3S5 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_ISEN 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_OCSET 67
U608 VR_CTL_PHY	VR_CTL				REG P3V3S5_FSET 67
U608 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_VOUT_R 67
U607 VR_CTL_PHY	VR_CTL				REG P3V3S5_FB 67
5V S3					
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG PHASE P5VS4 67
U616 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG B00T P5VS4 67
U618 VR_DIDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG B00T P5VS4_RC 67
U619 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG UGATE P5VS4 67
U620 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG LGATE P5VS4 67
U617 VR_DIDT_PHY	VR_SWITCH	12V	TRUE		REG SNUBBER P5VS4 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_ISEN 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_OCSET 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_FSET 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_VOUT_R 67
U616 VR_CTL_PHY	VR_CTL				REG P5VS4_FB 67
Output Bus					
U615 POWER	POWER	5V			PP5V_S5 67
U616 POWER	POWER	5V			PP5V_S4 67
U616 POWER	POWER	3.3V			PP3V3_S5 67
FET Switched					
U620 POWER	POWER	5V			PP5V_S0 67
U624 POWER	POWER	3.3V			PP3V3_S4 67
U620 POWER	POWER	3.3V			PP3V3_S0 67
U620 POWER	POWER	3.3V			PP3V3_S0_SSD 67
U620 POWER	POWER	3.3V			PP3V3_ENET 67
U620 POWER	POWER	3.3V			PP3V3_TBTLIC 67
Sensed					
U625 POWER	POWER	3.3V			PPSSD_S0 67
U627 POWER	POWER	3.3V			PP3V3_S4_AP 67

3.42V G3H

Physical	Spacing	Voltage	DIDT	NO_TEST	
3.42V G3H					
U630 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_BOOST 62
U630 POWER	VR_SWITCH	12V	TRUE		P3V42G3H_SW 62
U630 VR_CTL_PHY	VR_CTL				P3V42G3H_FB 62
U630 VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L 62
Output Bus					
U630 POWER	POWER	3.425V			PP3V42_G3H 67

3.3V G3

Physical	Spacing	Voltage	DIDT	NO_TEST	
U627 POWER	POWER	3.3V			PP3V3_G3 67

HDD S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
FET Switched					
U629 POWER	POWER	5V			PPHDD_S0 67
Sensed					
U630 POWER	POWER	5V			PP5V_S0_HDD 67

12V

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
U630 POWER	POWER	12V			PP12V_ACDC 67
FET Switched					
U630 POWER	POWER	12V			PP12V_S5 67
U630 POWER	POWER	12V			PP12V_S0 67
U630 POWER	POWER	12V			PP12V_S0_BLC 67
U630 POWER	POWER	12V			PPHDD12_S0 67
Sensed					
U630 POWER	POWER	12V			PP12V_G3H 67
U630 POWER	POWER	12V			PP12V_G3H_P3V42 62
U630 POWER	POWER	12V			PP12V_S0_GPUCORE 67
U630 POWER	POWER	12V			PP12V_S0_FBVDDQ 67
U630 POWER	POWER	12V			PP12V_S0_HDD 67

Ground/Common

Physical	Spacing	Voltage	DIDT	NO_TEST	
Common					
U630 GND	GND	0V			GND

SYNC_MASTER=J17_ULTIMATE

SYNC_DATE=05/16/2013

PAGE_TITLE

Platform VReg Constraints

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DRAWING NUMBER

051-9884

REVISION

A.0.0

BRANCH

PAGE

120 OF 127

SHEET

101 OF 106

8

7

6

5

4

3

2

1

Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTD_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_12C	*	=2x_DIELECTRIC	?
TBT_SPI	*	=2x_DIELECTRIC	?
TBTDP	*	=5x_DIELECTRIC	?
TBTDP	TOP, BOTTOM	=7x_DIELECTRIC	?
BGA_TBT_AREA	*	0. 075MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_TBT	BGA_TBT_AREA

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

Pairs should be within 100 mils of clock length.

Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.

DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

Electrical Constraint Set	Physical	Spacing		
H000	DP_850	DTSP1AYPORT	DP_TBTSNK0_ML_C_P<3...0>	28 83
H001	DP_850	DTSP1AYPORT	DP_TBTSNK0_ML_C_N<3...0>	28 83
H002	DP_TBTSNK0_ML	DP_850	DTSP1AYPORT	28
H003	DP_TBTSNK0_ML	DP_850	DTSP1AYPORT	28
H004	DP_850	DTSP1AYPORT	DP_TBTSNK0_AUXCH_C_P	28 83
H005	DP_850	DTSP1AYPORT	DP_TBTSNK0_AUXCH_C_N	28
H006	DP_TBTSNK0_AUX	DP_850	DTSP1AYPORT	28
H007	DP_TBTSNK0_AUX	DP_850	DTSP1AYPORT	28
H008	DP_850	DTSP1AYPORT	DP_TBTSNK1_ML_C_P<3...0>	28 83
H009	DP_850	DTSP1AYPORT	DP_TBTSNK1_ML_C_N<3...0>	28 83
H010	DP_TBTSNK1_ML	DP_850	DTSP1AYPORT	28
H011	DP_TBTSNK1_ML	DP_850	DTSP1AYPORT	28
H012	DP_850	DTSP1AYPORT	DP_TBTSNK1_AUXCH_C_P	28 83
H013	DP_850	DTSP1AYPORT	DP_TBTSNK1_AUXCH_C_N	28 83
H014	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28 83
H015	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H016	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H017	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H018	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H019	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H020	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H021	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H022	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H023	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H024	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H025	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H026	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H027	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H028	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H029	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H030	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H031	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H032	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H033	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H034	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H035	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H036	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H037	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H038	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H039	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H040	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H041	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H042	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H043	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H044	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H045	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H046	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H047	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H048	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H049	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H050	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H051	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H052	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H053	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H054	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H055	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H056	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H057	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H058	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H059	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H060	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H061	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H062	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H063	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H064	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H065	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H066	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H067	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H068	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H069	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	28
H070	DP_TBTSNK1_AUX	DP_850	DTSP1AYPORT	2

*: Only used on hosts supporting T29 video-in

DisplayPort

Electrical Constraint Set	Physical	Spacing	
Graphics Source			
FE00P DP_INTPNL_EG_ML_MUX	DP_B50	DISPLAYPORT	DP INT EG ML P<3..0> 43 83
FE00S DP_INTPNL_EG_ML_MUX	DP_B50	DISPLAYPORT	DP INT EG ML N<3..0> 43 83
FE00P DP_INTPNL_EG_AUX_MUX	DP_B50	DISPLAYPORT	DP INT EG AUX P 43 83
FE00S DP_INTPNL_EG_AUX_MUX	DP_B50	DISPLAYPORT	DP INT EG AUX N 43 83
FE00P DP_INTPNL_EG_AUX_MUX	DP_B50	DISPLAYPORT	DP INT EG AUX C P
FE00S DP_INTPNL_EG_AUX_MUX	DP_B50	DISPLAYPORT	DP INT EG AUX C N 43
Internal Panel			
FE00P	DP_B50	DISPLAYPORT	DP INTPNL ML C P<3..0> 43
FE00S	DP_B50	DISPLAYPORT	DP INTPNL ML C N<3..0> 43
FE00P DP_INTPNL_ML_CONN	DP_B50	DISPLAYPORT	DP INTPNL ML P<3..0> 42 43
FE00S DP_INTPNL_ML_CONN	DP_B50	DISPLAYPORT	DP INTPNL ML N<3..0> 42 43
FE00P DP_INTPNL_AUX_CONN	DP_B50	DISPLAYPORT	DP INTPNL AUX P 42 43
FE00S DP_INTPNL_AUX_CONN	DP_B50	DISPLAYPORT	DP INTPNL AUX N 42 43
Internal DP SPDIF			
FE00S		HDA	DP_INT SPDIF AUDIO 42 54

TBT/DP Net Properties

Electrical Constraint Set	Physical	Spacing	
Port A			
E621 TBT_A_R2D1	TBTDP_960	TBTDP	TBT_A_R2D_C_P<1>
E622 TBT_A_R2D1	TBTDP_960	TBTDP	TBT_A_R2D_C_N<1>
E623 TBT_A_R2D0	TBTDP_960	TBTDP	TBT_A_R2D_C_P<0>
E624 TBT_A_R2D0	TBTDP_960	TBTDP	TBT_A_R2D_C_N<0>
E625	TBTDP_960	TBTDP	TBT_A_R2D_P<1..0>
E626	TBTDP_960	TBTDP	TBT_A_R2D_N<1..0>
E627 DP_TBTPA_ML1	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_P<1>
E628 DP_TBTPA_ML1	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_N<1>
E629 DP_TBTPA_ML3	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_P<3>
E630 DP_TBTPA_ML3	DP_850	DISPLAYPORT	DP_TBTPA_ML_C_N<3>
E631	DP_850	DISPLAYPORT	DP_TBTPA_ML_P<1>
E632	DP_850	DISPLAYPORT	DP_TBTPA_ML_N<1>
E633	DP_850	DISPLAYPORT	DP_TBTPA_ML_P<3>
E634	DP_850	DISPLAYPORT	DP_TBTPA_ML_N<3>
E635 DP_A_LSX	DP_850	DISPLAYPORT	DP_A_LSX_ML_P<1>
E636 DP_A_LSX	DP_850	DISPLAYPORT	DP_A_LSX_ML_N<1>
E637	TBTDP_960	TBTDP	TBT_A_D2R_C_P<1..0>
E638	TBTDP_960	TBTDP	TBT_A_D2R_C_N<1..0>
E639 TBT_A_D2R1	TBTDP_960	TBTDP	TBT_A_D2R_P<1>
E640 TBT_A_D2R1	TBTDP_960	TBTDP	TBT_A_D2R_N<1>
E641 TBT_A_D2R0	TBTDP_960	TBTDP	TBT_A_D2R_P<0>
E642 TBT_A_D2R0	TBTDP_960	TBTDP	TBT_A_D2R_N<0>
E643 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_C_P
E644 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_C_N
E645	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_P
E646	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_N
E647 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP_A_AUXCH_DDC_P
E648 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP_A_AUXCH_DDC_N
E649	TBTDP_960	TBTDP	TBT_A_D2R1_AUXDDC_P
E650	TBTDP_960	TBTDP	TBT_A_D2R1_AUXDDC_N
Port B			
E651 TBT_B_R2D1	TBTDP_960	TBTDP	TBT_B_R2D_C_P<1>
E652 TBT_B_R2D1	TBTDP_960	TBTDP	TBT_B_R2D_C_N<1>
E653 TBT_B_R2D0	TBTDP_960	TBTDP	TBT_B_R2D_C_P<0>
E654 TBT_B_R2D0	TBTDP_960	TBTDP	TBT_B_R2D_C_N<0>
E655	TBTDP_960	TBTDP	TBT_B_R2D_P<1..0>
E656	TBTDP_960	TBTDP	TBT_B_R2D_N<1..0>
E657 DP_TBTPB_ML1	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_P<1>
E658 DP_TBTPB_ML1	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_N<1>
E659 DP_TBTPB_ML3	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_P<3>
E660 DP_TBTPB_ML3	DP_850	DISPLAYPORT	DP_TBTPB_ML_C_N<3>
E661	DP_850	DISPLAYPORT	DP_TBTPB_ML_P<1>
E662	DP_850	DISPLAYPORT	DP_TBTPB_ML_N<1>
E663	DP_850	DISPLAYPORT	DP_TBTPB_ML_P<3>
E664	DP_850	DISPLAYPORT	DP_TBTPB_ML_N<3>
E665 DP_B_LSX	DP_850	DISPLAYPORT	DP_B_LSX_ML_P<1>
E666 DP_B_LSX	DP_850	DISPLAYPORT	DP_B_LSX_ML_N<1>
E667	TBTDP_960	TBTDP	TBT_B_D2R_C_P<1..0>
E668	TBTDP_960	TBTDP	TBT_B_D2R_C_N<1..0>
E669 TBT_B_D2R1	TBTDP_960	TBTDP	TBT_B_D2R_P<1>
E670 TBT_B_D2R1	TBTDP_960	TBTDP	TBT_B_D2R_N<1>
E671 TBT_B_D2R0	TBTDP_960	TBTDP	TBT_B_D2R_P<0>
E672 TBT_B_D2R0	TBTDP_960	TBTDP	TBT_B_D2R_N<0>
E673 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_C_P
E674 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_C_N
E675	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_P
E676	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_N
E677 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP_B_AUXCH_DDC_P
E678 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP_B_AUXCH_DDC_N
E679	TBTDP_960	TBTDP	TBT_B_D2R1_AUXDDC_P
E680	TBTDP_960	TBTDP	TBT_B_D2R1_AUXDDC_N

SYNCH MASTER-J17 ULTIMATE		SYNCH DATE=05/16/2013	
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TBT/DP Constraints			
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		BRANCH	
		PAGE	121 OF 127
		SHEET	102 OF 106

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

Trace-to-Trace				Isolation					
Table	Micro	Design	Strip	Design	Micro	Design	Strip	Design	Comments
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Memory address (MA). Implemented 4.5 Gbps or less rules for K70.
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Address dynamic bus inversion (ADBI)
5-6/5-7	2:1	2:1	2:1	2:1	5:1	5:1	5:1	5:1	Control (CTRL)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Clock (CLK)
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data (DQ)
5-6/5-7	7:1	7:1	7:1	7:1	7:1	7:1	7:1	7:1	Error detection pins (EDC). Using larger isolation rules,
5-6/5-7	3:1	3:1	3:1	3:1	5:1	5:1	5:1	5:1	Data dynamic bus inversion (DBI)
5-6/5-7	5:1	5:1	5:1	5:1	5:1	5:1	5:1	5:1	Forwarded clock (WCK)

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3x_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3x_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints ($x \in \{A, B\}, y \in \{0, 1\}$)

Memory Address: M_{Axy}[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set		Physical	Spacing		
Memory Address					
MEM0	GDDR A0 MA	GDDR MA PHY	GDDR A 0 MA	FB A0 A<8..0>	NO TEST=TRUE
MEM1	GDDR A1 MA	GDDR MA PHY	GDDR A 1 MA	FB A1 A<8..0>	NO TEST=TRUE
Address Dynamic Bus Inv					
FE00	GDDR A0 ADBT	GDDR ADBT PHY	GDDR A 0 ADBT	FB A0 ABT L	NO TEST=TRUE
FE01	GDDR A1 ADBT	GDDR ADBT PHY	GDDR A 1 ADBT	FB A1 ABT L	NO TEST=TRUE
Control					
FE04	GDDR A0 CKE	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CKE L	NO TEST=TRUE
FE05	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CS L	NO TEST=TRUE
FE06	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 WE L	NO TEST=TRUE
FE07	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 CAS L	NO TEST=TRUE
FE08	GDDR A0 CTRL	GDDR CTRL PHY	GDDR A 0 CTRL	FB A0 RAS L	NO TEST=TRUE
FE09	GDDR A1 CKE	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CKE L	NO TEST=TRUE
FE0A	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CS L	NO TEST=TRUE
FE0B	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 WE L	NO TEST=TRUE
FE0C	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 CAS L	NO TEST=TRUE
FE0D	GDDR A1 CTRL	GDDR CTRL PHY	GDDR A 1 CTRL	FB A1 RAS L	NO TEST=TRUE
Clock					
FE0E	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK P	NO TEST=TRUE
FE0F	GDDR A0 CLK	GDDR CLK PHY	GDDR A 0 CLK	FB A0 CLK N	NO TEST=TRUE
FE10	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK P	NO TEST=TRUE
FE11	GDDR A1 CLK	GDDR CLK PHY	GDDR A 1 CLK	FB A1 CLK N	NO TEST=TRUE
Data					
FE12	GDDR A0 DQ BYTE0	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<7..0>	NO TEST=TRUE
FE13	GDDR A0 DQ BYTE1	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<15..8>	NO TEST=TRUE
FE14	GDDR A0 DQ BYTE2	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<23..16>	NO TEST=TRUE
FE15	GDDR A0 DQ BYTE3	GDDR DQ PHY	GDDR A 0 DQ	FB A0 DQ<31..24>	NO TEST=TRUE
FE16	GDDR A1 DQ BYTE0	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<7..0>	NO TEST=TRUE
FE17	GDDR A1 DQ BYTE1	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<15..8>	NO TEST=TRUE
FE18	GDDR A1 DQ BYTE2	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<23..16>	NO TEST=TRUE
FE19	GDDR A1 DQ BYTE3	GDDR DQ PHY	GDDR A 1 DQ	FB A1 DQ<31..24>	NO TEST=TRUE
Error Detection					
FE20	GDDR A0 EDC0	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<0>	NO TEST=TRUE
FE21	GDDR A0 EDC1	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<1>	NO TEST=TRUE
FE22	GDDR A0 EDC2	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<2>	NO TEST=TRUE
FE23	GDDR A0 EDC3	GDDR EDC PHY	GDDR A 0 EDC	FB A0 EDC<3>	NO TEST=TRUE
FE24	GDDR A1 EDC0	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<0>	NO TEST=TRUE
FE25	GDDR A1 EDC1	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<1>	NO TEST=TRUE
FE26	GDDR A1 EDC2	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<2>	NO TEST=TRUE
FE27	GDDR A1 EDC3	GDDR EDC PHY	GDDR A 1 EDC	FB A1 EDC<3>	NO TEST=TRUE
Data Dynamic Bus Inv					
FE28	GDDR A0 DBT0	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBT L<0>	NO TEST=TRUE
FE29	GDDR A0 DBT1	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<1>	NO TEST=TRUE
FE2A	GDDR A0 DBT2	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<2>	NO TEST=TRUE
FE2B	GDDR A0 DBT3	GDDR DBT PHY	GDDR A 0 DBT	FB A0 DBI L<3>	NO TEST=TRUE
FE2C	GDDR A1 DBT0	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<0>	NO TEST=TRUE
FE2D	GDDR A1 DBT1	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<1>	NO TEST=TRUE
FE2E	GDDR A1 DBT2	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<2>	NO TEST=TRUE
FE2F	GDDR A1 DBT3	GDDR DBT PHY	GDDR A 1 DBT	FB A1 DBI L<3>	NO TEST=TRUE
Forwarded Clock					
FE30	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<0>	NO TEST=TRUE
FE31	GDDR A0 WCK0	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<0>	NO TEST=TRUE
FE32	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK P<1>	NO TEST=TRUE
FE33	GDDR A0 WCK1	GDDR WCK PHY	GDDR A 0 WCK	FB A0 WCLK N<1>	NO TEST=TRUE
FE34	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<0>	NO TEST=TRUE
FE35	GDDR A1 WCK0	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<0>	NO TEST=TRUE
FE36	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK P<1>	NO TEST=TRUE
FE37	GDDR A1 WCK1	GDDR WCK PHY	GDDR A 1 WCK	FB A1 WCLK N<1>	NO TEST=TRUE

GPU

Electrical Constraint Set	Physical	Spacing	
Clocks			
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL 78
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG 78
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE 83
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM 78
PEX0	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID NO_TEST=TRUE 81
PEX0	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID NO_TEST=TRUE 82
PEX0	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK 83 85
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALOUT 83 84
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTALIN 83 84
SMB			
ES00	SMB_PHY	SMB	GPU_SMB_CLK 83 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT 83 84
ES00	SMB_PHY	SMB	GPU_SMB_CLK_R 49 84
ES00	SMB_PHY	SMB	GPU_SMB_DAT_R 49 84
PCIe Compensation			
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PD_VDDQ 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_PU_GND 80
ES00	PCIE_56S	COMP_PCIE	FB_CAL_TERM_GND 80

GDDR5 Frame Buffer B

Electrical Constraint Set		Physical	Spacing		
Memory Address					
H009	GDDR_B0_MA	GDDR_MA_PHY	GDDR_B_0_MA	FB_B0_A<8..0>	NO TEST=TRUE
H009	GDDR_B1_MA	GDDR_MA_PHY	GDDR_B_1_MA	FB_B1_A<8..0>	NO TEST=TRUE
Address Dynamic Bus Inv					
H009	GDDR_B0_ADBT	GDDR_ADBT_PHY	GDDR_B_0_ADBT	FB_B0_ABT_L	NO TEST=TRUE
H009	GDDR_B1_ADBT	GDDR_ADBT_PHY	GDDR_B_1_ADBT	FB_B1_ABT_L	NO TEST=TRUE
Control					
H009	GDDR_B0_CKE	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CKE_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CS_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_WE_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_CAS_L	NO TEST=TRUE
H009	GDDR_B0_CTRL	GDDR_CTRL_PHY	GDDR_B_0_CTRL	FB_B0_RAS_L	NO TEST=TRUE
H009	GDDR_B1_CKE	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CKE_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CS_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_WE_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_CAS_L	NO TEST=TRUE
H009	GDDR_B1_CTRL	GDDR_CTRL_PHY	GDDR_B_1_CTRL	FB_B1_RAS_L	NO TEST=TRUE
Clock					
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_P	NO TEST=TRUE
H009	GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_B_0_CLK	FB_B0_CLK_N	NO TEST=TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_P	NO TEST=TRUE
H009	GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_B_1_CLK	FB_B1_CLK_N	NO TEST=TRUE
Data					
H009	GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<7..0>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<15..8>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<23..16>	NO TEST=TRUE
H009	GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_0_DQ	FB_B0_DQ<31..24>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<7..0>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<15..8>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<23..16>	NO TEST=TRUE
H009	GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_B_1_DQ	FB_B1_DQ<31..24>	NO TEST=TRUE
Error Detection					
H009	GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<0>	NO TEST=TRUE
H009	GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<1>	NO TEST=TRUE
H009	GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<2>	NO TEST=TRUE
H009	GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_B_0_EDC	FB_B0_EDC<3>	NO TEST=TRUE
H009	GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<0>	NO TEST=TRUE
H009	GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<1>	NO TEST=TRUE
H009	GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<2>	NO TEST=TRUE
H009	GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_B_1_EDC	FB_B1_EDC<3>	NO TEST=TRUE
Data Dynamic Bus Inv					
H009	GDDR_B0_DBI0	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<0>	NO TEST=TRUE
H009	GDDR_B0_DBI1	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<1>	NO TEST=TRUE
H009	GDDR_B0_DBI2	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<2>	NO TEST=TRUE
H009	GDDR_B0_DBI3	GDDR_DBI_PHY	GDDR_B_0_DBI	FB_B0_DBI_L<3>	NO TEST=TRUE
H009	GDDR_B1_DBI0	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<0>	NO TEST=TRUE
H009	GDDR_B1_DBI1	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<1>	NO TEST=TRUE
H009	GDDR_B1_DBI2	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<2>	NO TEST=TRUE
H009	GDDR_B1_DBI3	GDDR_DBI_PHY	GDDR_B_1_DBI	FB_B1_DBI_L<3>	NO TEST=TRUE
Forwarded Clock					
H009	GDDR_B0_WCLK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<0>	NO TEST=TRUE
H009	GDDR_B0_WCLK0	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<0>	NO TEST=TRUE
H009	GDDR_B0_WCLK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_P<1>	NO TEST=TRUE
H009	GDDR_B0_WCLK1	GDDR_WCK_PHY	GDDR_B_0_WCK	FB_B0_WCLK_N<1>	NO TEST=TRUE
H009	GDDR_B1_WCLK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<0>	NO TEST=TRUE
H009	GDDR_B1_WCLK0	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<0>	NO TEST=TRUE
H009	GDDR_B1_WCLK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_P<1>	NO TEST=TRUE
H009	GDDR_B1_WCLK1	GDDR_WCK_PHY	GDDR_B_1_WCK	FB_B1_WCLK_N<1>	NO TEST=TRUE

Frame Buffer Reset

Electrical Constraint Set		Physical	Spacing	
Reset				
H442	GD0R_A0_RESET	GD0R_56S	GD0R_CTR1	FB_A0_RESET_L
H445	GD0R_A1_RESET	GD0R_56S	GD0R_CTR1	FB_A1_RESET_L
H446	GD0R_B0_RESET	GD0R_56S	GD0R_CTR1	FB_B0_RESET_L
H447	GD0R_B1_RESET	GD0R_56S	GD0R_CTR1	FB_B1_RESET_L
H448	GD0R_C0_RESET	GD0R_56S	GD0R_CTR1	FB_C0_RESET_L
H449	GD0R_C1_RESET	GD0R_56S	GD0R_CTR1	FB_C1_RESET_L
H450	GD0R_D0_RESET	GD0R_56S	GD0R_CTR1	FB_D0_RESET_L
H451	GD0R_D1_RESET	GD0R_56S	GD0R_CTR1	FB_D1_RESET_L

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PAGE TITLE		PAGE	
GDDR5/GPU Constraints			
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		REVISION A.0.0	
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		SHEET 103 OF 106	

D

C

B

A

Constraints

BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Physical		Spacing	Voltage	DIDT	NO_TEST	
Input	Bus					
H460	POWER	POWER	12V			PP12V_S0_BLC_VIN2
H477	POWER	POWER	12V			PP12V_S0_BLC_VINP
H365	POWER	POWER	14V			PRE_REG_OUT
H107	POWER	POWER	3.3V			BLC_P3V3S
H107	POWER	POWER	3.3V			BLC_P3V3_REF
H365	POWER	POWER	3.3V			BLC_P3V3
H107	POWER	POWER	3.3V			PP3V3_S0_BLC_R
H104	POWER	POWER	5V			SPTX_VIN
H101	POWER	POWER	5V			PP5V_BLC
H104	POWER	POWER	5V			BLC_VIN2
H107	POWER	POWER	14V			BOOST_FET_DRAIN
H105	POWER	POWER	12V			BOOST_VDD
H105	POWER	POWER	5V			PP5V_S0_BLC_R
H105	POWER	POWER	12V			PP12V_S0_BLC_F
Local Ground						
H460	BLC_CTL_PHY	BLC_PHASE	5V			BLC_GND_1
H105	BLC_CTL_PHY	BLC_PHASE	5V			BLC_GND_2
H105	BLC_CTL_PHY	BLC_PHASE	5V			BLC_GND_3
H105	GND	GND	5V			AGND_BLC
Backlight						
H460	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1
H460	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R
H460	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2
H460	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R
H105	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3
H460	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R
H750	BLC_CTL_PHY	BLC_CTL				LED_DRV_R_CS_RC_1
H750	BLC_CTL_PHY	BLC_CTL				LED_DRV_R_CS_RC_2
H460	BLC_CTL_PHY	BLC_CTL				LED_DRV_R_CS_RC_3
H105	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1
H105	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2
H105	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3
H107	BLC_CTL_PHY	BLC_CTL				LED_DRV_R_CS_C1
H107	BLC_CTL_PHY	BLC_CTL				LED_DRV_R_CS_C2
H107	BLC_CTL_PHY	BLC_CTL				LED_DRV_R_CS_C3
H107	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_1
H107	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_2
H107	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_3
H107	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK1
H107	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK2
H107	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK3
H107	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R
H107	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R
H107	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R
H107	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1
H107	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2
H107	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3
H107	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1
H107	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2
H107	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3
H460	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1
H460	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2
H460	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3
H107	BLC_CTL_PHY	BLC_CTL				BCOMP1
H107	BLC_CTL_PHY	BLC_CTL				BCOMP2
H107	BLC_CTL_PHY	BLC_CTL				BCOMP3
H107	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1
H107	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2
H107	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT3
H107	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1
H107	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2
H107	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3
H107	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R
H105	BLC_CTL_PHY	BLC_CTL				BOOST_FB
H105	BLC_CTL_PHY	BLC_CTL				BOOST_COMP
H105	BLC_CTL_PHY	BLC_CTL				BOOST_COMP_C
H105	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV
H105	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV_R
H105	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS
H105	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS_R
H460	BLC_CTL_PHY	BLC_HV	80V			LED_DRV_R_DRAIN_1
H107	BLC_CTL_PHY	BLC_HV	80V			LED_DRV_R_DRAIN_2
H460	BLC_CTL_PHY	BLC_HV	80V			LED_DRV_R_DRAIN_3
OUTPUT BUS						
H460	POWER_BLC_RET	BLC_HV	80V			IS1_BLC_F
H460	POWER_BLC_RET	BLC_HV	80V			IS2_BLC_F
H460	POWER_BLC_RET	BLC_HV	80V			IS3_BLC_F
H460	POWER_BLC_RET	BLC_HV	80V			IS1_BLC
H460	POWER_BLC_RET	BLC_HV	80V			IS2_BLC
H460	POWER_BLC_RET	BLC_HV	80V			IS3_BLC
H107	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_1
H107	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_1
H107	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_2
H107	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_2
H107	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_3
H107	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_3
H107	POWER_BLC	BLC_HV	80V			BLC_VOUT1
H107	POWER_BLC	BLC_HV	80V			BLC_VOUT2
H107	POWER_BLC	BLC_HV	80V			BLC_VOUT3

D

GPU CORE PHASES

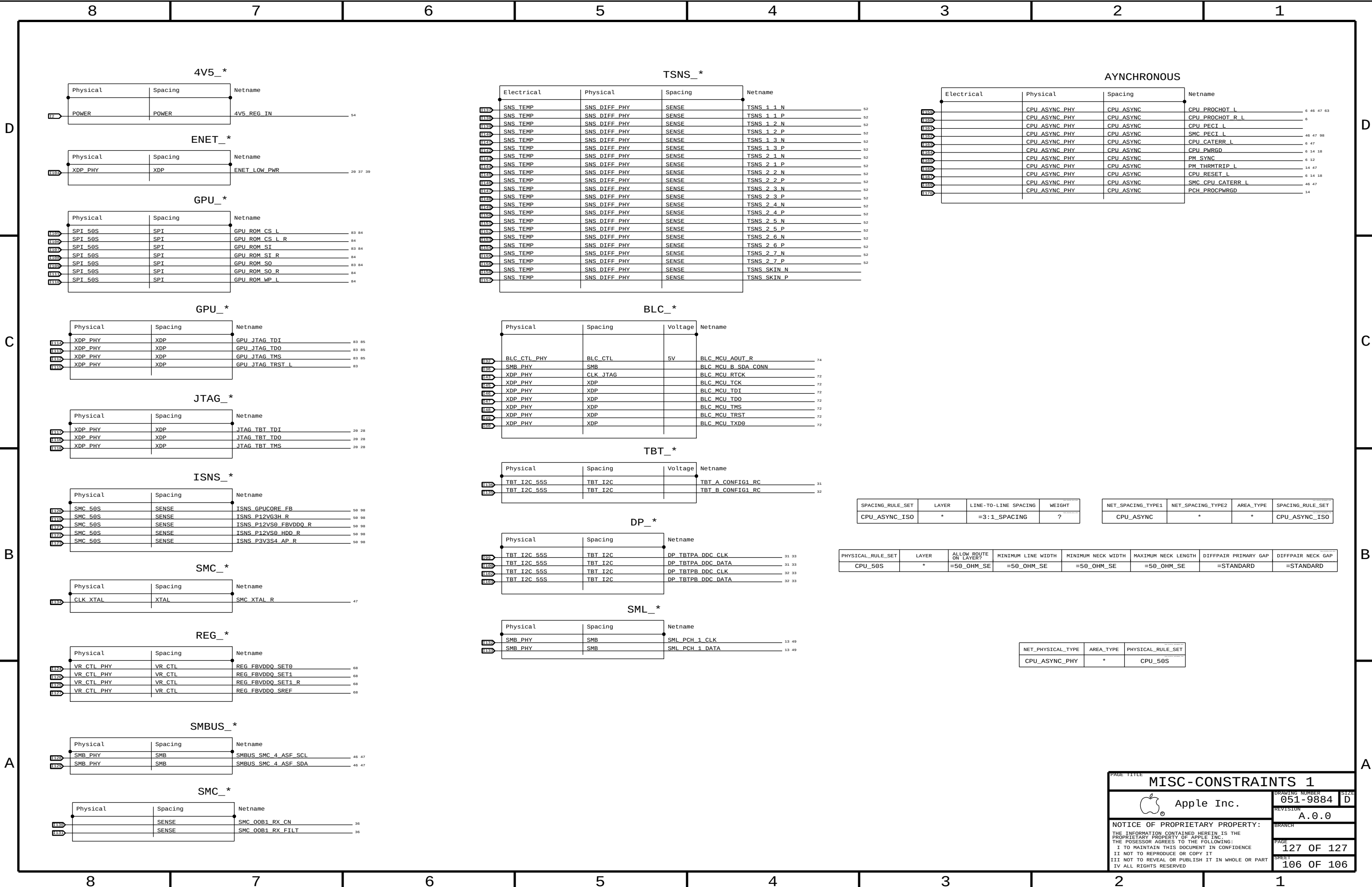
Electrical Contrainst Set		Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus							
1000		POWER	POWER	12V			PP12V_S0_GPUCORE_FLT
1010		POWER	POWER	5V			PP5V_S0_GPU_VCORE_VCC
Local Ground							
1020		GND	GND	0V			AGND_GPU
Phase 1							
1030		POWER	POWER	12V			REG_LVCC_UB510
1040		POWER	POWER	12V			REG_UVCC_UB510
1050		VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_1
1060		VR_CTL_PHY	VR_CTL				VR_GPU_PWM1_R
1070		POWER	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_1
1080		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_1
1090		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_1_RC
1100		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_1
1110		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_1
1120		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_1
1130		POWER	POWER	0.9V			PPGPUCORE_S0_SENSE_1
1140	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_1_P
1150	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_1_N
1160	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS1_R_P
1170	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS1_R_N
1180	TSNS_GPU_CORE		SENSE				VR_GPU_ISNS1_RR_2
Phase 2							
1190		POWER	POWER	12V			REG_LVCC_UB530
1200		VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_2
1210		VR_CTL_PHY	VR_CTL				VR_GPU_PWM2_R
1220		POWER	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_2
1230		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_2
1240		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_2_RC
1250		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_2
1260		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_2
1270		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_2
1280		POWER	POWER	0.9V			PPGPUCORE_S0_SENSE_2
1290	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_2_P
1300	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_2_N
1310	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS2_R_P
1320	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS2_R_N
1330	TSNS_GPU_CORE		SENSE				VR_GPU_ISNS2_RR_2
Phase 3							
1340		POWER	POWER	12V			REG_VCC_UB550
1350		POWER	POWER	12V			REG_UVCC_UB550
1360		POWER	POWER	12V			REG_LVCC_UB550
1370		VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_3
1380		VR_CTL_PHY	VR_CTL				VR_GPU_PWM3_R
1390		POWER	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_3
1400		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_3
1410		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_3_RC
1420		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_3
1430		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_3
1440		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_3
1450		POWER	POWER	0.9V			PPGPUCORE_S0_SENSE_3
1460	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_3_P
1470	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_3_N
1480	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS3_R_P
1490	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS3_R_N
1500	TSNS_GPU_CORE		SENSE				VR_GPU_ISNS3_RR_2
Phase 4							
1510		POWER	POWER	12V			REG_LVCC_UB650
1520		VR_CTL_PHY	VR_CTL				REG_PWM_GPUCORE_4
1530		VR_CTL_PHY	VR_CTL				VR_GPU_PWM4_R
1540		POWER	VR_SWITCH	12V	TRUE		REG_PHASE_GPUCORE_4
1550		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_GPUCORE_4
1560		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_GPUCORE_4_RC
1570		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_GPUCORE_4
1580		VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_GPUCORE_4
1590		VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_GPUCORE_4
1600		POWER	POWER	0.9V			PPGPUCORE_S0_SENSE_4
1610	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_4_P
1620	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				REG_ISEN_GCORE_4_N
1630	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS4_R_P
1640	TSNS_GPU_CORE	SNS_DIEF_PHY	SENSE				VR_GPU_ISNS4_R_N
1650	TSNS_GPU_CORE		SENSE				VR_GPU_ISNS4_RR_2

GPU CORE CONTROLLER

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
ISL6334					
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_COMP
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_COMP_R
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_COMP_RC
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_FB
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_FB_R
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_VDIFF
REG00	VR_CTL_PHY	VR_CTL			VR_VDF_R1
REG00	VR_CTL_PHY	VR_CTL			VR_VDF_R2
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_TCAMP
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_OFS
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_FS
REG00		VR_CTL			VR_GPU_EN_VTT
REG00		VR_CTL			PM_EN_REG_GPUCORE_S0
REG00		VR_CTL			PM_PGOOD_REG_GPUCORE_S0
REG00		VR_CTL			GPU_PSI_L
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_IOUT
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_IMON
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_FAN
REG00		VR_CTL			VR_GPU_VRDMOT
REG00		VR_CTL			VR_GPU_EN_PWR
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_SS
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_DAC
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_REF
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_TM
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_IMON
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_FAN
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_VRHOT
REG00		VR_CTL			GPU_PSI_R
REG00		VR_CTL			GPU_PSI_L_R
REG00	VR_CTL_PHY	VR_CTL			VR_GPU_IMON_R
VSNS_GPU_VDD	SNS_DIFF_PHY	SENSE			VSNS_GPU_VDD
VSNS_GPU_VSS	SNS_DIFF_PHY	SENSE			VSNS_GPU_VSS
SEN00	SNS_DIFF_PHY	SENSE			VR_GPU_VSEN
SEN00	SNS_DIFF_PHY	SENSE			VR_GPU_RGND
GPU_VIDS					
REG00		VR_VID			REG_GPUCORE_VID7
REG00		VR_VID			REG_GPUCORE_VID6
REG00		VR_VID			REG_GPUCORE_VID5
REG00		VR_VID			REG_GPUCORE_VID4
REG00		VR_VID			REG_GPUCORE_VID3
REG00		VR_VID			REG_GPUCORE_VID2
REG00		VR_VID			REG_GPUCORE_VID1
REG00		VR_VID			REG_GPUCORE_VID0
REG00		VR_VID			GPU_VCORE_VID6
REG00		VR_VID			GPU_VCORE_VID5
REG00		VR_VID			GPU_VCORE_VID4
REG00		VR_VID			GPU_VCORE_VID3
REG00		VR_VID			GPU_VCORE_VID2
REG00		VR_VID			GPU_VCORE_VID1
REG00		VR_VID			GPU_VCORE_VID0
Output Bus					
REG00	POWER	POWER	0.9V		PPVCORE_S0_GPU

GPU FBVDDQ

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
	POWER	POWER	5V			REG_VCC_U7750
	POWER	POWER	5V			REG_PVCC_U7750
Local Ground						
	GND	GND	0V			AGND_FBVDDQ
FBVDDQ						
	POWER	VR_SWITCH	12V	TRUE		REG_PHASE_FBVDDQ
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_FBVDDQ
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_FBVDDQ_RC
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_FBVDDQ
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_FBVDDQ_R
	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_FBVDDQ
	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_FBVDDQ
		SENSE				VSNS_FBVDDQ
	VSNS_GPU_VDDQ	SENSE				VSNS_FBVDDQ_P
	VSNS_GPU_VDDQ	SENSE				VSNS_FBVDDQ_N
		SENSE				FBVDDQ_SENSE_R
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_OCSET
	VR_CTL_PHY	VR_CTL				REG_FBVDDQ_VO
		SENSE				REG_FBVDDQ_FB
		SENSE				REG_FBVDDQ_RTN
		SENSE				REG_FBVDDQ_FB_R
		SENSE				REG_FBVDDQ_RTN_R
Output Bus						
	POWER	POWER	1.5V			PPFBVDDQ_S0_GPU



SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ASYNC_ISO	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_ASYNC	*	*	CPU_ASYNC_ISO

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_ASYNC_PHY	*	CPU_50S

MISC-CONSTRAINTS 1

Apple Inc.

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127 OF 127

106 OF 106